

An operational perturbation proteomics-based virtual cell model

Corresponding Author: Professor Tiannan Guo

Any redactions in this file are there to maintain patient confidentiality, the confidentiality of unpublished data, or to remove third-party material.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 2:

Reviewer comments:

Referee #1

(Remarks to the Author)

The manuscript by Guo and colleagues describes a massive amount of theoretical work constructing a digital cell using perturbation measurements in breast cancer cell lines, resulting in a “neural ordinary differential equation-based foundation model” called ProteinTalks.

Originality and significance: While not an original idea, the scale of the project is impressive (“quantity is a quality of its own”). The authors claim their model predicts drug efficacy and synergy in cell lines and xenografts. The main limitation is this still needs validation in clinical studies, as cancer is a multi/system-level disease, imperfectly modelled by cells in culture.

The methodology used by the authors is sound, and the quantity of data impressive. The relatively modest number of quantified proteins (around 5,000) is not surprising, as the authors used a 15-minute method. Large-scale proteomics efforts generally have to sacrifice some proteome depth for sample throughput. In general, the quality of the presentation is also sufficient to understand what the authors did and why.

Importantly for this kind of work, the raw data is deposited in the ProteomeXchange Consortium PRIDE repository, and the data analysis code available in GitHub. This means the data can be used by other researchers, replicating or expanding on the analyses presented in this manuscript.

The statistical approach appears solid. The two-sided unpaired Welch’s with Benjamini-Hochberg correction used to find differentially abundant proteins is, if anything, on the conservative side, but this can be motivated by the scale of the study.

The next step would be to validate the predicted prescription or combinations of drugs in actual clinical trials, or use data from past or ongoing trials to validate some aspects of the model. This would significantly raise the impact of this work.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

Review comments for “A perturbation proteomics-based foundation model for digital cell construction”

Summary

The paper presents a novel proteome foundation model named ProteinTalks, designed to understand biological processes within cells and predict protein network dynamics. This model is built on an extensive dataset of over 38 million protein measurements from 16,000 perturbed proteomic samples, making it the largest proteomics dataset to date. ProteinTalks

employs a neural ordinary differential equation (ODE) framework integrated with perturbation-aware neural networks to capture time-dependent behavior of cellular systems. The model demonstrates significant improvements in predicting drug efficacy, synergy, and resistance-associated proteins, and shows potential for enhancing prognosis predictions in clinical settings, particularly in breast cancer. While the model presents a very interesting advance in training proteome-based models for biological insights, its superiority and broad utility remains questionable.

Major comments

- The key main comment of a more general character is that the authors should perform more extensive comparisons to other models before claiming superiority, as also noted more specifically below.
- For comparison with other methods in benchmarking, there is no hyperparameter optimization for the relevant methods. The performance of other techniques is close to the ProteinTalks model (e.g., AUPRC of 0.854 for ProteinTalks vs 0.822 for RandomForest), which would warrant the best hyperparameter search for the other methods to assess if the proposed approach is indeed the best. It may be so that the other methods are superior if their hyperparameters are optimized.
- Furthermore, the benchmark models are only compared to the proposed approach in setting 1 (randomly divided), not in settings 2 (leave-one-cell_line-out) and 3(leave-one-drug-out), which is not reasonable since the proposed approach should also outperform the benchmark models in those data splits.
- The authors claim that ProteinTalks is a foundation model to model the perturbed proteomic networks, however, this neural ODE model seems to be very limited to certain tasks and does not exhibit potential to scale up and generalize to diverse tasks. Even within this paper's scope, when testing the pretrained model's performance on the newly acquired dataset (98 additional anti-cancer small-molecular 224 compounds), it cannot match the performance on the original test set. If the authors are to claim that this is a foundation model, they should demonstrate generalizability to numerous tasks.
- SHAP score was used to evaluate the interpretability of the model and performed some retrospective analysis to uncover new insights in drug resistance mechanisms, which is good. This approach can be enhanced by integrating the distinctive aspects of ODE parameter estimation, incorporating uncertainty quantification and prospective interpretation of model decisions, thereby enabling scalability and deployment for automated discovery of novel pathways or proteins of interest.
- The paper would be strengthened if the authors could demonstrate that the obtained insights are indeed obtained due to the massive proteome dataset, and not simply due to the constraints to known pathways.
- How did the authors manage proteins that fall outside of known pathways in the ODE?
- PertScore seems to be an ad-hoc metric used to quantify the protein expression changes over time. This score is dependent on determining if a protein expression is upgraded or downgraded, which could be subjective. In addition, given any cutoff of this score (10 used in this paper), what is the false discovery rate in the downstream applications?
- In the discussion of predicting chemotherapy 345 sensitivity in these biopsy tissue samples, it is okay to compare 60 proteins selected by ProteinTalks to random selection, but it is more interesting to see if this comparison is done with knowledge-driven and other benchmarking models' selection. In addition, among these 60 proteins, are they mostly previously known biomarkers? Any new findings? And if there are new findings, are they thanks to the model or the proteome-wide dataset, or both?

Minor comments

- The writing requires improvement, particularly in the Methods section, where sentence flow could be smoother and phrasing more concise.

(Remarks on code availability)

I'm of the opinion that this paper presents marginal advantages to existing models and poses exaggerated claims about having developed a "foundation model" without the data to back that up with broad and generalizable applications. Due to the lack of novel biological insights, I believe this paper would be more suitable for a journal like Nature Methods.

Referee #3

(Remarks to the Author)

Overall, this work presents a great data set that took a lot of work to produce.

However, we feel that the AI model is not that innovative and we worry that 3 data points are not enough to get a meaningful embedding.

Thus, while enthusiastic, we are not sure if this is over the bar for Nature.

Summary of the key results

- Summary of the model:

A neural ODE based foundational model, ProteinTalks, was built based on 38 million perturbed protein measurements in breast cancer cell lines. It learned cellular protein network dynamics and predicted drug efficacy, synergy, and patient responses to drugs.

- Summary of results:

- (1) Generated a large scale-perturbation proteome dataset from DIA-MS analytics of 18 breast cancer cell lines treated with 63 FDA-approved drugs at 6, 24, 48 hours.
 - (2) Proposed PertScore, a quantitative metric that captures drug-induced protein alterations, reflecting the drug's MOA.
 - (3) Longitudinal proteomic responses to drug perturbation distinguishes between chemotherapy-resistant and sensitive groups.
 - (4) A two-module foundation model - the first module ingests untreated proteome data, storing protein network dynamics, and predicts perturbed protein expression at 6, 24, 48 hours; the second module integrates the unperturbed proteomes, the predicted perturbed proteomes, and molecular structures of drugs to predict drug efficacy and combo synergy using MLP.
 - (5) ProteinTalks outperformed typical ML models in predicting drug responsiveness.
 - (6) The model's capability of predicting drug synergy was validated experimentally.
- SHAP values reflect the model's capability of identifying pathways and protein networks responsible for drug resistance.
- (8) Transfer learning of drug sensitivity prediction was performed on breast cancer patient-derived tumor cells and pan-cancer cells from PDTX models.
 - (9) Drug response was tested using clinical biopsy tissue samples both in short-term and long-term.

Originality and significance:

- Significance: this study is at least ten times larger in size compared to previous studies, enabling it to be directly used for training AI models

- Direct applications:

- (1) Assisting drug discovery by uncovering drug MOAs
- (2) Predicting the prognosis of breast cancer patients when integrated with clinical patient data

Data & methodology: validity of approach, quality of data, quality of presentation

- Question on quality of validity of approach:

The inherent complexity of DIA data can often lead to false-positive identifications, especially in the presence of perturbations. Did the authors consider and adopt any strategies to mitigate false-positives? How was the accuracy (or potential inaccuracy) of the DIA-NN accounted for?

- Question on model performance:

In Figure S4, the model performance on the test set appears to be poor for cell line DU4475 in all three sub figures (Synergy Accuracy, AUPRC, and AUROC) despite the excellent train and validation performance. Any explanation for this?

- Question on quality of presentation:

- (1) Figure S3 seems to be missing Figure caption for subfigure B.
- (2) The paper mixed up the description of results and methods, making it not straightforward for readers to understand. This is seen in the 9 subsections under Results.

Appropriate use of statistics and treatment of uncertainties

Strength:

- The authors conducted a massive perturbation proteomics study with over 38 million protein measurements. The authors also applied stringent quality control. For example, multiple control samples were used to account for batch effects and approximately 10% of randomly selected samples were run in replicates for quality control.
- The authors used a robust DE pipeline. They combined Welch's t-test and Benjamini-Hochberg adjustment to ensure false discovery control.
- SHAP values were used for interpreting model decisions, giving interpretation of protein importance in model outputs. The authors also performed several cross validation studies to ensure robustness of the method. Ablation studies were also performed to demonstrate the value of proteomic input vs. transcriptomic or metadata-only baselines.

Limitations:

- The specific p-values and effect sizes of protein alterations are not always reported in the main text. For example, the authors could provide log2 fold-change and give more biological relevance of the finding beyond saying it is statistically significant.
- The authors report AUROC, AUPRC, and accuracy, but don't show confidence intervals for those metrics.

Conclusions: robustness, validity, reliability

- The model has shown strong robustness in several validation settings: leave-one-cell-line-out and leave-one-drug-out. The

authors also performed transfer learning on transcriptomic data to confirm its robustness

- Experimental validation was performed. Specifically, the knockdown of AKR1C3 and CMPK1 increased drug sensitivity in expected ways. The model was also validated by discovering known drug-protein interactions.
- Model performance was consistent across various test datasets, achieving a decent AUROC score across settings.

Suggested improvements: experiments, data for possible revision

- The authors could try to include more time points as three time points are maybe too few to truly learn meaningful dynamics, especially in a high-dimensional proteomic space.
- The encoder input only takes categorical inputs like drug, cell line and time points. Is it better to also add some baseline omics to guide ODE dynamics better? (Both cell line and drug ID are one-hot embedding and time point is also one-hot or scalar, so including some omics data may be better for describing the initial state)
- The latent trajectory is not interpreted in the paper.

References: appropriate credit to previous work?

- Yes, the paper appropriately cites foundational and recent work. Citations are current and cover biological and computational domains.
- The authors used the software DIA-NN for DIA-MS data analytics, so we would recommend citing its original paper in addition to references 40 and 41 which described how to use the software. The original paper by the Francis-Crick and Cambridge group is: DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput (<https://www.nature.com/articles/s41592-019-0638-x>)

Clarity and context: lucidity of abstract/summary, appropriateness of abstract, introduction and conclusions

- The results and methods sections of the paper left the readers with a general impression that the ProteinTalks model contributes to drug discovery and predicting cancer patient responses to drugs. The quality and significance of the ML model is clear; however, the Abstract, Introduction, and Discussion, seem to emphasize how the model is a good foundation for constructing a digital twin to biological cancer cells in humans. Not sure if it is immediately obvious for the readers to arrive at this conclusion without providing more context on the digital cell concept. To be precise, a good digital cell should ideally integrate all types of variables relevant to pathogenesis, not just drug perturbations. What are the authors trying to convey by including this digital cell notion?
- An example of how people have provided context for the digital cell concept when trying to emphasize this in their paper: <https://genomemedicine.biomedcentral.com/articles/10.1186/s13073-022-01048-4>

(Remarks on code availability)

The codeocean repository has a 0B data directory. Was it supposed to be left empty or the authors meant to share something?

Version 3:

Reviewer comments:

Referee #1

(Remarks to the Author)

The authors have successfully addressed my suggestions. I have no further comments.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

The authors have addressed my comments. Experiments on organoids and biopsies are a nice addition.

I'm still hesitant to whether this paper represents a significant advance over current models that it warrants publication in Nature. The data is super nice, and it's obvious that the data produces a significant advantage in that proteins creates a closer proxy to drug response than transcriptome embeddings or other embeddings that the authors compare to. The innovation in terms of the AI model remains low, and calling it a "virtual cell model" is definitely a stretch.

(Remarks on code availability)

Referee #3

(Remarks to the Author)

Overall Assessment

The authors addressed the requested revisions, but we remain unconvinced by the proposed model and are therefore uncertain about recommending acceptance.

1. Summary of the key results

ProteinTalks is a neural ODE-based virtual cell model trained on 38 million perturbed protein measurements from 18 breast cancer cell lines treated with 63 FDA-approved drugs at three time points. The model outperforms transcriptomic foundation models (Geneformer, GeneCompass, UCE) by 63–213% in AUPRC for cross-cell-line validation. Key validations include: knockdown confirmation of resistance proteins (AKR1C3, CMPK1, TYMS), clinical stratification of 504 TNBC patients, and prospective PDO screening of ~3,000 compounds identifying three candidates (CAY10603, citarinstat, THZ-1) that outperformed cisplatin.

2. Originality and significance

Strengths: First proteome-based virtual cell model with temporal dynamics; dataset ~10-fold larger than comparable studies; ~1000-fold fewer parameters than transcriptomic foundation models while achieving superior performance.

Addressed: Three time points justified theoretically (dynamical systems analysis) and empirically (10-time-point benchmark confirming 6h/24h/48h capture early/intermediate/late phases). Scaling law analysis supports foundation model claims.

3. Data & methodology

Addressed: DIA-MS with 1% FDR appropriate; rigorous QC (Pearson $r > 0.9$, CV $< 10\%$); systematic hyperparameter optimization for all baselines (Tables S4A-B); Results-Methods mapping table added; figure captions corrected.

4. Appropriate use of statistics and treatment of uncertainties

Addressed: Performance metrics now include SE/SD (Tables S5-S7, S9, S11); effect sizes reported in main text (e.g., ATG3 0.15-fold, CKS2 7.71-fold). SHAP uncertainty quantified.

Minor limitation: PertScore sensitivity analysis (85% consistency) lacks formal FDR estimate.

5. Conclusions: robustness, validity, reliability

Robustness: Consistent superiority across seven baselines; scaling law confirmed; DU4475 poor performance explained by class imbalance.

Validity: Knockdown validations; drug synergies confirmed ($\Delta E_{\max}$ 0.28–0.74); clinical alignment with ongoing trials (NCT02286687); prospective PDO validation successful.

6. Suggested improvements

Experimental validation of at least one novel protein (15 identified without prior cancer links)

Show negative transcriptomics integration result mentioned but not presented

Formal FDR for PertScore

7. References

DIA-NN original paper now cited (reference 47). Appropriate coverage of recent foundation model critiques (Boiarsky et al., Ahlmann-Eltze et al.).

8. Clarity and context

Title appropriately revised to "virtual cell"; clear conceptual positioning; limitations acknowledged (no PTMs, cancer cell lines only, single-omics).

(Remarks on code availability)

Code reproducibility verified

Version 4:

Reviewer comments:

Referee #2

(Remarks to the Author)

The reviewers have addressed my concerns. While this is a very nice manuscript that proves the value of proteomics data for drug response cell modeling, I remain skeptical to if it reaches the novelty or impact bar required by Nature.

(Remarks on code availability)

Referee #3

(Remarks to the Author)

All my previous concerns have been adequately addressed. I recommend proceeding with publication.

(Remarks on code availability)
Code reproducibility verified previously

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Point-by-point reply

Referees' comments:

Referee #1 (Remarks to the Author):

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The methodology used by the authors is sound, and the quantity of data impressive. The relatively modest number of quantified proteins (around 5,000) is not surprising, as the authors used a 15-minute method. Large-scale proteomics efforts generally have to sacrifice some proteome depth for sample throughput. In general, the quality of the presentation is also sufficient to understand what the authors did and why.

Importantly for this kind of work, the raw data is deposited in the ProteomeXchange Consortium PRIDE repository, and the data analysis code available in GitHub. This means the data can be used by other researchers, replicating or expanding on the analyses presented in this manuscript.

The statistical approach appears solid. The two-sided unpaired Welch’s with Benjamini-Hochberg correction used to find differentially abundant proteins is, if anything, on the conservative side, but this can be motivated by the scale of the study.

Reply: We sincerely thank the reviewer for recognizing the theoretical nature of our study, the scale/quantity/quality of our study, and greatly appreciate the constructive suggestions in the following. During the past months, we have tried very hard to do more biological and clinical validation of the ProteinTalks model.

We are also grateful for the reviewer's recognition of our methodology, data quality, and presentation, as well as the acknowledgment of our efforts in ensuring transparency by depositing raw data in the public domain and making the analysis code openly available. We also appreciate the reviewer’s comments on our statistical approach, and we are pleased that the rationale for our choices was well understood.

The next step would be to validate the predicted prescription or combinations of drugs in actual clinical trials, or use data from past or ongoing trials to validate some aspects of the model. This would significantly raise the impact of this work.

Reply: We sincerely appreciate the reviewer's constructive feedback on the need for clinical validation. In this revision, we have expanded the results to include three complementary layers of evidence that address this point — literature/clinical-trial support and two new experimental validations using clinical samples, and prospective organoid validation. A summary is provided below:

1. Literature and clinical-trial evidence.

We added this in the revised manuscript “*ProteinTalks predicts synergistic drug combination*”, and “*ProteinTalks prioritizes candidate repurposed drugs from clinical biopsy proteomes*”, as copied below.

“Among the top five predicted drug combinations, cisplatin-gemcitabine is a first-line therapy for TNBC²⁸, while talazoparib-cisplatin is currently under clinical evaluation (NCT02286687).”

Additionally, we screened a repurposing compounds library containing 3000 repurposed drugs for three clinical TNBC patients guided by ProteinTalks. “The model ranked cisplatin among top candidates. Notably, cisplatin and the clinically-used carboplatin are both platinum-based agents. All three patients achieved Miller & Payne grade 5 complete pathological remission with carboplatin-paclitaxel-camrelizumab combination therapy.”

“Beyond standard therapeutics, ProteinTalks also highlighted three repurposing candidates, citarinostat (HDAC6 inhibitor, Phase II NCT02551185), THZ-1 (CDK7 inhibitor), and CAY10603 (HDAC6 inhibitor)^{32,33} for these three TNBC patients.”

2. Newly added experimental validation using proteomic data from clinical TNBC patients.

We collected tissue samples from 504 Triple-Negative Breast Cancer (TNBC) Patients treated with first-line therapies and analyzed the proteome data. After integrating the proteome data and drug structural information, ProteinTalks classified patients into high- and low-risk groups with significant differences in survival-free and metastasis-free outcomes (revised manuscript **Results** section *“ProteinTalks stratifies prognosis from clinical biopsy”* and *Figures 5C-E*).

Figure 5D

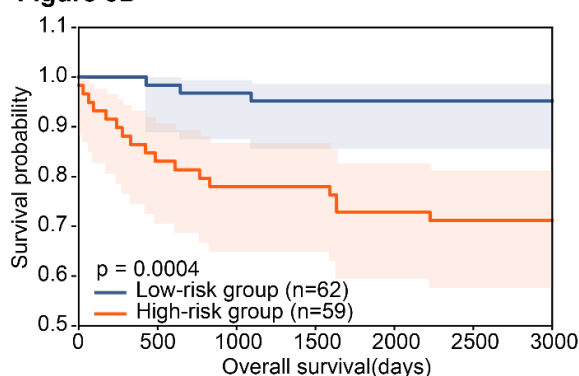


Figure 5E

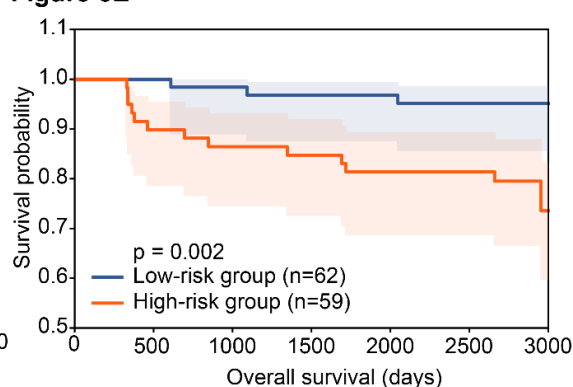


Figure S12C

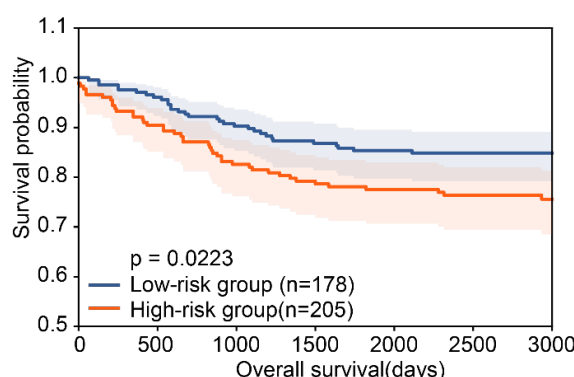
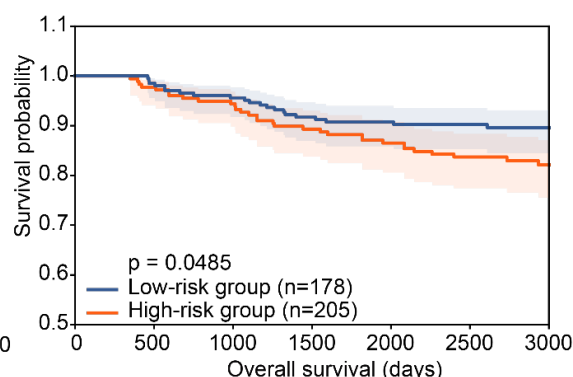


Figure S12D



Figures 5D–E show that ProteinTalks distinguishes high- and low-risk patients receiving two-drug combinations, and Figures S12C–D show that it also distinguishes those receiving three- or four-drug combinations.

3. Newly added prospective validation in three patient-derived TNBC organoids.

We validated the ProteinTalks’ prediction in a patient-derived TNBC organoid by screening ~3000 compounds from a repurposing library (including drugs in ongoing clinical trials or these already approved). “ProteinTalks screened 3000 marketed and clinical-stage drugs (Table S14) against baseline proteomes from three TNBC patient-derived organoids (PDOs: P855, P831, P865; Figure 5F). The model ranked cisplatin among top candidates. Notably, cisplatin and the clinically-used carboplatin are both platinum-based agents. All three patients achieved Miller & Payne grade 5 complete pathological remission with carboplatin-paclitaxel-camrelizumab combination therapy. Beyond standard therapeutics, ProteinTalks also highlighted three repurposing candidates, citarinostat (HDAC6 inhibitor, Phase II NCT02551185), THZ-1 (CDK7 inhibitor), and CAY10603 (HDAC6 inhibitor)^{40,41} for these three TNBC patients. To directly test these predictions, we established PDOs from these three patients and performed seven-point dose–response cytotoxicity assays. All three candidates exhibited significantly lower IC₅₀ values than cisplatin (platinum-based therapy) across all three patients: for P855, CAY10603 (0.35 nM), citarinostat (0.98 nM), and THZ-1 (0.19 nM) versus cisplatin (1.10 mM) (Figures 5); for P831, CAY10603 (0.43 μM), citarinostat (0.67 μM), and THZ-1 (10.13 μM) versus cisplatin (0.19 mM) (Figures 5H); for

P865, CAY10603 (5.78 nM) versus cisplatin (0.36 μM) (Figures 5I)." (Revised manuscript **Results** section "ProteinTalks prioritizes candidate repurposed therapies from clinical biopsy proteomes" and Figures 5F-I).

Figure 5F

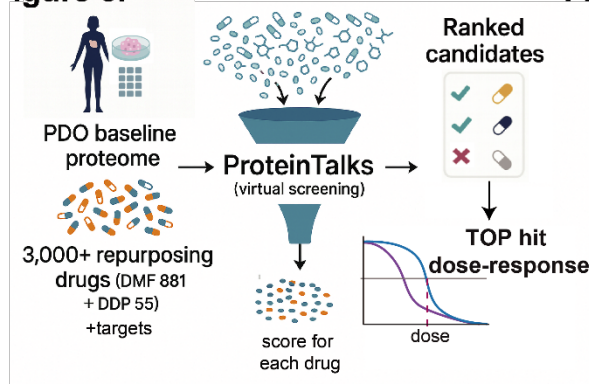


Figure 5G

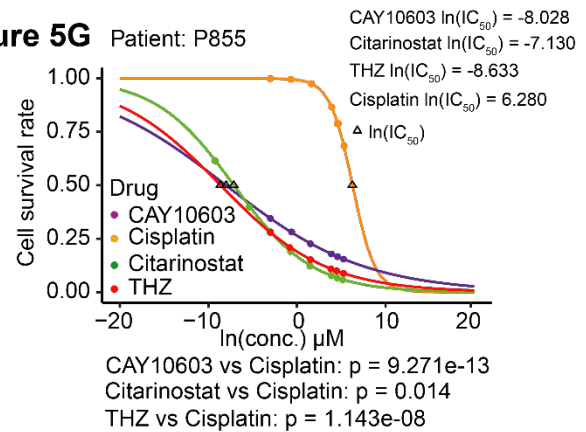


Figure 5H

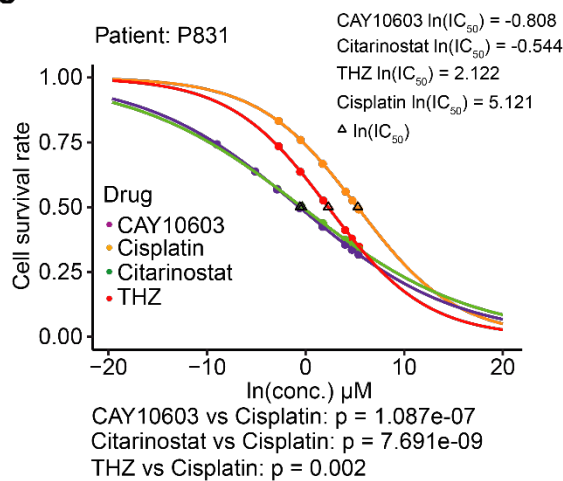


Figure 5I

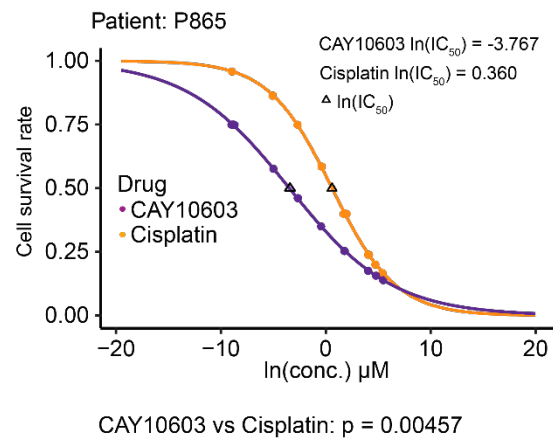


Figure 5F illustrates ProteinTalks' screening of effective drugs for TNBC from 3000 compounds, while Figure 5G-I demonstrates that the top-ranked drugs outperform standard platinum-based therapies.

Together, these newly added experimental validations, spanning retrospective clinical proteomic data and prospective PDO assays, substantially enhance the robustness, clinical credibility, and translational impact of ProteinTalks.

Referee #2 (Remarks to the Author):

Review comments for "A perturbation proteomics-based foundation model for digital cell construction"

Summary

The paper presents a novel proteome foundation model named ProteinTalks, designed to understand biological processes within cells and predict protein network dynamics. This model is built on an extensive dataset of over 38 million protein measurements from 16,000 perturbed proteomic samples, making it the largest proteomics dataset to date. ProteinTalks employs a neural ordinary differential equation (ODE) framework integrated with perturbation-aware neural networks to capture time-dependent behavior of cellular systems. The model demonstrates significant improvements in predicting drug efficacy, synergy, and resistance-associated proteins, and shows potential for enhancing prognosis predictions in clinical settings, particularly in breast cancer. While the model presents a very interesting advance in training proteome-based models for biological insights, its superiority and broad utility remains questionable.

Major comments

- The key main comment of a more general character is that the authors should perform more extensive comparisons to other models before claiming superiority, as also noted more specifically below.

Reply: We thank the reviewer for the constructive suggestion to demonstrate the superiority of our model by comparing to existing ones. In this revision, we have compared ProteinTalks with three existing foundation models including Geneformer (Nature, 2023), GeneCompass (Cell Res, 2024), and UCE (bioRxiv, 2024). We selected these models because they are representative in the field. Specifically, Geneformer is the first omics foundation model; GeneCompass is a widely recognized large model that incorporates prior biological knowledge, and UCE is a popular large model designed for cross-species analysis. Additionally, all of these models have publicly available code.

ProteinTalks was systematically evaluated across diverse predictive tasks, including cross-cell-line and cross-drug prediction, novel-drug generalization, and pan-cancer patient-derived xenograft (PDX) response prediction. Across all tasks, ProteinTalks consistently achieved superior or comparable performance, indicating that its advantage is robust, generalizable, and not confined to a single dataset or task.

Moreover, we highlight the distinct advantages of the proteomics-centered and multi-timepoint design of ProteinTalks, which delivers strong predictive performance with 1000-fold fewer model parameters and datapoints than transcriptomics-based foundation models.

For transparency and clarity, we provide detailed results, datasets, and performance metrics in the responses to subsequent reviewer comments and in the **Results** of the revised manuscript, including “*Prediction of responsiveness of drugs*”, “*Predicting effective repurposed compounds beyond standard therapies*”, “*Prediction of drug sensitivity across multiple cancer types*”, “*Predicting therapeutic outcomes in patient-derived tumor models*”.

- For comparison with other methods in benchmarking, there is no hyperparameter optimization for the relevant methods. The performance of other techniques is close to the ProteinTalks model (e.g., AUPRC of 0.854 for ProteinTalks vs 0.822 for RandomForest), which would warrant the best hyperparameter search for the other methods to assess if the proposed approach is indeed the best. It may be so that the other methods are superior if their hyperparameters are optimized.

Reply: We thank the reviewer for highlighting the importance of fair benchmarking with proper hyperparameter optimization. In response, we conducted systematic hyperparameter searches for all baseline models to ensure fair comparison. Specifically, for each model, we performed grid or randomized searches across commonly used parameter spaces, using nested cross-validation to avoid overfitting. As noted above, we also refined ProteinTalks and expanded the benchmark to include a broader panel of competitive baselines. However, even after fine-tuning, none of them achieved performance matched ProteinTalks, which remained to achieve superior performance across tasks robustly. Full details of the hyperparameter search spaces and selected configurations are now included in the revised **Results** and **Methods** “*Hyperparameters search for ProteinTalks and other methods*” and *Tables S4A-B* show beam search of hyperparameters.

Table S4A. Beam search of hyperparameters for ProteinTalks

No.	Hidden Layer Size	Dropout Rate	Optimizer	Learning Rate	Batch Size	Weight Decay	ACC	F1 Score	Matthews CC	Pheno AUROC	Loss
1	128	0	sgd	0.0005	128	0.0001	0.8856	0.7799	0.706	0.901	0.292
2	32	0	sgd	0.0005	128	0.0001	0.8856	0.7799	0.706	0.905	0.289
3	32	0	sgd	0.0005	64	0.0001	0.8856	0.7799	0.706	0.904	0.290
4	64	0	sgd	0.0005	128	0.0001	0.8856	0.7799	0.706	0.899	0.289
5	64	0	sgd	0.0005	64	0.0001	0.8856	0.7799	0.706	0.901	0.287
6	128	0.1	sgd	0.0005	64	0.0001	0.8856	0.7799	0.706	0.911	0.287
7	64	0.2	sgd	0.0005	64	0.0001	0.8856	0.7799	0.706	0.898	0.288
8	64	0	adamw	0.0005	64	0.0001	0.8824	0.775	0.698	0.927	0.266
9	128	0	sgd	0.0005	64	0.0001	0.8824	0.775	0.698	0.895	0.294
10	128	0	sgd	0.0005	128	0.0001	0.8824	0.7722	0.697	0.899	0.292
11	32	0	adamw	0.0005	128	0.0001	0.8758	0.7711	0.686	0.909	0.277
12	32	0.1	adamw	0.0001	64	0.0001	0.8758	0.7683	0.685	0.913	0.270
13	32	0	adamw	0.0005	64	0.0001	0.8725	0.7636	0.677	0.912	0.279
14	64	0	adamw	0.0001	128	0.0001	0.8725	0.7607	0.675	0.901	0.274
15	128	0	adamw	0.0001	128	0.0001	0.8725	0.7607	0.675	0.911	0.269
16	32	0	adamw	0.0001	64	0.0001	0.8693	0.7561	0.668	0.910	0.281
17	128	0	adamw	0.0001	64	0.0001	0.8693	0.7561	0.668	0.912	0.269
18	128	0	adamw	0.0005	64	0.0001	0.8693	0.7561	0.668	0.923	0.261
19	128	0.2	adamw	0.0001	64	0.0001	0.8693	0.7561	0.668	0.905	0.282
20	128	0.2	adamw	0.0001	128	0.0001	0.8693	0.7561	0.668	0.900	0.284
21	32	0	adamw	0.0001	128	0.0001	0.8693	0.7561	0.668	0.906	0.283
22	128	0	adamw	0.0005	128	0.0001	0.866	0.7515	0.661	0.922	0.260
23	64	0	adamw	0.0005	128	0.0001	0.866	0.7515	0.661	0.913	0.267
24	64	0.2	adamw	0.0005	128	0.0001	0.866	0.7485	0.659	0.914	0.269
25	32	0	sgd	0.0001	128	0.0001	0.8725	0.7451	0.669	0.882	0.303
26	128	0.2	sgd	0.0001	128	0.0001	0.8693	0.7333	0.659	0.873	0.311
27	128	0.1	sgd	0.0001	128	0.0001	0.8693	0.7333	0.659	0.873	0.311
28	128	0.2	sgd	0.0001	64	0.0001	0.8693	0.7333	0.659	0.873	0.311
29	128	0.1	sgd	0.0001	64	0.0001	0.8693	0.7333	0.659	0.873	0.311
30	128	0	sgd	0.0001	64	0.0001	0.8693	0.7333	0.659	0.873	0.311
31	32	0	sgd	0.0001	64	0.0001	0.8627	0.7237	0.642	0.862	0.317
32	128	0	sgd	0.0001	64	0.0001	0.8627	0.72	0.6412	0.863	0.320
33	128	0	sgd	0.0001	128	0.0001	0.8627	0.72	0.6412	0.863	0.320

34	64	0	sgd	0.0001	64	0.0001	0.8562	0.6944	0.6216	0.876	0.308
35	64	0	sgd	0.0001	64	0.0001	0.7222	0	0	0.679	0.476

Table S4B. Beam search of hyperparameters for other models

Split	ACC	AUROC	AUPRC	Models	Pearson	R2 score	Params
1	0.758	0.536	0.266	Bootstrap (Bagging with Decision Tree)	0.881	0.687	{'n_estimators': 100, 'estimator': DecisionTreeRegressor(max_depth=5, random_state=42), 'bootstrap': True, 'n_jobs': -1, 'random_state': 42}
1	0.765	0.708	0.505	Random Forest	0.873	0.681	{'n_estimators': 100, 'max_depth': 20, 'min_samples_split': 2, 'max_features': 'log2', 'n_jobs': -1, 'random_state': 42}
1	0.725	0.660	0.344	KNN	0.847	0.639	{'n_neighbors': 3, 'weights': 'distance', 'n_jobs': -1}
2	0.765	0.697	0.457	Deepsynergy	0.077	-16.579	{'hidden_dim': 4096, 'dropout_rate': 0.3733880257343942, 'lr': 0.007669023607261605}
2	0.778	0.565	0.267	Bootstrap (Bagging with Decision Tree)	0.883	0.689	{'n_estimators': 10, 'estimator': DecisionTreeRegressor(max_depth=5, random_state=42), 'bootstrap': True, 'n_jobs': -1, 'random_state': 42}
2	0.810	0.739	0.532	Random Forest	0.873	0.692	{'n_estimators': 10, 'max_depth': 20, 'min_samples_split': 2, 'max_features': 'sqrt', 'n_jobs': -1, 'random_state': 42}
2	0.732	0.697	0.367	KNN	0.857	0.673	{'n_neighbors': 3, 'weights': 'distance', 'n_jobs': -1}
3	0.784	0.779	0.485	Deepsynergy	0.176	-13.382	{'hidden_dim': 9192, 'dropout_rate': 0.3202480383647589, 'lr': 0.0091000628587735}
3	0.719	0.595	0.334	Bootstrap (Bagging with Decision Tree)	0.878	0.682	{'n_estimators': 100, 'estimator': DecisionTreeRegressor(max_depth=5, random_state=42), 'bootstrap': True, 'n_jobs': -1, 'random_state': 42}
3	0.752	0.713	0.541	Random Forest	0.867	0.678	{'n_estimators': 10, 'max_depth': None, 'min_samples_split': 2, 'max_features': 'sqrt', 'n_jobs': -1, 'random_state': 42}
3	0.706	0.682	0.407	KNN	0.848	0.644	{'n_neighbors': 3, 'weights': 'distance', 'n_jobs': -1}
4	0.752	0.802	0.634	Deepsynergy	0.131	-16.613	{'hidden_dim': 9192, 'dropout_rate': 0.4165060616954387, 'lr': 0.0077417841617481815}
4	0.778	0.613	0.268	Bootstrap (Bagging with Decision Tree)	0.877	0.674	{'n_estimators': 100, 'estimator': DecisionTreeRegressor(max_depth=5, random_state=42), 'bootstrap': True, 'n_jobs': -1, 'random_state': 42}
4	0.830	0.781	0.543	Random Forest	0.870	0.676	{'n_estimators': 100, 'max_depth': 20, 'min_samples_split': 2, 'max_features': 'log2', 'n_jobs': -1, 'random_state': 42}
4	0.725	0.746	0.376	KNN	0.867	0.681	{'n_neighbors': 5, 'weights': 'uniform', 'n_jobs': -1}
5	0.804	0.782	0.535	Deepsynergy	0.077	-16.860	{'hidden_dim': 4096, 'dropout_rate': 0.3928978216673674, 'lr': 0.007458111810176773}
5	0.745	0.614	0.307	Bootstrap (Bagging with Decision Tree)	0.877	0.675	{'n_estimators': 100, 'estimator': DecisionTreeRegressor(max_depth=5, random_state=42), 'bootstrap': True, 'n_jobs': -1, 'random_state': 42}
5	0.758	0.770	0.579	Random Forest	0.873	0.684	{'n_estimators': 100, 'max_depth': 20, 'min_samples_split': 2, 'max_features': 'log2', 'n_jobs': -1, 'random_state': 42}
5	0.771	0.745	0.429	KNN	0.847	0.641	{'n_neighbors': 3, 'weights': 'distance', 'n_jobs': -1}
5	0.752	0.813	0.577	Deepsynergy	0.086	-15.599	{'hidden_dim': 4096, 'dropout_rate': 0.28078754989555926, 'lr': 0.007385342854481402}

The detailed comparative results are presented as follows.

(1) Leave-one-cell line-out cross-validation: ProteinTalks improved AUPRC, AUROC, and accuracy by approximately 13%–213%, 10%–98%, and 3%–17%, respectively (revised manuscript: the **Results** section “Prediction of responsiveness of drugs”; Figures S5C-E, Table S5).

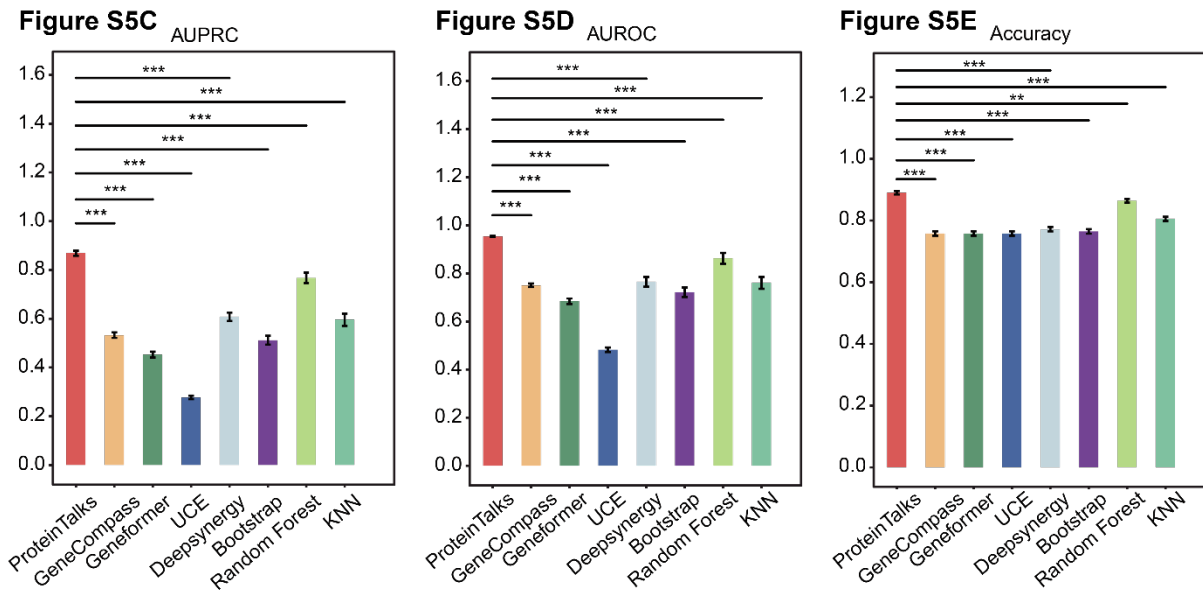


Figure S5C-E. AUPRC, accuracy, and AUROC of the leave-one-cell line-out cross validation and compared with other seven models.

Figures S5C–E illustrate the superior performance of ProteinTalks compared with other foundation models in leave-on-cell line-out cross validation, with detailed quantitative results summarized in Tables S5A–B.

Table S5A. Performance comparison of ProteinTalks and other seven models in leave-one-cell line-out cross validation

Models	AUPRC (mean ± SD)	AUROC (mean ± SD)	Accuracy (mean ± SD)
ProteinTalks	0.869±0.0106	0.953±0.00200	0.89±0.00558
GeneCompass	0.533±0.0112	0.751±0.00700	0.758±0.00732
Geneformer	0.453±0.0118	0.684±0.0120	0.758±0.00732
UCE	0.277±0.00704	0.482±0.00900	0.758±0.00732
KNN	0.596±0.0253	0.760±0.0250	0.805±0.00686
Random Forest	0.768±0.0216	0.863±0.0230	0.864±0.00600
Bootstrap	0.512±0.0183	0.722±0.0200	0.765±0.00728
DeepSynergy	0.608±0.0174	0.764±0.0200	0.772±0.00704

Table S5B. Statistical comparison of ProteinTalks against other seven models in leave-one-cell line-out cross validation

Models' comparison	Index	p-value	Significant
ProteinTalks vs GeneCompass	Accuracy	6.02E-31	*** (p < 0.001)
ProteinTalks vs GeneCompass	AUROC	1.73E-51	*** (p < 0.001)
ProteinTalks vs GeneCompass	AUPRC	6.84E-49	*** (p < 0.001)
ProteinTalks vs Geneformer	Accuracy	6.02E-31	*** (p < 0.001)
ProteinTalks vs Geneformer	AUROC	1.51E-61	*** (p < 0.001)
ProteinTalks vs Geneformer	AUPRC	5.03E-40	*** (p < 0.001)

ProteinTalks vs UCE	Accuracy	6.02E-31	*** ($p < 0.001$)
ProteinTalks vs UCE	AUROC	2.41E-92	*** ($p < 0.001$)
ProteinTalks vs UCE	AUPRC	6.02E-69	*** ($p < 0.001$)
ProteinTalks vs KNN	Accuracy	8.16E-18	*** ($p < 0.001$)
ProteinTalks vs KNN	AUPRC	2.30E-17	*** ($p < 0.001$)
ProteinTalks vs KNN	AUROC	1.45E-11	*** ($p < 0.001$)
ProteinTalks vs Random Forest	Accuracy	0.00167	** ($p < 0.01$)
ProteinTalks vs Random Forest	AUPRC	0.0000486	*** ($p < 0.001$)
ProteinTalks vs Random Forest	AUROC	0.000129	*** ($p < 0.001$)
ProteinTalks vs Bootstrap	Accuracy	4.37E-29	*** ($p < 0.001$)
ProteinTalks vs Bootstrap	AUPRC	6.49E-36	*** ($p < 0.001$)
ProteinTalks vs Bootstrap	AUROC	8.84E-20	*** ($p < 0.001$)
ProteinTalks vs Deepsynergy	Accuracy	1.29E-27	*** ($p < 0.001$)
ProteinTalks vs Deepsynergy	AUPRC	1.03E-25	*** ($p < 0.001$)
ProteinTalks vs Deepsynergy	AUROC	1.47E-14	*** ($p < 0.001$)

(2) Leave-one-drug-out cross-validation: ProteinTalks delivered consistent gains across evaluation metrics, outperforming these models by 39%–55% in AUPRC, 26%–57% in AUROC, 15-31% in accuracy (revised manuscript: the **Results** section “ProteinTalks outperforms baselines for drug response prediction across cell lines and drugs”; Figures S5F-G, Table S6)

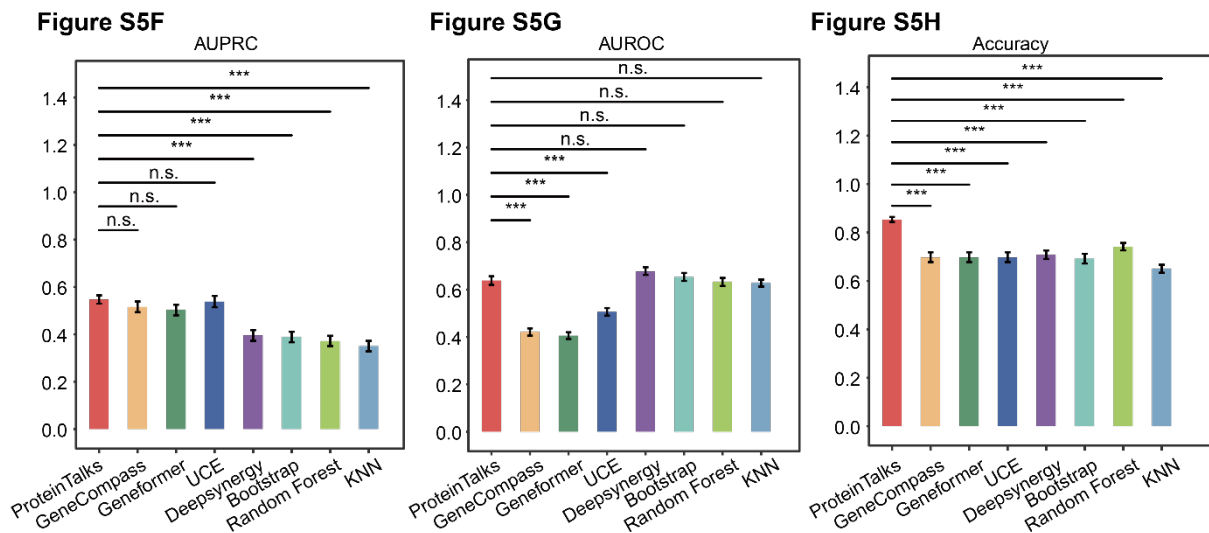


Figure S5F-H. AUPRC, AUROC, and Accuracy of the leave-one-drug-out cross validation and compared with other seven models.

Figures 5F-H illustrate the superior performance of ProteinTalks compared with other foundation models in leave-on-drug-out cross validation, with detailed quantitative results summarized in Tables S6A-B.

Table S6A. Performance comparison of ProteinTalks and other seven models in leave-one-drug-out cross validation

Models	AUPRC (mean $\pm$ SE)	AUROC (mean $\pm$ SE)	Accuracy (mean $\pm$ SE)
ProteinTalks	0.547$\pm$0.0179	0.638$\pm$0.0180	0.853$\pm$0.0105
Geneformer	0.503$\pm$0.0223	0.406$\pm$0.0140	0.698$\pm$0.0203
GeneCompass	0.516$\pm$0.0227	0.421$\pm$0.0150	0.698$\pm$0.0203

<i>UCE</i>	<i>0.538±0.0240</i>	<i>0.506±0.0160</i>	<i>0.698±0.0203</i>
<i>Bootstrap</i>	<i>0.389±0.0223</i>	<i>0.653±0.0170</i>	<i>0.692±0.0201</i>
<i>KNN</i>	<i>0.351±0.0219</i>	<i>0.627±0.0150</i>	<i>0.650±0.0168</i>
<i>RandomForest</i>	<i>0.372±0.0218</i>	<i>0.632±0.0170</i>	<i>0.741±0.0151</i>
<i>DeepSynergy</i>	<i>0.395±0.0224</i>	<i>0.678±0.0160</i>	<i>0.708±0.0176</i>

Table S6B. Statistical comparison of ProteinTalks against other seven models in leave-one-drug-out cross validation

<i>Models' comparison</i>	<i>Index</i>	<i>p-value</i>	<i>Significant</i>
<i>ProteinTalks vs GeneCompass</i>	<i>AUPRC</i>	<i>0.284</i>	<i>n.s.</i>
<i>ProteinTalks vs GeneCompass</i>	<i>AUROC</i>	<i>2.79E-18</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs GeneCompass</i>	<i>Accuracy</i>	<i>2.82E-11</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs Geneformer</i>	<i>AUPRC</i>	<i>0.119</i>	<i>n.s.</i>
<i>ProteinTalks vs Geneformer</i>	<i>AUROC</i>	<i>2.39E-21</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs Geneformer</i>	<i>Accuracy</i>	<i>2.82E-11</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs UCE</i>	<i>AUPRC</i>	<i>0.768</i>	<i>n.s.</i>
<i>ProteinTalks vs UCE</i>	<i>AUROC</i>	<i>9.26E-08</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs KNN</i>	<i>Accuracy</i>	<i>2.82E-11</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs KNN</i>	<i>AUPRC</i>	<i>9.45E-12</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs KNN</i>	<i>AUROC</i>	<i>0.649</i>	<i>n.s.</i>
<i>ProteinTalks vs Bootstrap</i>	<i>Accuracy</i>	<i>1.29E-22</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs Bootstrap</i>	<i>AUPRC</i>	<i>4.41E-08</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs Bootstrap</i>	<i>AUROC</i>	<i>0.541</i>	<i>n.s.</i>
<i>ProteinTalks vs Random.Forest</i>	<i>Accuracy</i>	<i>4.08E-12</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs Random.Forest</i>	<i>AUPRC</i>	<i>9.67E-10</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs Random.Forest</i>	<i>AUROC</i>	<i>0.813</i>	<i>n.s.</i>
<i>ProteinTalks vs DeepSynergy</i>	<i>Accuracy</i>	<i>2.05E-09</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs DeepSynergy</i>	<i>AUPRC</i>	<i>0.000000161</i>	*** (<i>p</i> < 0.001)
<i>ProteinTalks vs DeepSynergy</i>	<i>AUROC</i>	<i>0.104</i>	<i>n.s.</i>

● Furthermore, the benchmark models are only compared to the proposed approach in setting 1 (randomly divided), not in settings 2 (leave-one-cell_line-out) and 3 (leave-one-drug-out), which is not reasonable since the proposed approach should also outperform the benchmark models in those data splits.

Reply: We thank the reviewer for the thoughtful comment regarding the need to provide stronger support for the practical utility and robustness of our methods. As the reviewer notes, comparisons under only random splits may inflate performance by favoring memorization over true generalizability. To address this concern and ensure a comprehensive and fair evaluation, we compared ProteinTalks with three state-of-the-art transcriptomics-based foundation models (Geneformer, GeneCompass, and UCE) under leave-one-cell-line-out and leave-one-drug-out settings. The detailed comparative results are presented as follows:

(1) Leave-one-breast cancer cell line-out cross-validation: Compared with three existing foundation models (GeneCompass, Geneformer, and UCE), **ProteinTalks improved AUPRC, AUROC, and accuracy by approximately 63%–213%, 27%–98%, and 17%–17%**, respectively (revised manuscript: the **Results** section “Prediction of responsiveness of drugs”; Figures 3B, S5A–B, Table S5).

Figure 3B

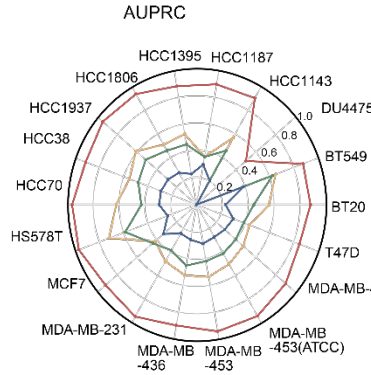


Figure S5A

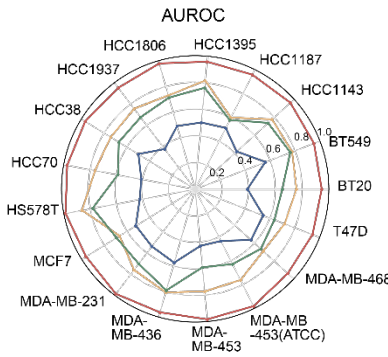
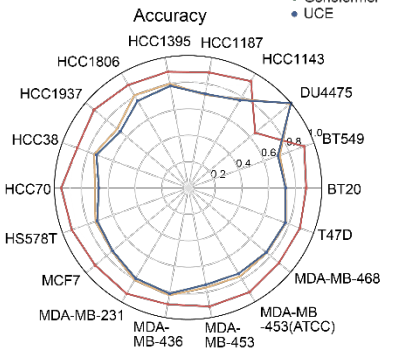


Figure S5B



Figures 3B, S5A–B AUPRC, AUROC, and accuracy of the leave-one-cell line-out cross validation and compared with GeneCompass, Geneformer, and UCE model.

Figures 3B and S5A–B illustrate the superior performance of ProteinTalks compared with other foundation models in leave-on-cell line-out cross validation, with detailed quantitative results summarized in Tables S5A–B.

Table S5A. Performance comparison of ProteinTalks and three foundation models in leave-one-cell line-out cross validation

Models	AUPRC (mean $\pm$ SD)	AUROC (mean $\pm$ SD)	Accuracy (mean $\pm$ SD)
ProteinTalks	0.869$\pm$0.0106	0.953$\pm$0.00200	0.890$\pm$0.00558
GeneCompass	0.533 $\pm$ 0.0112	0.751 $\pm$ 0.00700	0.758 $\pm$ 0.00732
Geneformer	0.453 $\pm$ 0.0118	0.684 $\pm$ 0.0120	0.758 $\pm$ 0.00732
UCE	0.277 $\pm$ 0.00704	0.482 $\pm$ 0.00900	0.758 $\pm$ 0.00732

Table S5B. Statistical comparison of ProteinTalks against three foundation models in leave-one-cell line-out cross validation

Models' comparison	Index	p-value	Significant
ProteinTalks vs GeneCompass	Accuracy	6.02E-31	*** ($p < 0.001$)
ProteinTalks vs GeneCompass	AUROC	1.73E-51	*** ($p < 0.001$)
ProteinTalks vs GeneCompass	AUPRC	6.84E-49	*** ($p < 0.001$)
ProteinTalks vs Geneformer	Accuracy	6.02E-31	*** ($p < 0.001$)
ProteinTalks vs Geneformer	AUROC	1.51E-61	*** ($p < 0.001$)
ProteinTalks vs Geneformer	AUPRC	5.03E-40	*** ($p < 0.001$)
ProteinTalks vs UCE	Accuracy	6.02E-31	*** ($p < 0.001$)
ProteinTalks vs UCE	AUROC	2.41E-92	*** ($p < 0.001$)
ProteinTalks vs UCE	AUPRC	6.02E-69	*** ($p < 0.001$)

(2) Leave-one-drug-out cross-validation: ProteinTalks outperformed transcriptome foundation-based models by 26%–57% in AUROC and 22% in accuracy, with comparable AUPRC performance (revised manuscript: the **Results** section “Prediction of responsiveness of drugs”; Figures S5F–H, Table S6). Although its improvement on this task was modest, the model achieved strong results on subsequent independent test sets (see later results).

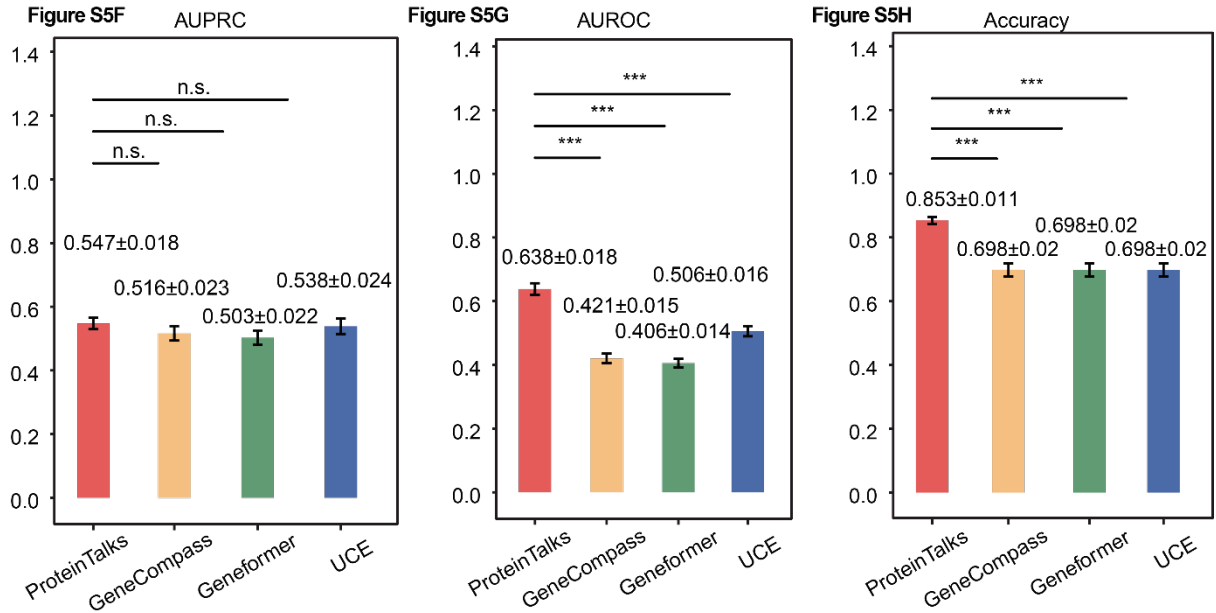


Figure S5F-H. AUPRC, AUROC, and Accuracy of the leave-one-drug-out cross validation and compared with Geneformer, GeneCompass, and UCE model.

Figures 5F-H illustrate the superior performance of ProteinTalks compared with other foundation models in leave-one-drug-out cross validation, with detailed quantitative results summarized in Tables S6A-B.

Table S6A. Performance comparison of ProteinTalks and three foundation models in leave-one-drug-out cross validation

Models	AUPRC (mean ± SE)	AUROC (mean ± SE)	Accuracy (mean ± SE)
ProteinTalks	0.547±0.0179	0.638±0.0180	0.853±0.0105
Geneformer	0.503±0.0223	0.406±0.0140	0.698±0.0203
GeneCompass	0.516±0.0227	0.421±0.0150	0.698±0.0203
UCE	0.538±0.0240	0.506±0.0160	0.698±0.0203

Table S6B. Statistical comparison of ProteinTalks against three foundation models in leave-one-drug-out cross validation

Models' comparison	Index	p-value	Significant
ProteinTalks vs GeneCompass	AUPRC	0.284	n.s.
ProteinTalks vs GeneCompass	AUROC	2.79E-18	*** ($p < 0.001$)
ProteinTalks vs GeneCompass	Accuracy	2.82E-11	*** ($p < 0.001$)
ProteinTalks vs Geneformer	AUPRC	0.119	n.s.
ProteinTalks vs Geneformer	AUROC	2.39E-21	*** ($p < 0.001$)
ProteinTalks vs Geneformer	Accuracy	2.82E-11	*** ($p < 0.001$)
ProteinTalks vs UCE	AUPRC	0.768	n.s.
ProteinTalks vs UCE	AUROC	9.26E-08	*** ($p < 0.001$)
ProteinTalks vs UCE	Accuracy	2.82E-11	*** ($p < 0.001$)

• The authors claim that ProteinTalks is a foundation model to model the perturbed proteomic networks, however, this neural ODE model seems to be very limited to certain tasks and does not exhibit potential to scale up and generalize to diverse tasks. Even within this paper's scope, when

testing the pretrained model's performance on the newly acquired dataset (98 additional anti-cancer small-molecular 224 compounds), it cannot match the performance on the original test set. If the authors are to claim that this is a foundation model, they should demonstrate generalizability to numerous tasks.

Reply: We thank the reviewer for raising this important point regarding the scalability and generalizability of ProteinTalks. We fully agree that a defining feature of a foundation model is its ability to scale and generalize across diverse tasks. Meanwhile, we would like to emphasize our pretraining rationale is to let the model learn the intrinsic patterns of proteomic network through temporal measurements, as to serve as the foundation for downstream tasks with this prior knowledge.

To address these in further details, we have made three major improvements:

1. **Scalability analysis:** We performed scaling law analysis to explicitly evaluate the model's scaling behavior. The results confirm that ProteinTalks follows scaling law and can effectively leverage increased training data, demonstrating its scalability.
2. **Expanded evaluation on new datasets:** We further optimized ProteinTalks through a carefully designed training protocol emphasizing stable convergence, calibrated predictions, and reproducibility without increasing model capacity. On the newly expanded PTNC dataset (now comprising 84 anti-cancer small molecules, refined from the previous 98 compounds by excluding 15 drugs with novel structural scaffolds), ProteinTalks achieved an accuracy of 0.881 and an AUROC of 0.896—representing a modest improvement over the previously reported version (0.844 and 0.840, respectively). More importantly, these values are 74–193% higher than those achieved by other transcriptomic foundation models, underscoring the model's substantial comparative advantage even on previously unseen compounds.
3. **Demonstration of generalizability:** Beyond benchmark tasks, we extended ProteinTalks to diverse application scenarios, including a larger set of drugs, patient-derived xenografts, FFPE clinical specimens, and patient-derived organoids. These applications collectively illustrate the model's ability to generalize to multiple biological and translational tasks.

We have incorporated these results into the revised manuscript, and the relevant sections are updated accordingly. The key supporting evidence is detailed below:

(1) Scalability analysis: We added scaling law analysis to demonstrate that ProteinTalks exhibits foundation model characteristics. As shown in **Figure S4E-G**, test performance follows a predictable trend of consistent improvement as the proportion of training data increases from 10% to 70% of the full dataset. We have added the following statement to the manuscript: *“Performance scaled predictably as training data increased from 10% to 70% (Figure S4E-G).”*

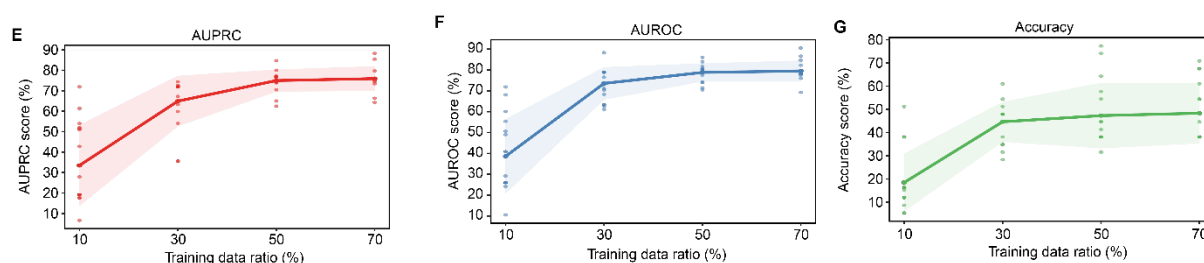


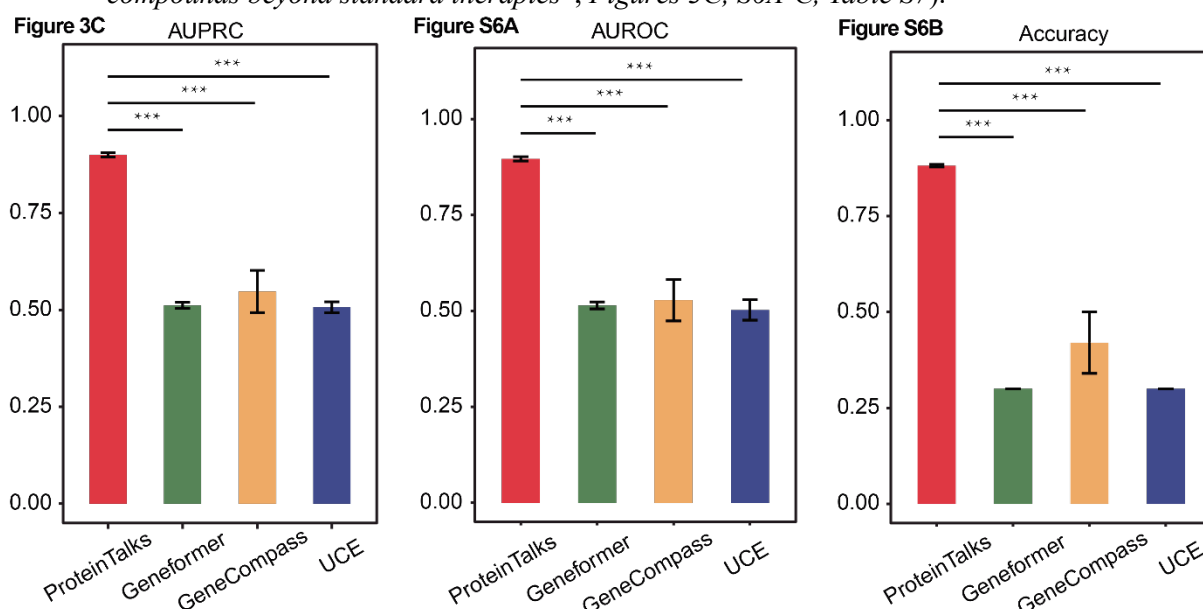
Figure S4 (E)-(G) Learning curves showing the performance of ProteinTalks evaluated by AUPRC (E), AUROC (F), and accuracy (G) increasing with the training data size.

(2) Model optimization: Guided by extensive hyperparameter sweeps and theoretical considerations, we adopted a training protocol that prioritizes stable convergence, calibrated predictions, and reproducibility without increasing model capacity. Concretely, we selected AdamW with a moderate batch size, a cosine-annealed learning-rate schedule, early stopping, and a late-phase SWAG approximation. We further decoupled the convergence phase from posterior averaging, enforced strict

seeding, and leveraged multi-timepoint supervision as an implicit regularizer for time-dependent drug-synergy prediction. These details have been added to the **Methods** section: “*ProteinTalks pretraining on PTDS-I*”

(3) Broad generalizability across tasks: We provide detailed results of the model’s comparisons and broad applicability across multiple biological and clinical contexts. The detailed comparative results are presented as follows:

- **Independent testing for novel drugs (PTNC): Averaged across the three transcriptomic-based foundation models, ProteinTalks delivered improvements of 74% in AUPRC, 75% in AUROC, and 144% in accuracy**, underscoring its strong generalization for newly compounds (revised manuscript: the **Results** section “*Predicting effective repurposed compounds beyond standard therapies*”; Figures 3C, S6A-C, Table S7).



Figures 3C, S6A-C. AUPRC, AUROC, and Accuracy of the novel drugs predictions using PTNC and compared with Geneformer, GeneCompass, and UCE model.

Figures 3C, S6A-C illustrate the superior performance of ProteinTalks compared with other foundation models in predicting efficacy of novel repurposed compounds, with detailed quantitative results summarized in Tables S7A-B.

Table S7A. Performance comparison between ProteinTalks and three foundation models in predicting efficacy of novel repurposed compounds

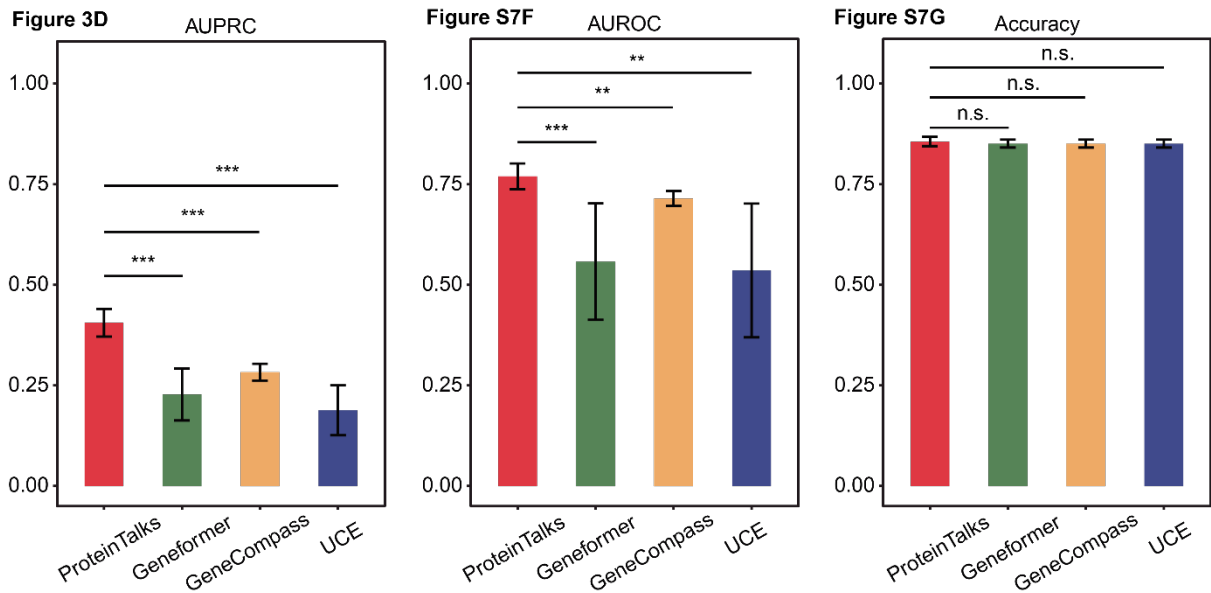
Models	AUPRC (mean $\pm$ SE)	AUROC (mean $\pm$ SE)	Accuracy (mean $\pm$ SE)
ProteinTalks	0.901 $\pm$ 0.00600	0.896 $\pm$ 0.00600	0.881 $\pm$ 0.00300
Geneformer	0.511 $\pm$ 0.00800	0.515 $\pm$ 0.00900	0.300 $\pm$ 0
GeneCompass	0.548 $\pm$ 0.0540	0.528 $\pm$ 0.0540	0.420 $\pm$ 0.0800
UCE	0.507 $\pm$ 0.0140	0.502 $\pm$ 0.0270	0.300 $\pm$ 0

Table S7B. Statistical comparison of ProteinTalks against three foundation models in predicting efficacy of novel repurposed compounds

Comparison	Index	p-value	Significant
ProteinTalks vs Geneformer	AUPRC	0.000000	*** ($p < 0.001$)
ProteinTalks vs GeneCompass	AUPRC	0.000199	*** ($p < 0.001$)

<i>ProteinTalks vs UCE</i>	<i>AUPRC</i>	<i>0.000000</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs Geneformer</i>	<i>AUROC</i>	<i>0.000000</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs GeneCompass</i>	<i>AUROC</i>	<i>0.000138</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs UCE</i>	<i>AUROC</i>	<i>0.000001</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs Geneformer</i>	<i>Accuracy</i>	<i>0.000000</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs GeneCompass</i>	<i>Accuracy</i>	<i>0.000423</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs UCE</i>	<i>Accuracy</i>	<i>0.000000</i>	<i>*** (p < 0.001)</i>

- **Few-shot transfer learning for pan-cancer cell lines (PTPC): Averaged across the three transcriptomic-based foundation models, ProteinTalks achieved markedly higher performance, with improvements of 74% in AUPRC and 32% in AUROC**, indicating its utility for broader pan-cancer drug-efficacy prediction (revised manuscript: the **Results** section “Prediction of drug sensitivity across multiple cancer types”; Figures 3D, S7A-C, Table S9).



Figures 3D, S7A-C. Comparative performance of ProteinTalks and foundation models (Geneformer, GeneCompass, and UCE) on the few-shot transfer learning task for pan-cancer drug efficacy prediction, evaluated by AUPRC, AUROC, and accuracy.

Figures 3D, S7A-C illustrate the superior performance of ProteinTalks compared with other foundation models in predicting drug sensitivity in pan-cancer cell lines, with detailed quantitative results summarized in Tables S9A-B.

Table S9A. Performance comparison between ProteinTalks and three foundation models in predicting drug sensitivity in pan-cancer cell lines

<i>Models</i>	<i>AUPRC (mean ± SD)</i>	<i>AUROC (mean ± SD)</i>	<i>Accuracy (mean ± SD)</i>
<i>ProteinTalks</i>	<i>0.406 ± 0.0350</i>	<i>0.770 ± 0.0320</i>	<i>0.857 ± 0.0120</i>
<i>Geneformer</i>	<i>0.228 ± 0.0650</i>	<i>0.558 ± 0.145</i>	<i>0.851 ± 0.00100</i>
<i>GeneCompass</i>	<i>0.283 ± 0.0210</i>	<i>0.715 ± 0.0190</i>	<i>0.851 ± 0.00100</i>
<i>UCE</i>	<i>0.188 ± 0.0620</i>	<i>0.536 ± 0.166</i>	<i>0.851 ± 0.00100</i>

Table S9B. Statistical comparison of ProteinTalks against three foundation models in predicting drug sensitivity in pan-cancer cell lines

<i>Comparison</i>	<i>Index</i>	<i>p-value</i>	<i>Significant</i>
<i>ProteinTalks vs Geneformer</i>	<i>AUPRC</i>	<i>0</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs Geneformer</i>	<i>AUROC</i>	<i>0.000143</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs Geneformer</i>	<i>Accuracy</i>	<i>0.310</i>	<i>n.s. (p >= 0.05)</i>
<i>ProteinTalks vs GeneCompass</i>	<i>AUPRC</i>	<i>1.60E-5</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs GeneCompass</i>	<i>AUROC</i>	<i>1.43E-4</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs GeneCompass</i>	<i>Accuracy</i>	<i>0.310</i>	<i>n.s. (p >= 0.05)</i>
<i>ProteinTalks vs UCE</i>	<i>AUPRC</i>	<i>1.00E-6</i>	<i>*** (p < 0.001)</i>
<i>ProteinTalks vs UCE</i>	<i>AUROC</i>	<i>0.00654</i>	<i>** (p < 0.01)</i>
<i>ProteinTalks vs UCE</i>	<i>Accuracy</i>	<i>0.310</i>	<i>n.s. (p >= 0.05)</i>

Furthermore, to strengthen the translational relevance of our model, we have added:

- **Pan-cancer PDX fine-tuning task (PDX-2015nm):** **Without any additional training, ProteinTalks substantially outperformed all three transcriptomic-based foundation models, yielding on average a 118% improvement in AUPRC and a 32% improvement in AUROC** (revised manuscript: the **Results** section “Predicting therapeutic outcomes in patient-derived tumor models”; Figures S11A-C, Table S11)

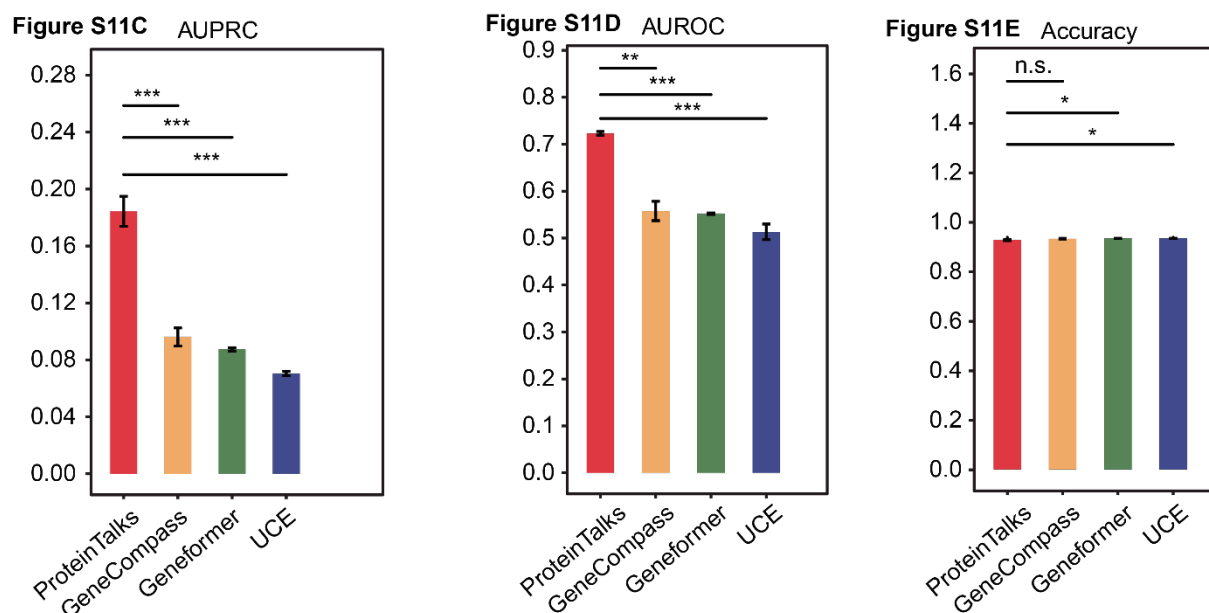


Figure S11C-E. Comparative performance of ProteinTalks and existing foundation models (Geneformer, GeneCompass, and UCE) on the pan-cancer PDX drug sensitivity prediction task, evaluated by AUPRC, AUROC, and accuracy.

Figures S11C-E illustrate the superior performance of ProteinTalks compared with other foundation models in predicting for pan-cancer PDXs, with detailed quantitative results summarized in Tables S11A-B.

Table S11A. Performance comparison between ProteinTalks and three foundation models on the pan-cancer PDX drug sensitivity prediction task.

<i>Models</i>	<i>AUROC (mean ± SE)</i>	<i>AUPRC (mean ± SE)</i>	<i>Accuracy (mean ± SE)</i>
<i>ProteinTalks</i>	<i>0.723 ± 0.00376</i>	<i>0.184 ± 0.0105</i>	<i>0.929 ± 0.00237</i>
<i>Geneformer</i>	<i>0.552 ± 0.00161</i>	<i>0.0874 ± 0.00119</i>	<i>0.935 ± 0</i>
<i>GeneCompass</i>	<i>0.558 ± 0.0206</i>	<i>0.0961 ± 0.00636</i>	<i>0.934 ± 0.00142</i>

<i>UCE</i>	0.513 ± 0.0165	0.0704 ± 0.00156	0.935 ± 0
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Table S11B. Statistical comparison of ProteinTalks against three foundation models on the pan-cancer PDX drug sensitivity prediction task.

<i>Comparison</i>	<i>Index</i>	<i>p-value</i>	<i>Significant</i>
<i>ProteinTalks vs Geneformer</i>	<i>AUPRC</i>	0.000228	*** ($p < 0.001$)
<i>ProteinTalks vs Geneformer</i>	<i>AUROC</i>	2.30E-09	*** ($p < 0.001$)
<i>ProteinTalks vs Geneformer</i>	<i>Accuracy</i>	0.0352	* ($p < 0.05$)
<i>ProteinTalks vs GeneCompass</i>	<i>AUPRC</i>	9.53E-05	*** ($p < 0.001$)
<i>ProteinTalks vs GeneCompass</i>	<i>AUROC</i>	0.00105	** ($p < 0.01$)
<i>ProteinTalks vs GeneCompass</i>	<i>Accuracy</i>	0.088	n.s. ($p \geq 0.05$)
<i>ProteinTalks vs UCE</i>	<i>AUPRC</i>	9.47E-05	*** ($p < 0.001$)
<i>ProteinTalks vs UCE</i>	<i>AUROC</i>	0.000134	*** ($p < 0.001$)
<i>ProteinTalks vs UCE</i>	<i>Accuracy</i>	0.0352	* ($p < 0.05$)

• SHAP score was used to evaluate the interpretability of the model and performed some retrospective analysis to uncover new insights in drug resistance mechanisms, which is good. This approach can be enhanced by integrating the distinctive aspects of ODE parameter estimation, incorporating uncertainty quantification and prospective interpretation of model decisions, thereby enabling scalability and deployment for automated discovery of novel pathways or proteins of interest.

Reply: We thank the reviewer for this insightful comment highlighting the potential of integrating ODE-based uncertainty quantification and prospective interpretation to enhance model interpretability. We have carefully considered these suggestions and expanded our analysis accordingly.

As described in “*ProteinTalks prioritizes proteins responsible for drug resistance*”, we used SHapley Additive exPlanations (SHAP) to quantify the contribution of each protein to drug sensitivity or drug-combination synergy. For each cell line and drug condition, SHAP values were computed under sensitive or synergistic settings, using resistant or non-synergistic samples as background references. This framework systematically measures how each protein influences the model’s prediction.

To address the reviewer’s suggestion on uncertainty quantification, we explicitly analyzed the variability of SHAP estimations. In our framework, SHAP uncertainty arises mainly from (1) sampling variability, due to heterogeneous background selection across cell lines and drugs, and (2) model variability, introduced by stochastic optimization and ODE integration in the Neural-ODE architecture. We quantified these uncertainties by calculating the standard deviation (s.d.)/standard error (s.e.) of SHAP values across multiple cell lines for the same drug or combination (*Supplementary Table 10A*). We have also added a detailed description of the metrics calculation in the **Methods** “*SHAP value filtration*”. This variance reflects the consistency of protein importance attribution under repeated perturbations of similar pharmacological contexts. In addition, all other quantitative results in the study are reported with corresponding uncertainty metrics (s.d. or s.e.) in *Tables S5–S7, S9, and S11*, and all relevant figures (*Figures 3C–D, S5C–H, S6A–B, S7F–G, S11C–E*). We have also added a detailed description of the metrics calculation in the **Methods** “*Metrics calculation*”. Therefore, the small s.d. of SHAP values provides additional support for the robustness of our analyses.

To incorporate the reviewer’s suggestion on prospective interpretation and dynamic modeling, we newly introduced time-dependent SHAP analysis, which evaluates the contribution of baseline (0 h) protein abundances to model predictions of downstream proteomic states at later time points (see

Methods “Dynamic SHAP calculation”, and **Results** “ProteinTalks delineates dynamic protein networks”). This enables prospective linkage between early perturbation signals and subsequent proteomic trajectories, effectively integrating temporal ODE dynamics into model interpretability. We have added these results to the manuscript, as copied below.

“To elucidate how drug perturbations reshape protein regulatory networks over time, we computed dynamic SHapley Additive exPlanations (SHAP) based on ProteinTalks’ predictions of perturbed protein expression at different time points (6, 24, 48 hours).

Drug-specific, time-dependent SHAP divergences between responsive and non-responsive samples revealed distinct temporal dynamics. Alkylating agents, HDAC inhibitors, and topoisomerase inhibitors showed minimal divergence at 6 hours but marked separation by 24 hours (Figures S8A-C), reflecting delayed protein-level manifestation of DNA damage and epigenetic remodeling through regulatory cascades. Conversely, CDK inhibitors, hormonal agents, and RXR inhibitors exhibited pronounced divergence at 6 hours, indicating immediate protein-level signaling responses (Figures S8D-F). Together, these findings indicate that latent trajectories could recapitulate biologically relevant temporal dynamics.”

- The paper would be strengthened if the authors could demonstrate that the obtained insights are indeed obtained due to the massive proteome dataset, and not simply due to the constraints to known pathways.

Reply: We thank the reviewer for raising this important point. We would like to clarify that our modeling framework is built entirely on whole-proteome input, without relying on prior knowledge of known pathways to constrain the model structure or training. In fact, we specifically highlight proteins outside canonical pathways that were selected based on their dynamic perturbation patterns in response to drug treatment. These proteins were prioritized based on their significant changes under perturbation and subsequently validated through biological interpretation and consistency with prior studies. Thus, the insights are not simply a by-product of pathway enrichment, but rather emerge directly from the large-scale proteomic measurements that enable us to capture both pathway-associated and pathway-independent regulatory signals. The scale of our dataset—comprising over 38 million perturbed protein measurements across multiple time points—provides the statistical power and dynamic resolution necessary to identify these biologically meaningful but previously uncharacterized regulatory patterns. As discussed in the **Results**, we highlight examples where such proteins exhibit functional relevance to drug response, as illustrated below:

“Multiple proteins maintained high expression levels in the resistant group while significantly decreased in the sensitive group (Figure 2D, ANOVA $p < 0.05$), suggesting their involvement in drug resistance. Upon treatment with 5-fluorouracil (an anti-metabolic agent), ATG3 showed the greatest divergence at 48 hours (fold change = 0.15; resistant vs. sensitive). As an autophagy-related protein previously implicated in chemotherapy resistance in hepatocellular carcinoma through regulation of autophagy processes¹⁹, the elevated ATG3 observed here suggests that enhanced autophagy may contribute to 5-fluorouracil resistance. For docetaxel (an anti-mitotic agent), the most pronounced divergence was observed for BTF3 at 48 hours (0.12-fold). The upregulation of BTF3 indicates that the epithelial cancer cells may possess stemness features²⁰, and cancer stem cells have been reported to be chemotherapy resistance²¹. These findings suggest that BTF3 may enhance docetaxel resistance by maintaining stemness. In erlotinib-treated cells (an EGFR inhibitor), PAK1 and NDE1 showed the largest differences at 48 hours (0.21-fold and “infinite”, respectively, the latter reflecting non-detection in one group). PAK1 has been associated with resistance to tyrosine kinase inhibitors in EGFR-mutant lung cancer²², likely via activation of the AKT pathway, while NDE1 is known to

interact with EGFR²³, potentially attenuating erlotinib efficacy. Following exposure to talazoparib (a PARP inhibitor), ELAVL2 displayed the largest difference at 48 hours (0.30-fold). Consistent with prior reports implicating ELAVL2 in regulation of glycolysis²⁴, this suggests that metabolic reprogramming may mediate resistance to PARP inhibition. Conversely, under alpelisib treatment (a PI3K inhibitor), CKS2 expression diverged in the opposite direction—elevated in the sensitive group but reduced in the resistant group—with a maximal difference of 7.71-fold at 48 hours. This aligns with previous studies showing that CKS2 antagonizes PI3K/AKT signaling²⁵, in contrast to the resistance-promoting proteins described above. Together, these protein-specific dynamics may provide potential biomarkers and therapeutic targets for overcoming drug resistance in breast cancer.”

The ProteinTalks model learns temporal protein dynamics across the full proteomic space, which enables unbiased exploration of drug-induced responses, including those involving proteins not annotated in canonical pathways. Moreover, our study identifies novel putative drug targets, which further demonstrates that the insights gained are not limited to proteins in known pathways, but are driven by the broad and dynamic coverage of the proteomic data. Critically, the massive scale of our perturbation proteomics dataset—far exceeding previous resources in both breadth and temporal resolution—directly enables the discovery of these pathway-independent regulatory mechanisms that would remain undetectable in smaller or static datasets.

In addition, as shown in **Figure S4E-G**, model performance consistently improves as the number of training proteomes increases. This indicates that the insights and predictive power of ProteinTalks are directly enhanced by the scale and diversity of the proteome dataset, rather than by overfitting to limited known biological constraints. We have added the following statement to the manuscript: *“Performance scaled predictably as training data increased from 10% to 70% (**Figure S4E-G**).”*

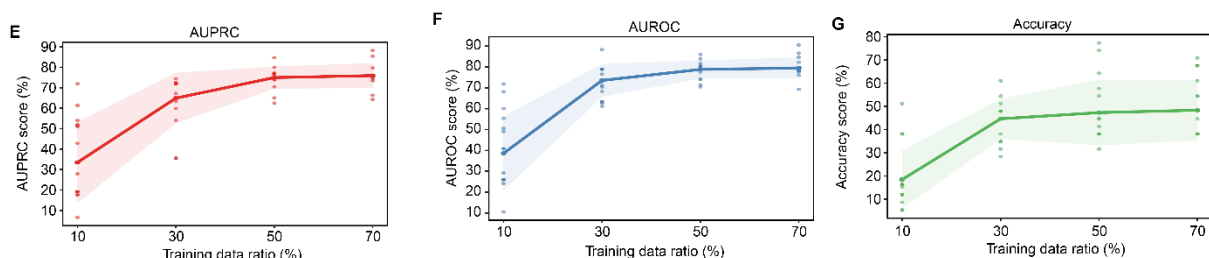


Figure S4 (E)-(G) Learning curves showing the performance of ProteinTalks evaluated by AUPRC (E), AUROC (F), and accuracy (G) increasing with the training data size.

Together, these additions clarify that the biological insights revealed by ProteinTalks arise from the richness and dynamics of the large-scale proteome data, rather than reliance on pre-defined pathways.

● How did the authors manage proteins that fall outside of known pathways in the ODE?

Reply: We appreciate the reviewer’s question. In ProteinTalks model, the ODE framework was constructed using all detectable proteins, rather than being restricted to predefined pathways. This fully data-driven design enables the system to capture temporal dynamics in an unbiased manner and, importantly, allows us to uncover proteins and interactions outside of known pathways. To better demonstrate this capability, we have removed pathway enrichment analysis in the revised manuscript to avoid confusion, instead we analyzed novel proteins emerging from our unbiased modeling. For example, we validated the role of AKR1C3 in regulating breast cancer cell sensitivity to microtubule-targeting drugs through gene knockdown. Furthermore, through manual literature curation, we further investigated the 15 proteins identified by ProteinTalks that had no prior cancer associations. We summarized their biological functions and potential clinical relevance beyond known oncogenic pathways. The detailed biological interpretation of these novel candidates has been incorporated into

the supplementary information to illustrate how ProteinTalks can reveal previously unrecognized yet mechanistically relevant biomarkers, *“manual curation of the top 100 proteins prioritized by ProteinTalks revealed that 60 have established roles in breast cancer diagnosis, treatment, or prognosis, and 22 have been implicated in other cancers. Notably, 15 proteins lacked prior cancer associations to our knowledge, including six involved in cellular component organization, five linked to nucleic acid metabolism, and four zinc-finger proteins, representing putative novel candidates for future investigation (Table S10). Although not previously reported in cancer, their core biological functions might be mechanistically connected to cancer biology, particularly in the context of chemotherapy. Chemotherapy primarily disrupted proteins governing cellular component organization through its mechanisms of action. For example, ZFYVE19, a cytokinesis checkpoint regulator that delays abscission to prevent DNA damage accumulation²⁷, might exhibit altered expression upon exposure to alkylating agents. Similarly, SYNE2, which serves as an actin cytoskeleton linker that maintains subcellular spatial organization²⁸, could plausibly be perturbed by antimitotic agents. Proteins linked to nucleic acid metabolic processes are likely affected by alkylating agents²⁹, given their DNA-targeting activity. Moreover, zinc-finger proteins, the largest class of DNA-binding proteins³⁰, participate in DNA damage recognition and repair—pathways that are critical for resistance to alkylating agents. Collectively, these 15 proteins constitute mechanistically plausible candidates whose perturbation could modulate chemotherapy efficacy, warranting further experimental validation. These findings suggest that the ProteinTalks model effectively identifies clinically relevant biomarkers.”*

- PertScore seems to be an ad-hoc metric used to quantify the protein expression changes over time. This score is dependent on determining if a protein expression is upgraded or downgraded, which could be subjective. In addition, given any cutoff of this score (10 used in this paper), what is the false discovery rate in the downstream applications?

Reply: We thank the reviewer for this insightful comment and fully agree that the design and evaluation of PertScore warrant clarification. We would like to clarify that PertScore is derived from systematic differential expression analysis. Specifically, it summarizes the number and directionality (up- or down-regulation) of significantly altered proteins across time points, based on predefined and consistently applied fold-change thresholds. This makes it an objective and reproducible metric that is grounded in observed data and assessed for statistical significance using Benjamini–Hochberg adjusted Student’s *t*-tests, ensuring that only statistically significant differences were included. The potential misunderstanding may have arisen because we described differential expression analysis and PertScore calculation separately in the original manuscript. To avoid ambiguity, we have rephrased the **Methods** “*PertScore Calculation*” as follows:

*“Prior to differential expression analysis, proteins that were absent in more than 80% of the samples across both groups were excluded. Group comparisons were performed using a two-sided unpaired Welch’s *t*-test, with *p*-values corrected for multiple testing using the Benjamini–Hochberg (B-H) method. Differentially expressed proteins were identified based on a combination of fold change and either raw or B-H adjusted *p*-values.*

*Based on these significantly altered proteins, we designed the PertScore to summarize how consistently a protein responds to perturbation across different conditions. For each drug treatment in each cell line and at each time point, significantly upregulated proteins were assigned +1, downregulated proteins –1, and unchanged proteins 0. By aggregating these values across all experiments, the PertScore highlights proteins that repeatedly and robustly change in response to drug perturbations, capturing recurrent biological signals rather than one-off fluctuations. To evaluate the robustness of this metric, we performed sensitivity analysis across different PertScore thresholds (7, 8, 9, 10, 11, 12) and found that 85% of identified perturbation-related proteins were consistent across thresholds (**Figure S3E**), indicating that PertScore is highly robust and not sensitive to arbitrary cutoff choices.”*

We also acknowledge the reviewer’s concern regarding potential dependence on cutoff selection. As there is currently no gold standard set of truly perturbed proteins to serve as ground truth, we performed this sensitivity analysis under different threshold settings to assess robustness, and added this in the **Methods** “*PertScore Calculation*”: *To evaluate the robustness of this metric, we performed sensitivity analysis across different PertScore thresholds (7, 8, 9, 10, 11, 12) and found that 85% of identified perturbation-related proteins were consistent across thresholds (Figure S3E), indicating that PertScore is highly robust and not sensitive to arbitrary cutoff choices.*

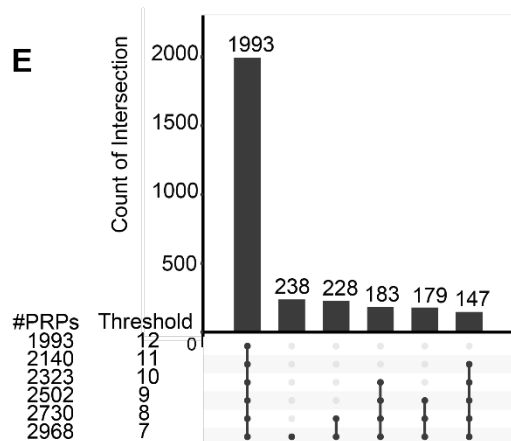


Figure S3E. The number of overlapping perturbation-related proteins (PRPs) identified with different PertScore threshold (7, 8, 9, 10, 11, 12).

● In the discussion of predicting chemotherapy 345 sensitivity in these biopsy tissue samples, it is okay to compare 60 proteins selected by ProteinTalks to random selection, but it is more interesting to see if this comparison is done with knowledge-driven and other benchmarking models’ selection. In addition, among these 60 proteins, are they mostly previously known biomarkers? Any new findings? And if there are new findings, are they thanks to the model or the proteome-wide dataset, or both?

Reply: We thank the reviewer for this constructive suggestion. To enhance robustness and avoid sign-related cancellations, we revised the feature-selection procedure to rank proteins by the mean absolute SHAP value and expanded the panel from the top 60 to the top 100 proteins identified by ProteinTalks. Analyses based on this updated Top-100 set are reported in the revised manuscript.

Regarding the composition of the selected proteins, *“Manual curation of the top 100 proteins prioritized by ProteinTalks revealed that 60 have established roles in breast cancer diagnosis, treatment, or prognosis, and 22 have been implicated in other cancers. Notably, 15 proteins lacked prior cancer associations to our knowledge, including six involved in cellular component organization, five linked to nucleic acid metabolism, and four zinc-finger proteins, representing putative novel candidates for future investigation (Table S10). Although not previously reported in cancer, their core biological functions might be mechanistically connected to cancer biology, particularly in the context of chemotherapy. Chemotherapy primarily disrupts proteins governing cellular component organization through its mechanisms of action. For example, ZFYVE19, a cytokinesis checkpoint regulator that delays abscission to prevent DNA damage accumulation²⁷, might exhibit altered expression upon exposure to alkylating agents. Similarly, SYNE2, which serves as an actin cytoskeleton linker that maintains subcellular spatial organization²⁸, could plausibly be perturbed by antimitotic agents. Proteins linked to nucleic acid metabolic processes are likely affected by alkylating agents²⁹, given their DNA-targeting activity. Moreover, zinc-finger proteins, the largest class of DNA-binding proteins³⁰, participate in DNA damage recognition and repair—pathways that are critical for resistance to alkylating agents. Collectively, these 15 proteins constitute*

mechanistically plausible candidates whose perturbation could modulate chemotherapy efficacy, warranting further experimental validation. These findings suggest that the ProteinTalks model effectively identifies clinically relevant biomarkers.”

Regarding whether new findings arose from the model or the dataset: We emphasize that the integration of dataset scale and model design was essential for uncovering these new findings. We have added the following to the **Discussion**, as copied below

“The availability of extensive time-course proteomics data was essential for implementing a neural ODE architecture capable of learning mechanistic cellular trajectories from temporal dynamics. Such dynamic modeling is inaccessible to transcriptome-based foundation models: public transcriptomic datasets lack time-series measurements, constraining these models to large-language-model-inspired Transformer architectures that learn static correlations rather than dynamic processes. Crucially, our large-scale temporal measurements capture the complex nonlinear dynamics inherent to real biological systems, enabling ProteinTalks to achieve superior generalization with ~1000-fold fewer parameters than transcriptomic foundation models. Recent benchmarks confirm that such parameter-heavy foundation models remain insufficient for perturbation response prediction⁷⁻⁹. By contrast, ProteinTalks—leveraging biologically grounded dynamics rather than parameter scale—exhibited robust and broad applicability across predictive tasks spanning 18 breast cancer cell lines, PDX models, and a 504-patient cohort, extending from 63 training drugs to 3000 compounds and validating clinical translatability from cell lines to patient biopsies.

This mechanistic grounding enabled discovery of actionable biological insights: experimentally validated drug synergies (e.g., AKR1C3 mediating toremifene-afatinib response), and 15 chemotherapy-associated proteins with no prior cancer links (including ZFYVE19, SYNE2, and zinc finger proteins). These findings emerged from synergy between large-scale temporal proteomics and dynamics-aware modeling, showing ProteinTalks' capacity to reveal previously inaccessible biological patterns.”

Minor comments

- The writing requires improvement, particularly in the Methods section, where sentence flow could be smoother and phrasing more concise.

Reply: We appreciate the reviewer's suggestion regarding the clarity and conciseness of the Methods section. In the revision, we have made the following improvements:

- (1) Enhanced sentence flow and conciseness throughout the Methods section. We carefully revised the Methods to improve readability by shortening complex sentences, removing redundant phrases, and using more direct phrasing where possible.
- (2) Added a navigation aid. We included *Supplementary Information*, “Mapping of Results sections to Methods descriptions,” which provides a clear correspondence between each Results subsection and its associated Methods subsection(s), facilitating easier navigation for readers.

Referee #2 (Remarks on code availability):

I'm of the opinion that this paper presents marginal advantages to existing models and poses exaggerated claims about having developed a “foundation model” without the data to back that up with broad and generalizable applications. Due to the lack of novel biological insights, I believe this paper would be more suitable for a journal like *Nature Methods*.

Reply: We sincerely thank the reviewer for the constructive and thoughtful comments. We acknowledge the reviewer's concerns about the generalizability and novelty of our work, and fully

agree that broad and robust evidence is essential for supporting the claims of a foundation model. In response, we have further optimized the model, incorporated additional benchmarking comparisons, and extended the validation to multiple datasets, including pan-cancer cell lines, clinical specimens, and patient-derived organoids. These revisions have yielded more robust and comprehensive results, as detailed in our responses to earlier points. We believe that these new analyses help to make our conclusions more convincing and address the reviewer's concerns.

First, our model demonstrates consistent and substantial advantages over existing foundation models, including Geneformer, GeneCompass, and UCE, across diverse benchmark tasks. In the cross-cell prediction task, ProteinTalks improved AUPRC, AUROC, and accuracy by approximately 63%–213%, 27%–98%, and 17%, respectively. In the cross-drug setting, ProteinTalks outperformed these models by 26%–57% in AUROC, 22% in accuracy, while achieving comparable AUPRC values. Notably, in predicting the efficacy of 708 newly repurposed compounds across five non-TNBC cell lines, ProteinTalks achieved its strongest performance, with average improvements of 74% in AUPRC and 32% in AUROC over these foundation models. Furthermore, without any additional training, ProteinTalks generalized effectively to pan-cancer PDX models, outperforming all foundation model baselines with an average gain of 118% in AUPRC and 32% in AUROC. These results demonstrate broad and robust generalization across heterogeneous cells, drugs, and tumor systems, substantially **beyond “marginal advantages”**.

Second, our architectural innovation moves beyond conventional Transformer-based models with large parameters. We introduce a neural-ODE framework that learns effective dynamical embeddings from perturbation proteomics datasets. This allows us to capture temporal and mechanistic aspects of protein network reorganization, enabling more biologically grounded and predictive modeling of cellular responses. Notably, recent benchmarking studies suggest that state-of-the-art transcriptomics-based foundation models remain highly insufficient for perturbation response prediction¹⁻³, highlighting the unique contribution of our dynamics-aware proteomics-based approach.

Third, regarding the concern about broad applicability, in the revision we provide extensive evidence across multiple biological and clinical dimensions:

- Pan-cancer extension: 2217 perturbation profiles from five additional non-breast cancer cell lines with 708 compounds.
- Clinical relevance: 504 longitudinal proteomes from TNBC patient biopsies treated with standard regimens.
- Drug repurposing: A library of ~3000 compounds screened in three patient-derived TNBC organoids, with functional validation identifying compounds (e.g., CAY10603, citarinstat, THZ-1) outperforming cisplatin.

Fourth, regarding novel biological insights, ProteinTalks uncovers previously unknown mechanisms and provides experimental validation across multiple levels. We have added the following to the

Discussion:

“This mechanistic grounding enabled discovery of actionable biological insights: experimentally validated drug synergies (e.g., AKRIC3 mediating toremifene-afatinib response), and 15 chemotherapy-associated proteins with no prior cancer links (including ZFYVE19, SYNE2, and zinc finger proteins). These findings emerged from synergy between large-scale temporal proteomics and dynamics-aware modeling, showing ProteinTalks' capacity to reveal previously inaccessible biological patterns.”

Taken together, we respectfully contend that ProteinTalks qualifies as a foundation model by virtue of its scale (millions of perturbation protein measurements), generalization (cross-cell, cross-drug, pan-cancer, and clinical settings), mechanistic interpretability (dynamical embeddings), and ability to generate new, clinically relevant biological insights. Unlike prior transcriptomics-based models mainly used in basic research, ProteinTalks bridges toward translational applications in precision

oncology. We believe the combination of (1) substantial performance improvements over existing models, (2) novel architectural innovations addressing fundamental limitations of current foundation models, (3) extensive validation across diverse biological contexts, and (4) discovery of previously uncharacterized cancer-relevant proteins demonstrates both methodological advances and biological impact that merit consideration for a broader audience.

References

- 1 Boiarsky, R. *et al.* Deeper evaluation of a single-cell foundation model. *Nature Machine Intelligence* **6**, 1443-1446 (2024).
- 2 Kedzierska, K. Z., Crawford, L., Amini, A. P. & Lu, A. X. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biol* **26**, 101 (2025).
- 3 Ahlmann-Eltze, C., Huber, W. & Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nat Methods* **22**, 1657-1661 (2025).

Referee #3 (Remarks to the Author):

Overall, this work presents a great data set that took a lot of work to produce. However, we feel that the AI model is not that innovative and we worry that 3 data points are not enough to get a meaningful embedding.

Reply: We sincerely thank the reviewer for recognizing the scale of our dataset and the effort behind its generation, and for raising thoughtful comments about the novelty of our model and the adequacy of three time points. We address both points below.

1. Novelty and significance of the ProteinTalks framework. Our study introduces the first virtual cell model grounded in large-scale perturbation proteomics, capturing dynamic protein responses across diverse perturbations and time points. This framework grounds virtual cell modeling in proteomic function and achieves transcriptomics-level performance with far fewer measurements through a protein-centric, perturbation-aware design. The model performs robustly across ten downstream tasks—from cancer cell lines to PDX and patient cohorts—and generalizes from 63 training drugs to 3000 compounds, highlighting both methodological robustness and translational potential. We elaborate on these aspects in detail below.
- (1) **Novelty of the concept:** While virtual cell modeling has recently attracted significant attention, most existing efforts are built almost exclusively on transcriptomic profiles, which reflect only a cell's transcriptional potential rather than its functional dynamics. Moreover, few of these models incorporate perturbation data, and even when they do, the datasets are typically limited to just two states—pre- and post-perturbation. Such designs inherently lack the temporal resolution and diversity needed to capture the dynamic evolution of cellular systems, resulting in models with limited ability to represent true cellular dynamics. Additionally, proteins—rather than mRNAs—are the direct executors of cellular functions, and their abundance, localization, and activity frequently diverge from transcript levels, especially under perturbations.
- (2) **Novelty of the model:** Recent benchmarking studies¹⁻³ suggest that cutting-edge foundation models, which are extensively trained on public transcriptomics data, despite their huge parameters based on large-language-model-like structures, remain highly insufficient in perturbation response prediction task. It is also noteworthy that, instead of using a Transformer-based architecture with huge parameters to optimize, we proposed an alternative, neural-ODE based framework to learn effective dynamical embeddings from large-scale perturbation proteomics datasets. This enables a more mechanistic and predictive representation of cellular responses, moving closer to the complex, multi-system nature of diseases such as cancer and addressing a key limitation highlighted by the reviewer.

- (3) **Novelty of the application scope:** Our study is also novel in that we extend the framework from cell lines to PDX models and patient cohorts, thereby demonstrating its translational relevance across multiple biological and clinical contexts.

Therefore, by pretraining on perturbation proteomics data with a dynamics-aware architecture, the ProteinTalks provides a new avenue for foundation model construction and have the potential to inspire future works along this direction. We have modified the **Introduction** and **Discussion**, as copied below.

In the **Introduction**,

“In silico drug discovery represents a transformative frontier in biomedical research. It is built on predictive cellular models that simulate molecular and cellular responses to therapeutic perturbations. In this context, the virtual cell concept has emerged as a promising paradigm in drug discovery, where the central challenge lies in accurately modeling cellular responses and dynamics. Given the inherent nonlinearity of cellular processes, virtual cells require training on real-world dynamic data to capture the true complexity of cellular behavior¹⁻³. Recently, foundation models trained on large-scale omics datasets are increasingly viewed as a critical path toward virtual cell. However, existing foundation models, such as Geneformer⁴, scFoundation⁵, and STATE⁶, have predominantly relied on single-cell transcriptomic data. Even those trained on perturbation transcriptomic data generated by Perturb-seq remain inherently limited to pre- and post-perturbation snapshots, lacking temporal resolution to capture true dynamics. Transcriptomic readouts capture only an intermediate layer of the central dogma, which may underlie the poor performance of transcriptomics-based foundation models on new datasets⁷⁻⁹. Proteins, as the primary executors of cellular functions, are more directly tied to phenotype —yet proteome-based virtual cell models remain underdeveloped due to the scarcity of large-scale, time-resolved proteomic datasets.

To address this gap, we established a systematic temporal perturbation proteomics resource focused on drug response. Recent advances in high-throughput proteomics enabled by data-independent acquisition mass spectrometry (DIA-MS)^{10,11}, have made large-scale perturbation proteomics data generation more feasible¹². Utilizing a 96-well-based high-throughput platform, we perturbed breast cancer cell lines with clinically relevant drugs and their combinations, yielding more than 38 million perturbed protein measurements alongside cell viability data.

Leveraging this large-scale dataset, we developed ProteinTalks, a virtual cell model that explicitly embeds time-evolving protein network trajectories using a neural ordinary differential equation (neural ODE)^{13,14} framework (Figure 1). Despite using a thousandfold fewer parameters and data points than existing static transcriptomic foundation models, ProteinTalks achieves superior predictive performance across diverse tasks, including cross-cell-line and cross-drug evaluation, independent drug tests, cross-disease transfer, drug synergy prediction, and fine-tuning for patient-derived xenograft (PDX) response prediction. Ultimately, it stratifies clinical outcomes in 504 breast cancer patients and prioritizes candidates from a 3000-repurposed-drug library for triple-negative breast cancer (TNBC) patient-derived organoids (PDOs). Interpretability analyses highlight previous unknown candidate biomarkers and mechanisms that align with experimental and clinical validation. Together, this work establishes ProteinTalks as a proteome-based virtual cell with cellular functional capability for drug screening.”

In the **Discussion**,

“Understanding cellular mechanisms and predicting drug responses remain central challenges in biology and drug discovery. We shed new light on this challenge through ProteinTalks, a proteome-based virtual cell model built on large-scale, time-resolved perturbation proteomics. This temporal foundation enabled dynamic modeling of cellular state transitions, allowing translation from controlled cell line experiments to patient-derived tissues.

Our dataset, comprising over 38 million protein measurements from ~16,000 perturbed proteomic samples that we produced in-house, represents the largest AI-ready proteomics resource to date, over an order of magnitude larger than previous perturbation proteomics datasets³⁴⁻³⁷. Critically, unlike static pre/post-perturbation snapshots common in existing datasets, our time-series design captures the dynamic evolution of cellular responses. This temporal resolution, combined with proteomic measurement, fundamentally distinguishes ProteinTalks from transcriptome-based foundation models⁴⁻⁶. Unlike transcriptomics, which reflects transcriptional potential rather than functional dynamics, proteomics directly captures the functional executors of cellular processes. Importantly, transcript and protein abundances frequently diverge under perturbations, making proteome-level measurements critical for understanding actual cellular responses.

The availability of extensive time-course proteomics data was essential for implementing a neural ODE architecture capable of learning mechanistic cellular trajectories from temporal dynamics. Such dynamic modeling is inaccessible to transcriptome-based foundation models: public transcriptomic datasets lack time-series measurements, constraining these models to large-language-model-inspired Transformer architectures that learn static correlations rather than dynamic processes. Crucially, our large-scale temporal measurements capture the complex nonlinear dynamics inherent to real biological systems, enabling ProteinTalks to achieve superior generalization with ~1000-fold fewer parameters than transcriptomic foundation models. Recent benchmarks confirm that such parameter-heavy foundation models remain insufficient for perturbation response prediction⁷⁻⁹. By contrast, ProteinTalks—leveraging biologically grounded dynamics rather than parameter scale—exhibited robust and broad applicability across predictive tasks spanning 18 breast cancer cell lines, PDX models, and a 504-patient cohort, extending from 63 training drugs to 3000 compounds and validating clinical translatability from cell lines to patient biopsies.

This mechanistic grounding enabled discovery of actionable biological insights: experimentally validated drug synergies (e.g., AKRIC3 mediating toremifene-afatinib response), and 15 chemotherapy-associated proteins with no prior cancer links (including ZFYVE19, SYNE2, and zinc finger proteins). These findings emerged from synergy between large-scale temporal proteomics and dynamics-aware modeling, showing ProteinTalks' capacity to reveal previously inaccessible biological patterns.”

2. Theoretical and empirical justification for three time points. We agree that temporal resolution is critical for the design of a new dynamical foundation model and address this “3 data points” concern in detail from two complementary aspects. (1) From a theoretical perspective, we provide a theoretical justification, grounded in dynamical systems and neural ODE analysis, showing that three post-treatment time points (plus baseline) are sufficient to identify both first-order dynamics and leading nonlinear corrections of drug–proteome trajectories. By leveraging local Taylor expansions, global over-determination across diverse cell lines and drugs, and stability arguments, we demonstrate that this sparse sampling still enables reliable learning of a stable and generalizable vector field. (2) To further validate this empirically, we generated a benchmark dataset with ten time points across three cell lines under drug treatment. Analysis of this dataset showed that the selected three time points effectively represent the early, intermediate, and late phases of cellular response to perturbation. Moreover, through a stepwise timepoint expansion approach, we found that three time points provide an optimal balance, enabling the model to achieve its best performance without unnecessary redundancy. We provide a detailed explanation in the following response.

References

1. Boiarsky, R. *et al.* Deeper evaluation of a single-cell foundation model. *Nature Machine Intelligence* **6**, 1443-1446 (2024).
2. Kedzierska, K. Z., Crawford, L., Amini, A. P. & Lu, A. X. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biol* **26**, 101 (2025).
3. Ahlmann-Eltze, C., Huber, W. & Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nat Methods* **22**, 1657-1661 (2025).

Thus, while enthusiastic, we are not sure if this is over the bar for Nature.

Summary of the key results

- Summary of the model:

A neural ODE based foundational model, ProteinTalks, was built based on 38 million perturbed protein measurements in breast cancer cell lines. It learned cellular protein network dynamics and predicted drug efficacy, synergy, and patient responses to drugs.

– Summary of results:

- (1) Generated a large scale-perturbation proteome dataset from DIA-MS analytics of 18 breast cancer cell lines treated with 63 FDA-approved drugs at 6, 24, 48 hours.
- (2) Proposed PertScore, a quantitative metric that captures drug-induced protein alterations, reflecting the drug's MOA.
- (3) Longitudinal proteomic responses to drug perturbation distinguishes between chemotherapy-resistant and sensitive groups.
- (4) A two-module foundation model - the first module ingests untreated proteome data, storing protein network dynamics, and predicts perturbed protein expression at 6, 24, 48 hours; the second module integrates the unperturbed proteomes, the predicted perturbed proteomes, and molecular structures of drugs to predict drug efficacy and combo synergy using MLP.
- (5) ProteinTalks outperformed typical ML models in predicting drug responsiveness.
- (6) The model's capability of predicting drug synergy was validated experimentally. SHAP values reflect the model's capability of identifying pathways and protein networks responsible for drug resistance.
- (8) Transfer learning of drug sensitivity prediction was performed on breast cancer patient-derived tumor cells and pan-cancer cells from PDTX models.
- (9) Drug response was tested using clinical biopsy tissue samples both in short-term and long-term.

Originality and significance:

- Significance: this study is at least ten times larger in size compared to previous studies, enabling it to be directly used for training AI models

Reply: We thank the reviewer for the clear and concise summary of our study. Indeed, ProteinTalks is the first neural ODE-based foundational model trained on a large-scale longitudinal perturbation proteomics dataset (>38 million protein measurements). Unlike prior efforts that typically relied on static snapshots of proteomes or transcriptomes, our model leverages temporal protein dynamics across multiple drugs, time points, and cell lines. This unique design enables the model to directly

learn cellular protein network reorganization under pharmacological perturbations and to generalize to multiple clinically relevant tasks.

As highlighted in the Results, ProteinTalks achieved several key advances:

1. Massive perturbation proteomics dataset. To our knowledge, this is the largest pharmacoproteomic dataset so far (18 breast cancer cell lines $\times$ 63 FDA-approved drugs $\times$ 3 time points), which is over an order of magnitude larger than existing studies.
2. PertScore metric a systematic, quantitative measurement of protein-level drug responses, allowing mode-of-action profiling.
3. Neural ODE modeling uniquely suited to capture temporal trajectories of proteomic dynamics, which cannot be resolved by conventional ML models.
4. Generalizable predictive capacity. Beyond drug efficacy in cell lines, ProteinTalks predicts drug synergy (validated experimentally) and patient responses (PDOs, PDTXs, and clinical biopsies).
5. Interpretability. SHAP analysis and downstream enrichment revealed both canonical pathways and novel proteins associated with drug sensitivity, including proteins outside known pathways.
6. Clinical significance. successful transfer learning to patient-derived datasets strengthens the translational potential of ProteinTalks.

In summary, we believe the originality and significance of our work lie in (i) establishing the first foundation model for perturbed proteomic dynamics, (ii) generating the largest longitudinal pharmacoproteomics dataset to date, and (iii) demonstrating broad applicability across drug response prediction, synergy discovery, and clinical patient stratification.

- Direct applications:

(1) Assisting drug discovery by uncovering drug MOAs

(2) Predicting the prognosis of breast cancer patients when integrated with clinical patient data

Reply: We thank the reviewer for pointing out the potential direct applications of our work. We fully agree that elucidating drug mechanisms of action (MOAs) and predicting patient prognosis through integration with clinical data represent two highly valuable directions. To further strengthen the translational relevance of ProteinTalks, we expanded our study in this revision to two potential clinical scenarios. First, using proteomic profiles from 504 TNBC patients who received first-line therapy, ProteinTalks accurately stratified patients into high- and low-risk groups with significant differences in survival and metastasis outcomes (revised manuscript: the **Results** section “*ProteinTalks stratifies prognosis from clinical biopsies*” and *Figures 5C-E*). Second, by integrating three TNBC organoid proteome data with a 3000-compound library, ProteinTalks prioritized candidate drugs, and three of the top-ranked agents showed superior efficacy over cisplatin in IC50 assays (revised manuscript: the **Results** section “*ProteinTalks prioritizes candidate repurposed drugs from clinical biopsy proteomes*” and *Figures 5F-G*). These results underscore the potential of our model to guide both prognosis and therapeutic decision-making in precision oncology.

Data & methodology: validity of approach, quality of data, quality of presentation

- Question on quality of validity of approach:

The inherent complexity of DIA data can often lead to false-positive identifications, especially in the presence of perturbations. Did the authors consider and adopt any strategies to mitigate false-positives? How was the accuracy (or potential inaccuracy) of the DIA-NN accounted for?

Reply: We thank the reviewer for this valuable comment highlighting the potential for false-positive identifications in DIA-based analyses. Indeed, DIA data are intrinsically complex due to the co-fragmentation of multiple precursor ions, which can increase the likelihood of ambiguous identifications, particularly under perturbation conditions.

In this study, DIA was chosen primarily for its suitability in large-scale and high-throughput proteomic profiling, where consistent peptide detection across runs is essential. Compared with isotope-labeled DDA, DIA requires less sample preparation time, facilitates automation, exhibits lower batch effects, and avoids the need for matched control samples across heterogeneous cell lines (Guzman et al., *Nat. Biotechnol.*, 2024; Guo et al., *Nature*, 2025). Several **large-scale proteomic studies** have successfully adopted DIA for similar experimental designs (e.g., Messner et al., *Cell*, 2023; Oostrum et al., *Cell*, 2023; Clark et al., *Cell*, 2019), and **longitudinal studies** have also demonstrated its robustness under dynamic or perturbation conditions (e.g., Bouhaddou et al., *Cell*, 2020; Ding et al., *Cell*, 2025; Muenzner et al., *Nature*, 2024).

To mitigate false-positives and ensure data accuracy, we implemented a multi-layered quality control (QC) framework:

- Sample-level filtering: We excluded any samples with fewer than 1000 proteins identified (see **Methods** “Proteomics data processing and quality control”).
- Protein-level filtering: We ensured that the missing value rate for each protein was below 90% across all samples (see **Methods** “Proteomics data processing and quality control”).
- Correlation analysis: The median Pearson correlation between replicates was greater than 0.9, further confirming reproducibility (see **Figures S1B, S3B**).
- Reproducibility assessment: We confirmed that the median coefficient of variation (CV) for biological replicates, pre-processing technical replicates, and LC-MS acquisition replicates was consistently below 10% (see **Figures S1C, S3C**).
- Batch effect evaluation: We observed no significant batch effects across different LC-MS/MS Machines (see **Figures S1D, S3D**).

These QC procedures are now described in greater detail in the **Methods** and **Results** sections. We believe that the combination of a statistically robust DIA analysis pipeline and strict QC filtering ensures the accuracy and reproducibility of our proteomic quantification, even under perturbation conditions.

For data processing, DIA-NN implements strict false discovery rate (FDR) control at both peptide and protein levels ($q < 0.01$) through a machine learning-based scoring strategy and a two-species spectral library approach. The software’s FDR calibration and benchmarking against alternative tools such as OpenSWATH, Skyline, and Spectronaut were rigorously validated in the original publication (Demichev et al., *Nat. Methods*, 2019), demonstrating reliable identification accuracy and low false-positive rates under diverse experimental conditions. DIA-NN (version 1.7.15) was used following the parameters and recommendations established by Messner et al. (*Cell*, 2023), who benchmarked this approach under perturbation conditions using a similar LC-MS setup. Detailed analysis parameters and cutoffs have been added to the **Methods** section to enhance transparency and reproducibility.

“Proteomics data processing and quality control

Raw DIA data files were initially converted to profile mode mzML format using msConvert. Subsequent analysis of mzML files was performed with DIA-NN software⁴⁸ (version 1.7.15), as previously described^{49,50}. Analysis parameters were meticulously set: MS2 and MS1 mass accuracies and scan window size set to auto; trypsin/P as the protease with a maximum of two missed cleavages allowed; N-terminal methionine excision, cysteine carbamidomethylation, and methionine oxidation as the specified modifications; peptide length restricted to a range of 7 to 30 amino acids; and an m/z

range of 400 to 2000 for precursor ions and 100 to 2000 for fragment ions. Precursors and peptides were restricted to q -values < 0.01 , retaining only proteotypic peptides. Protein abundances were then quantified with the MaxLFQ algorithm (as implemented in the DIA-NN R package), with statistical confidence ensured by applying a q -value cutoff of 0.01 at the protein level. Data were processed using the single-pass mode of the neural network classifier.

From the initial dataset, 689 samples with identification counts below 1000 were discarded, leaving 16,311 samples for analysis. No proteins exhibited a missing rate greater than 90% in the complete dataset, with an overall missing rate of 51.7%. The initial step in data preprocessing involved imputing missing values with a value equivalent to 0.8 times the minimum detected intensity. The reproducibility of biological replicates, technical replicates, and pooled samples was assessed using Pearson correlation and the coefficient of variation (CV) as metrics.”

While DIA-NN provides a statistically controlled framework for DIA data analysis, the interpretation of DIA spectra remains a developing area. The analytical accuracy of DIA-based proteomics continues to depend on algorithmic design and training data coverage. We have expanded the **Discussion** section to acknowledge these limitations and outline future directions, as follows:

“Our DIA-MS strategy prioritized throughput over depth, yielding fewer proteins than ultra-deep methods³⁴⁻³⁶, though ongoing AI-driven advances^{38,39} are enhancing DIA data analysis.”

- Question on model performance:

In Figure S4, the model performance on the test set appears to be poor for cell line DU4475 in all three sub figures (Synergy Accuracy, AUPRC, and AUROC) despite the excellent train and validation performance. Any explanation for this?

Reply: Thanks for the comments. We have added explanation for this result, as copied below. “For DU4475 with exclusively negative labels, ProteinTalks achieved the highest AUPRC (Figure 3B), suggesting robustness to class imbalance.”

- Question on quality of presentation:

(1) Figure S3 seems to be missing Figure caption for subfigure B.

Reply: Thanks for notification. We have replaced this figure and checked all caption for each figure.

(2) The paper mixed up the description of results and methods, making it not straightforward for readers to understand. This is seen in the 9 subsections under Results.

Reply: We thank the reviewer for pointing this out. We agree that in the current version, to improve clarity, we have thoroughly restructured the manuscript: (1) methodological descriptions have been consolidated and moved into the **Methods** section, with each method clearly labeled with a subsection title that corresponds to the relevant **Results** section; and (2) we have streamlined the Results text to remove procedural details while maintaining scientific rigor. We believe this revised organization will make it much easier for readers to follow the logic of our study.

To further facilitate navigation, we have added **Supplementary Information**, “Mapping of Results sections to Methods descriptions”, which provides a clear correspondence between each Results subsection and its associated Methods subsection(s), enabling readers to quickly locate the methodological details for any result presented.

Mapping of results sections to methods descriptions

<i>Results subsection</i>	<i>Corresponding Methods subsection(s)</i>
1. Generation of perturbation proteome datasets for building a proteomic-based virtual cell model	- Cell line panel - Cell viability assay - Drug treatment for proteomics preparation - LC-MS/MS analysis - Proteomics data processing and quality control
2. The perturbation proteomics dataset captures drug-induced modulation of protein networks	- PertScore calculation - Pathway enrichment - mFuzz analysis
3. Proteome dynamics is associated with drug resistance	- PertScore calculation - Pathway enrichment - mFuzz analysis
4. Development of a virtual cell model ProteinTalks	- Data preprocessing for ProteinTalks pretraining - SMILES generation - Drug molecular fingerprints - ProteinTalks model architecture and parameters - Multi-task learning - ProteinTalks pretraining on PTDS-1
5. Balancing temporal resolution with dataset diversity	- Drug treatment for proteomics preparation - LC-MS/MS - Proteomics data processing and quality control - Eleven-time-point perturbation proteomics experiment and model training
6. Prediction of responsiveness of drugs	- Implementation of bagged decision trees, random forests, KNN, and DeepSynergy - Benchmark architecture for Geneformer, GeneCompass, and UCE - Leave-one-cell line-out evaluation - Leave-one-drug-out evaluation - Hyperparameters search for ProteinTalks, bagged decision trees, random forests, KNN, and DeepSynergy - Metrics calculation for model performance evaluation
7. Predicting effective repurposed compounds beyond standard therapies	- Cell viability assay - Independent testing on PTDS-2 - Benchmark architecture for Geneformer, GeneCompass, and UCE - Metrics calculation for model performance evaluation
8. Prediction of drug sensitivity across multiple cancer types	- Drug treatment for proteomics preparation - LC-MS/MS analysis - Proteomics data processing and quality control - Few-shot transfer on the PTDS-3

Results subsection	Corresponding Methods subsection(s)
	- Benchmark architecture for Geneformer, GeneCompass, and UCE - Metrics calculation for model performance evaluation
9. ProteinTalks predicts synergistic drug combinations	- In silico drug–drug combination library generation and scoring - Cell viability assay
10. ProteinTalks delineates dynamic protein networks remodeling	- Dynamic SHAP calculation
11. ProteinTalks prioritizes proteins responsible for drug resistance	- Drug efficacy SHAP value calculation for each protein - SHAP value filtration - Gene interference in different cell lines - Drug treatment for proteomics preparation - LC-MS/MS analysis - Proteomics data processing and quality control - Cell viability assay - Validation of the top 100 proteins identified by SHAP values in a 16-patient proteomic dataset
12. Predicting therapeutic outcomes in patient-derived tumor models	- Transfer learning for the transcriptomics data of PDTC model - Fine-tuning with a pan-cancer PDX cohort (PDX-2015nm) - Benchmark architecture for Geneformer, GeneCompass, and UCE - Metrics calculation for model performance evaluation
13. ProteinTalks stratifies prognosis from clinical biopsy	- FFPE clinical sample collection and proteomics preparation - LC-MS/MS analysis - Proteomics data processing and quality control - Survival analysis in a cohort of 504 TNBC patients - Validation of the top 100 proteins identified by SHAP values in a cohort of 504 TNBC patients
14. ProteinTalks prioritizes candidate repurposed therapies from clinical biopsy proteomes	- PDO culture and drug efficacy assay - LC-MS/MS analysis - Proteomics data processing and quality control - Drug library generation for PDO drug efficacy prediction - Drug efficacy prediction for a TNBC PDO

Appropriate use of statistics and treatment of uncertainties

Strength:

- The authors conducted a massive perturbation proteomics study with over 38 million protein measurements. The authors also applied stringent quality control. For example, multiple control

samples were used to account for batch effects and approximately 10% of randomly selected samples were run in replicates for quality control.

- The authors used a robust DE pipeline. They combined Welch's t-test and Benjamini-Hochberg adjustment to ensure false discovery control.

- SHAP values were used for interpreting model decisions, giving interpretation of protein importance in model outputs. The authors also performed several cross validation studies to ensure robustness of the method. Ablation studies were also performed to demonstrate the value of proteomic input vs. transcriptomic or metadata-only baselines.

Reply: We sincerely thank the reviewer for the careful reading of our manuscript and for recognizing our treatment of statistics and uncertainties. We appreciate the acknowledgement of the study's scale (>38 million measurements), our stringent QC strategy (including multiple controls and ~10% replicate runs), the robust differential expression pipeline (Welch's t-test with Benjamini-Hochberg adjustment), and the use of SHAP for interpretable model decisions. We are also grateful for the positive assessment of our cross-validation and ablation studies. We have further clarified these procedures in the Methods/Results to enhance transparency and reproducibility.

Limitations:

- The specific p-values and effect sizes of protein alterations are not always reported in the main text. For example, the authors could provide log2 fold-change and give more biological relevance of the finding beyond saying it is statistically significant.

Reply: We thank the reviewer for this valuable suggestion. Previously, detailed parameters such as p-values and effect sizes were provided in the **Methods** section or Supplementary Information. In the revised manuscript, we have now incorporated explicit reporting of p-values, fold changes and biological relevance in the main text of the **Results** section. The updated descriptions are copied below.

*"Next, we narrowed down to the consistently dysregulated proteins resulting from drug subtype perturbations through mFuzz clustering. Both chemotherapy and targeted drugs increase the expression levels of proteins related to nucleic acid synthesis and fatty acid metabolism, while decreasing the expression levels of proteins involved in RNA processing, RNA metabolism, and cell cycle progression ($p < 0.05$, **Figure 2B**). Kinase inhibitors specifically upregulated proteins linked to MET, NOTCH4, and mTOR signaling pathways ($p < 0.05$) (**Figure 2B**). Therefore, each compound interacts with one or more protein targets and induces proteome-level changes via both common and perturbagen-specific modulation of cellular processes that reflect its MOA. These perturbagen-induced abundance changes vary in magnitude from protein to protein."*

"A total of 113 proteins exhibited differentially dynamic expression in response to drug treatment, effectively distinguishing between the chemotherapy-resistant and sensitive groups ($ANOVA p < 0.05$, Table S3). Specifically, 75 proteins exhibited an increase in expression following perturbation in the resistant group but decreased in the sensitive group ($ANOVA p < 0.05$, Figure 2C). These proteins were mainly enriched in cell cycle associated pathways, protein localization, purine ribonucleotide metabolic process, and carbon metabolism ($p < 0.05$, Figure 2C). Conversely, 38 proteins showed an inverse dynamic expression pattern ($ANOVA p < 0.05$), mainly associated with small molecule transport, endomembrane system organization, and the AKT signaling pathway ($p < 0.05$, Figure 2C)."

"Upon treatment with 5-fluorouracil (an anti-metabolic agent), ATG3 showed the greatest divergence at 48 hours (fold change = 0.15; resistant vs. sensitive). As an autophagy-related protein previously implicated in chemotherapy resistance in hepatocellular carcinoma through regulation of autophagy processes¹⁹, the elevated ATG3 observed here suggests that enhanced autophagy may contribute to 5-fluorouracil resistance. For docetaxel (an anti-mitotic agent), the most pronounced divergence was observed for BTF3 at 48 hours (0.12-fold). The upregulation of BTF3 indicates that the epithelial cancer cells may possess stemness features²⁰, and cancer stem cells have been reported to be

*chemotherapy resistance*²¹. These findings suggest that BTF3 may enhance docetaxel resistance by maintaining stemness. In erlotinib-treated cells (an EGFR inhibitor), PAK1 and NDE1 showed the strongest differences at 48 hours (0.21-fold and “infinite”, respectively, the latter reflecting non-detection in one group). PAK1 has been associated with resistance to tyrosine kinase inhibitors in EGFR-mutant lung cancer²², likely via activation of the AKT pathway, while NDE1 is known to interact with EGFR²³, potentially attenuating erlotinib efficacy. Following exposure to talazoparib (a PARP inhibitor), ELAVL2 displayed the largest difference at 48 hours (0.30-fold). Consistent with prior reports implicating ELAVL2 in regulation of glycolysis²⁴, this suggests that metabolic reprogramming may mediate resistance to PARP inhibition. Conversely, under alpelisib treatment (a PI3K inhibitor), CKS2 expression diverged in the opposite direction—elevated in the sensitive group but reduced in the resistant group—with a maximal difference of 7.71-fold at 48 hours. This aligns with previous studies showing that CKS2 antagonizes PI3K/AKT signaling²⁵, in contrast to the resistance-promoting proteins described above. Together, these protein-specific dynamics may provide potential biomarkers and therapeutic targets for overcoming drug resistance in breast cancer.”

- The authors report AUROC, AUPRC, and accuracy, but don't show confidence intervals for those metrics.

Reply: We thank the reviewer for raising this important point. In the revised manuscript, we now report—for each task—the mean performance over independent repeats together with its standard error (SE) or standard deviation (SD). The rationale is to characterize the stability of the learning procedure under our protocol (random seeds and data splits), rather than the sampling uncertainty of a single, fixed model on a given test set. Using this approach, SE provides a direct measure of the precision of the mean performance across repeats and offers a transparent, comparable, and low-variance summary of algorithmic variability. We have updated all relevant **Figures** (Figures 3C-D, S5C-H, S6A-B, S7E-G, S11C-E) and tables to include these SE/SD values (as shown in Table S5-S7, S9, S11), providing a quantitative measure of variability across runs. We have also added a detailed description of the metrics calculation in the **Methods** “Metrics calculation”. The detailed results are presented as follows:

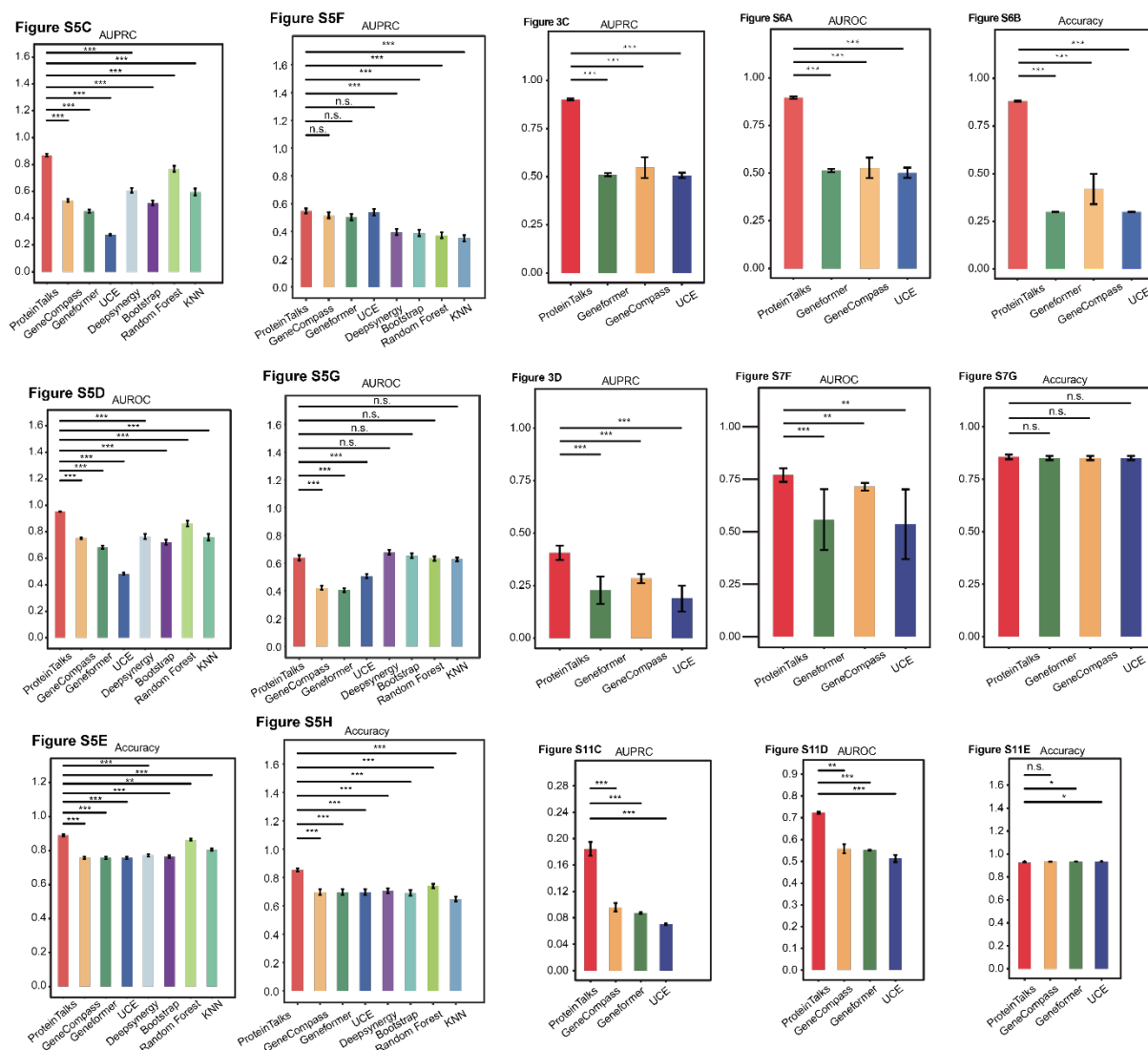


Figure S5C-E. AUPRC, accuracy, and AUROC of the leave-one-cell line-out cross validation and compared with other seven models. **Figure S5F-H.** AUPRC, AUROC, and Accuracy of the leave-one-drug-out cross validation and compared with other seven models. **Figures 3C, S6A-C.** AUPRC, AUROC, and Accuracy of the novel drugs predictions using PTNC and compared with Geneformer, GeneCompass, and UCE model. **Figures 3D, S7A-C.** Comparative performance of ProteinTalks and foundation models (Geneformer, GeneCompass, and UCE) on the few-shot transfer learning task for pan-cancer drug efficacy prediction, evaluated by AUPRC, AUROC, and accuracy. **Figure S11C-E.** Comparative performance of ProteinTalks and existing foundation models (Geneformer, GeneCompass, and UCE) on the pan-cancer PDX drug sensitivity prediction task, evaluated by AUPRC, AUROC, and accuracy.

Conclusions: robustness, validity, reliability

- The model has shown strong robustness in several validation settings: leave-one-cell-line-out and leave-one-drug-out. The authors also performed transfer learning on transcriptomic data to confirm its robustness
- Experimental validation was performed. Specifically, the knockdown of AKR1C3 and CMPK1 increased drug sensitivity in expected ways. The model was also validated by discovering known drug-protein interactions.
- Model performance was consistent across various test datasets, achieving a decent AUROC score across settings.

Reply: We sincerely thank the reviewer for their thoughtful evaluation and for recognizing the robustness, validity, and reliability of our study. We greatly appreciate the acknowledgment of our multi-level validation strategies, experimental confirmations, and consistent model performance, which further encourages our confidence in the significance and applicability of this work.

Suggested improvements: experiments, data for possible revision

- The authors could try to include more time points as three time points are maybe too few to truly learn meaningful dynamics, especially in a high-dimensional proteomic space.

Reply: We appreciate the reviewer's insightful suggestion regarding the number of time points. (1) theoretical justification demonstrating that four time points (baseline plus three post-treatment measurements) are mathematically sufficient to capture meaningful dynamics. (2) empirical validation using a benchmark dataset with ten time points. We address this concern in detail from two complementary aspects.

1. Theoretical justification for sufficiency of four time points. We provide a theoretical justification in **Supplementary Information**, grounded in dynamical systems and neural ODE analysis, showing that three post-treatment time points (plus baseline) are sufficient to identify both first-order dynamics and leading nonlinear corrections of drug-proteome trajectories. Let the proteomic state evolve as $\dot{x}(t) = f_\theta(x(t), u)$, where $x \in \mathbb{R}^p$ and u is a time-invariant embedding of the perturbing drug. For each trajectory we observe $x(0)$ (pre-treatment) and three post-treatment states $x(t_{6h}), x(t_{24h}), x(t_{48h})$. Although sparse in time, this design is sufficient to learn a stable and predictive vector field under mild regularity, for three reasons:

(1) Local flow identification via time-Taylor constraints.

By the variation-of-constants formula, $x(t) = x(0) + \int_0^t f_\theta(x(s), u) ds$. Around $t = 0$, the flow admits the series:

$$x(t) = x(0) + tv + \frac{t^2}{2}a + O(t^3), \quad v = f_\theta(x(0), u), \quad a = J_x f_\theta(x(0), u)v.$$

Three distinct $t_k > 0$ provide linearly independent constraints that identify v (the instantaneous response) and a (local curvature driven by the Jacobian) via a well-posed least-squares fit; the truncation error is $O(t^3)$ and is bounded when f_θ is locally Lipschitz with bounded Jacobian. Thus, even with three post-baseline points, the model learns both first-order dynamics and leading nonlinear corrections that govern short-to-intermediate-horizon behavior.

(2) Global constraints across initial conditions and controls.

To further enhance the generalization capability and identifiability of model, in ProteinTalks, each trajectory is sparsely sampled, the dynamical parameters θ are shared across many cell-line baselines, which are determined by the prescribed initial condition $x_i(0)$ and the embedded drug controls u_j . Training couples all trajectories through the single vector field f_θ and enforces multi-time collocation:

$$x_{ij}(t_k) \approx x_{ij}(0) + \int_0^{t_k} f_\theta(x_{ij}(s), u_j) ds, \quad k = 1, 2, 3.$$

Across all cell lines i and drugs j , each trajectory adds several time-point constraints. Together, these constraints form a large, over-determined set of integral equations. Fitting them jointly sharply narrows the admissible parameters and effectively identifies θ . This holds if the vector field f_θ is not over-parameterized. It also requires diverse coverage of initial states and perturbation $(x(0), u)$. Our

data satisfy these conditions: many cell lines provide distinct baselines, and many drugs provide distinct perturbations.

(3) Stability and additional supervision.

Under Picard-Lindelöf conditions, the flow is unique and perturbation-stable (Grönwall inequality). Training at three times jointly regularizes the vector field over the entire interval $[0, t_3]$; moreover, our model is simultaneously supervised to predict the binary efficacy label, which imposes an additional functional constraint on the trajectory, further reducing the admissible hypothesis space for f_θ .

Beyond these ODE-based properties, the neural-ODE parameterization further eases solution: the network encodes a smooth, shared vector field that renders the integral constraints well-conditioned and enables stable, efficient adjoint-based optimization even with sparse time sampling.

In summary, while denser sampling can further improve identifiability, four time points (baseline + three post-treatment) provide mathematically informative constraints—capturing instantaneous response and leading nonlinear dynamics—and, when aggregated across many initial conditions and drug controls, are sufficient to learn a stable neural-ODE that generalizes to unseen cell-line-drug pairs.

1. **Empirical validation using a ten-time-point benchmark dataset.** To further clarify the sufficiency of time points, we conducted a benchmark experiment in this revision using the multiple-timepoint ProteinTalks dataset (mpPTDS), which contains proteomic measurements from three cell lines treated with 60 single drugs and 20 drug combinations across **ten time points**. This comprehensive design allowed us to directly compare the representational capacity and learning performance of models trained on the full ten-time-point dataset versus a simplified three-time-point version. We evaluated the impact of data from different time points on the results from two aspects: temporal landscape visualization through dimensionality reduction features of high-dimensional data and the stepwise timepoint expansion benchmark.
- (1) Temporal landscape visualization: *“UMAP revealed time as the dominant source of variation, with three temporal clusters: early (0, 2, 4, 6 hours), intermediate (8, 10, 12, 24 hours), and late (36, 48, 60 hours) (Figure S2C). We then selected 6, 24, and 48 hours as representative time points to capture early, intermediate, and late proteomic responses.”* In AI modeling, *“The model captured gradual, ordered transition across ten post-treatment states (Figure S4A), effectively encodes the time-dependent evolution of protein expression following perturbation.”*

Figure S2E

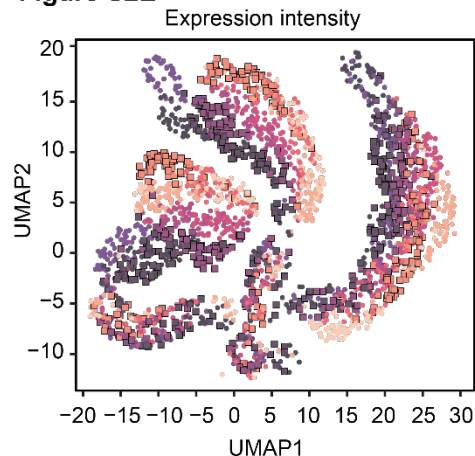


Figure S4A

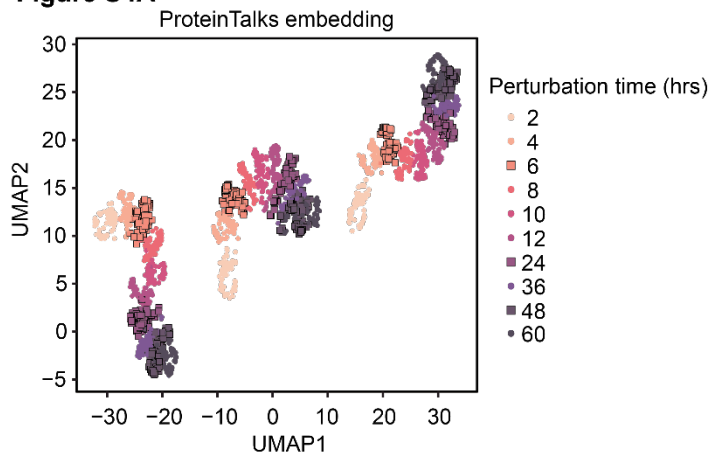


Figure S2E UMAP shows whole proteomics grouping by perturbation durations.

Figure S4A UMAP shows whole proteomics embedding by ProteinTalks grouping by different perturbation durations.

(2) Stepwise timepoint expansion benchmark: *Increasing time points modestly improved accuracy ($p=0.031$), but decreased AUPRC ($p=0.02$) (Figure S4B-D), reflecting a trade-off between discrimination (accuracy) and precision–recall (AUPRC). We selected four time points (0, 6, 24, 48 hours) to balance these metrics (with theoretical justification provided in supplementary information), consistent with biological dimension reduction results (Figure S2E), which captured early-, intermediate-, and late-phase proteome responses to drug perturbations while maintained sufficient dataset diversity and broad applications.”*

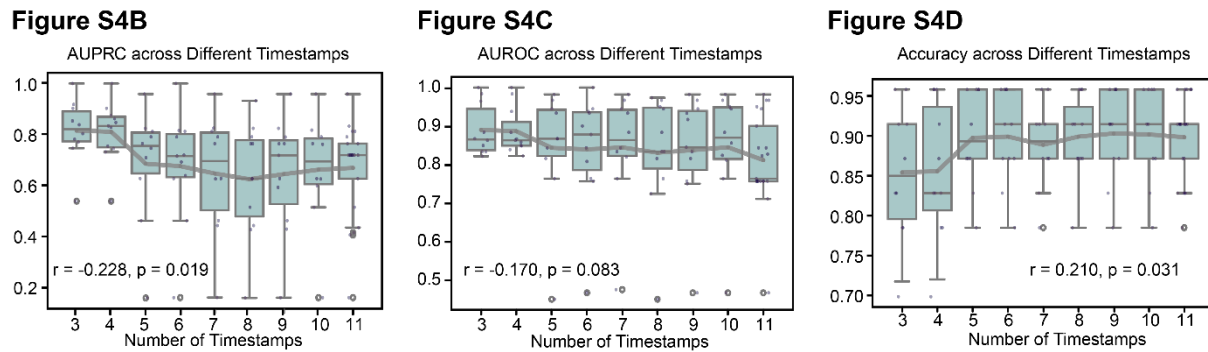


Figure S4 (B)-(D) Boxplots show the performance of ProteinTalks evaluated by AUPRC (B) and AUROC (C) decreasing with the number of time points, accuracy (D) increasing with the number of time points.

Altogether, our analysis revealed that the baseline plus three strategically selected post-treatment time points (0h, 6 h, 24 h, and 48 h) effectively captured the major temporal patterns observed across the full-time course for downstream modeling, while maintaining sufficient dataset diversity and generalizability.

- The encoder input only takes categorical inputs like drug, cell line and time points. Is it better to also add some baseline omics to guide ODE dynamics better? (Both cell line and drug ID are one-hot embedding and time point is also one-hot or scalar, so including some omics data may be better for describing the initial state)

Reply: We sincerely thank the reviewer for this thoughtful and constructive suggestion. We agree that incorporating additional baseline omics data such as transcriptomics or proteomics could, in principle, provide a richer description of the initial cellular state and potentially guide the ODE dynamics more effectively. However, we would like to clarify that our model already incorporates baseline proteomic profiles as a key component of the initial state representation. We address this point from three important aspects:

Firstly, regarding drug and cell line input, we clarify that these identities are categorical variables rather than raw one-hot vectors. Instead, we employ learnable embeddings to encode both the drug structures and the baseline proteomic profile of each cell line. This embedding strategy enables the model to capture nuanced biological relationships beyond what one-hot encoding can provide, offering a compact but information-rich representation of drug and cell-specific properties. We have clarified this point in **Methods** “ProteinTalks architecture and parameters”, as copied below.

“All inputs are real-valued feature vectors, not one-hot encodings. P_0 is the baseline proteomic profile measured in the treated cell line before perturbation. D is the perturbation descriptor; it encodes the drug-target profile and can reflect zero, one, or multiple targets. D_1 and D_2 are the two drug feature vectors capturing physicochemical properties of the respective agents.”

Second, incorporating additional baseline omics (e.g., transcriptomics) would face practical and performance limitations. This is rarely feasible in downstream applications, particularly in clinical or

large-scale screening contexts where multi-omics data are often incomplete or unavailable. Moreover, RNA is more prone to degradation and less stable than proteins, limiting its practical application. Importantly, recent benchmarking studies suggest that the cutting-edge transcriptomics-based foundation models, despite being trained on massive dataset with huge parameters, still perform poorly in perturbation response prediction task ¹⁻³.

Third, we empirically tested transcriptome integration without performance gain. We have explored the use of transcriptome data as part of a pretraining strategy. However, our benchmarking against representative transcriptome-based foundation model (Geneformer, geneCompass, and UCE) showed no performance gain when pretraining with transcriptomics data for downstream drug response prediction. These results suggest that simply adding omics (especially transcriptomics) input does not necessarily improve generalization in this context.

We hope this clarifies our design choices and the trade-offs we considered. We appreciate the reviewer's suggestion and agree that more flexible or modular omics integration strategies may be a promising direction for future work. We have revised the **Discussion** to reflect this point, as copied below.

“Despite these advances, this study has several limitations.” ... “Post-translational modifications and multi-omics integration (transcriptomics, metabolomics, epigenomics) are not included, and our dataset focused on cancer cell lines, particularly TNBC. Future expansion to diverse biological contexts and harmonization of heterogeneous, temporally resolved multi-omics data will further improve mechanistic interpretability and virtual cell completeness.”

References

- 1 Boiarsky, R. *et al.* Deeper evaluation of a single-cell foundation model. *Nature Machine Intelligence* **6**, 1443-1446 (2024).
- 2 Kedzierska, K. Z., Crawford, L., Amini, A. P. & Lu, A. X. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biol* **26**, 101 (2025).
- 3 Ahlmann-Eltze, C., Huber, W. & Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nat Methods* **22**, 1657-1661 (2025).

- The latent trajectory is not interpreted in the paper.

Reply: We appreciate the reviewer's valuable comment regarding the interpretation of the latent trajectory. To address this point, we have added comprehensive analysis of how the learned latent trajectories encode biologically meaningful temporal dynamics. Specifically, we performed a time-resolved dynamic SHAP analysis, which revealed time-varying regulatory effects of baseline proteins and distinct temporal patterns across drug classes. These results highlight the importance of capturing both network topology and response latency in our modeling framework. We have now added the corresponding methodology and results in the manuscript, as copied below:

“ProteinTalks delineates dynamic protein network remodeling

To elucidate how drug perturbations reshape protein regulatory networks over time, we computed dynamic SHapley Additive exPlanations (SHAP) based on ProteinTalks' predictions of perturbed protein expression at different time points (6, 24, 48 hours).

Drug-specific, time-dependent SHAP divergences between responsive and non-responsive samples revealed distinct temporal dynamics. Alkylating agents, HDAC inhibitors, and topoisomerase inhibitors showed minimal divergence at 6 hours but marked separation by 24 hours (Figures S8A-C), reflecting delayed protein-level manifestation of DNA damage and epigenetic remodeling through regulatory cascades. Conversely, CDK inhibitors, hormonal agents, and RXR inhibitors exhibited pronounced divergence at 6 hours, indicating immediate protein-level signaling responses (Figures S8D-F). Together, these findings indicate that latent trajectories could recapitulate biologically

relevant temporal dynamics.”

References: appropriate credit to previous work?

- Yes, the paper appropriately cites foundational and recent work. Citations are current and cover biological and computational domains.
- The authors used the software DIA-NN for DIA-MS data analytics, so we would recommend citing its original paper in addition to references 40 and 41 which described how to use the software. The original paper by the Francis-Crick and Cambridge group is: DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput (<https://www.nature.com/articles/s41592-019-0638-x>)

Reply: Thanks for your suggestions. We have added this paper in reference 47.

Clarity and context: lucidity of abstract/summary, appropriateness of abstract, introduction and conclusions

- The results and methods sections of the paper left the readers with a general impression that the ProteinTalks model contributes to drug discovery and predicting cancer patient responses to drugs. The quality and significance of the ML model is clear; however, the Abstract, Introduction, and Discussion, seem to emphasize how the model is a good foundation for constructing a digital twin to biological cancer cells in humans. Not sure if it is immediately obvious for the readers to arrive at this conclusion without providing more context on the digital cell concept. To be precise, a good digital cell should ideally integrate all types of variables relevant to pathogenesis, not just drug perturbations. What are the authors trying to convey by including this digital cell notion?
- An example of how people have provided context for the digital cell concept when trying to emphasize this in their paper: <https://genomemedicine.biomedcentral.com/articles/10.1186/s13073-022-01048-4>

Reply: We thank the reviewer for this insightful constructive suggestion and for pointing out the need for clearer context around the virtual/digital cell concept. In our view, the **digital twin**, literally meaning the digital "twin" of a cell, represents the ultimate and most advanced goal in computational biology—a personalized and continuously updated model of a biological system (e.g., a patient's cell or tumor) that evolves in real time with new experimental or clinical data, enabling individualized prediction and intervention. The **digital cell**, as a foundational layer toward this goal, refers to a comprehensive, data-centric collection of molecular and cellular information that integrates multi-omics measurements (e.g., genomics, transcriptomics, proteomics, and metabolomics) to describe the molecular composition and state of a cell. The more practical and currently achievable form, the **virtual cell**, focuses on simulating and predicting specific cellular functions and responses to perturbations. Unlike a digital cell or digital twin, a virtual cell does not require the integration of every molecular variable; rather, it aims to capture key cellular functions sufficient to reproduce and forecast biological behaviors under defined conditions.

In another word, the digital twin represents a long-term goal in computational biology, *i.e.* a continuously updated, personalized model of a living system. The digital cell refers to an integrative framework that ultimately incorporates multiple detailed molecular layers and cellular variables to reflect such biological complexity. Within this spectrum, the virtual cell represents a more attainable step that prioritizes modeling the dynamic behaviors and functions of cells, particularly the proteomic changes that directly govern function to reconstruct and forecast cellular responses under defined perturbations.

Although building a true digital twin is a long-term goal, current virtual cell models have already leveraged single-cell transcriptomic data to develop foundation models that predict cellular behaviors. As Stephen Quake has noted, “In the future, biologists will spend less time wielding pipettes and more time relying on computational tools.”¹ Many ongoing efforts, such as those led by CZI and Google DeepMind, have focused on constructing AIVCs based on single-cell RNA sequencing data, which provide snapshots of gene expression under specific conditions². Similarly, Arc Institute’s first virtual cell model, State, also utilizes single-cell transcriptomics to predict cellular responses to drug or genetic perturbation³. The recent US\$175,000 virtual-cell competition launched by the Arc Institute, which challenges researchers to predict how human stem cells respond to genetic alterations⁴.

ProteinTalks belongs to the “virtual cell” category. It is a proteome-based AI virtual cell designed to capture the temporal trajectories of protein networks in response to drugs, establishing the dynamic foundation where future digital cells can be constructed. Our intent is not to claim a fully realized digital twin, but rather to demonstrate that modeling proteomic dynamics is a necessary and foundational step toward achieving it, especially in the drug screening domain using large-scale dynamical proteomics datasets.

To make our intention clearer, we have revised the title to explicitly “*ProteinTalks: an operational perturbation proteomics-based virtual cell model for drug discovery*”, and we have revised the **Title**, **Introduction**, and **Discussion** to clarify the novelty of our study, explicitly presenting ProteinTalks as a virtual cell model built on large-scale perturbation proteomics in cancer cells.

In the **Introduction**,

“In silico drug discovery represents a transformative frontier in biomedical research. It is built on predictive cellular models that simulate molecular and cellular responses to therapeutic perturbations. In this context, the virtual cell concept has emerged as a promising paradigm in drug discovery, where the central challenge lies in accurately modeling cellular responses and dynamics. Given the inherent nonlinearity of cellular processes, virtual cells require training on real-world dynamic data to capture the true complexity of cellular behavior¹⁻³. Recently, foundation models trained on large-scale omics datasets are increasingly viewed as a critical path toward virtual cell. However, existing foundation models, such as Geneformer⁴, scFoundation⁵, and STATE⁶, have predominantly relied on single-cell transcriptomic data. Even those trained on perturbation transcriptomic data generated by Perturb-seq remain inherently limited to pre- and post-perturbation snapshots, lacking temporal resolution to capture true dynamics. Transcriptomic readouts capture only an intermediate layer of the central dogma, which may underlie the poor performance of transcriptomics-based foundation models on new datasets⁷⁻⁹. Proteins, as the primary executors of cellular functions, are more directly tied to phenotype —yet proteome-based virtual cell models remain underdeveloped due to the scarcity of large-scale, time-resolved proteomic datasets.

To address this gap, we established a systematic temporal perturbation proteomics resource focused on drug response. Recent advances in high-throughput proteomics enabled by data-independent acquisition mass spectrometry (DIA-MS)^{10,11}, have made large-scale perturbation proteomics data generation more feasible¹². Utilizing a 96-well-based high-throughput platform, we perturbed breast cancer cell lines with clinically relevant drugs and their combinations, yielding more than 38 million perturbed protein measurements alongside cell viability data.

Leveraging this large-scale dataset, we developed ProteinTalks, a virtual cell model that explicitly embeds time-evolving protein network trajectories using a neural ordinary differential equation

(neural ODE) ^{13,14} framework (Figure 1). Despite using a thousandfold fewer parameters and data points than existing static transcriptomic foundation models, ProteinTalks achieves superior predictive performance across diverse tasks, including cross-cell-line and cross-drug evaluation, independent drug tests, cross-disease transfer, drug synergy prediction, and fine-tuning for patient-derived xenograft (PDX) response prediction. Ultimately, it stratifies clinical outcomes in 504 breast cancer patients and prioritizes candidates from a 3000-repurposed-drug library for triple-negative breast cancer (TNBC) patient-derived organoids (PDOs). Interpretability analyses highlight previous unknown candidate biomarkers and mechanisms that align with experimental and clinical validation. Together, this work establishes ProteinTalks as a proteome-based virtual cell with cellular functional capability for drug screening.”

In the **Discussion**,

“Understanding cellular mechanisms and predicting drug responses remain central challenges in biology and drug discovery. We shed new light on this challenge through ProteinTalks, a proteome-based virtual cell model built on large-scale, time-resolved perturbation proteomics. This temporal foundation enabled dynamic modeling of cellular state transitions, allowing translation from controlled cell line experiments to patient-derived tissues.

Our dataset, comprising over 38 million protein measurements from ~ 16,000 perturbed proteomic samples that we produced in-house, represents the largest AI-ready proteomics resource to date, over an order of magnitude larger than previous perturbation proteomics datasets ³⁴⁻³⁷. Critically, unlike static pre/post-perturbation snapshots common in existing datasets, our time-series design captures the dynamic evolution of cellular responses. This temporal resolution, combined with proteomic measurement, fundamentally distinguishes ProteinTalks from transcriptome-based foundation models ⁴⁻⁶. Unlike transcriptomics, which reflects transcriptional potential rather than functional dynamics, proteomics directly captures the functional executors of cellular processes. Importantly, transcript and protein abundances frequently diverge under perturbations, making proteome-level measurements critical for understanding actual cellular responses.

The availability of extensive time-course proteomics data was essential for implementing a neural ODE architecture capable of learning mechanistic cellular trajectories from temporal dynamics. Such dynamic modeling is inaccessible to transcriptome-based foundation models: public transcriptomic datasets lack time-series measurements, constraining these models to large-language-model-inspired Transformer architectures that learn static correlations rather than dynamic processes. Crucially, our large-scale temporal measurements capture the complex nonlinear dynamics inherent to real biological systems, enabling ProteinTalks to achieve superior generalization with ~1000-fold fewer parameters than transcriptomic foundation models. Recent benchmarks confirm that such parameter-heavy foundation models remain insufficient for perturbation response prediction ⁷⁻⁹. By contrast, ProteinTalks—leveraging biologically grounded dynamics rather than parameter scale—exhibited robust and broad applicability across predictive tasks spanning 18 breast cancer cell lines, PDX models, and a 504-patient cohort, extending from 63 training drugs to 3000 compounds and validating clinical translatability from cell lines to patient biopsies.

This mechanistic grounding enabled discovery of actionable biological insights: experimentally validated drug synergies (e.g., AKRIC3 mediating toremifene-afatinib response), and 15 chemotherapy-associated proteins with no prior cancer links (including ZFYVE19, SYNE2, and zinc finger proteins). These findings emerged from synergy between large-scale temporal proteomics and dynamics-aware modeling, showing ProteinTalks' capacity to reveal previously inaccessible biological patterns.”

References

- 1 Callaway, E. Can AI build a virtual cell? Scientists race to model life's smallest unit. *Nature* **643**, 13-14 (2025).
- 2 Leslie, M. The silicon cell. *Science* **390**, 436-439 (2025).
- 3 Abhinav K. Adduri, D. G., Beatrice Bevilacqua, Alishba Imran, Rohan Shah, Mohsen Naghipourfar, Noam Teyssier, Rajesh Ilango, Sanjay Nagaraj, Mingze Dong, Chiara Ricci-Tam, Christopher Carpenter, Vishvak Subramanyam, Aidan Winters, Sravya Tirukkovular, Jeremy Sullivan, Brian S. Plosky, Basak Eraslan, Nicholas D. Youngblut, Jure Leskovec, Luke A. Gilbert, Silvana Konermann, Patrick D. Hsu, Alexander Dobin, Dave P. Burke, Hani Goodarzi, Yusuf H. Roohani. Predicting cellular responses to perturbation across diverse contexts with State. *bioRxiv* (2025).
- 4 Roohani, Y. H. *et al.* Virtual Cell Challenge: Toward a Turing test for the virtual cell. *Cell* **188**, 3370-3374 (2025).

Referee #3 (Remarks on code availability):

The codeocean repository has a 0B data directory. Was it supposed to be left empty or the authors meant to share something?

Reply: We thank the reviewer for the careful reading and valuable suggestion. We have uploaded a demo dataset to the CodeOcean repository to facilitate verification of the model's correctness and functionality. In addition, the full implementation code has been made available on GitHub (<https://github.com/guomics-lab/PTV-1>) to ensure the reproducibility of our analyses.

Point-by-point reply

Referees' comments:

Although the large proteomics dataset itself is a major contribution, neural ODE-based approaches have been previously applied to model cellular dynamics and perturbation responses (e.g., PRESCIENT, Yeo et al., 2021; CellOT, Bunne et al., Nature Methods 2023), and the application to proteomics data, while valuable, represents an extension of this established framework rather than a methodological breakthrough. Hence, we are not sure if the modeling contribution could meet Nature's novelty bar.

Reply: We thank the reviewer for recognizing the significance of our proteomic dataset, a perspective also shared by the other reviewers. We respectfully clarify that the core innovation of our paper lies neither in the proteomic dataset nor in the model architecture in isolation, but *rather in their synergistic integration, which yields qualitative breakthroughs unattainable by either component alone*. In contrast to major *in silico* cell models built on single-cell transcriptomic data, our study offers an alternative and promising approach using large-scale perturbation proteomics.

That said, we would also like to clarify that our modeling framework is indeed not a simple replication nor adaptation of existing models. We provide further clarification below on the key differences between our ProteinTalks model and existing modeling approaches.

The ProteinTalks model learns a *shared dynamic embedding space* from large-scale temporal perturbation proteomics through *pretraining*, together with a perturbation-conditioned continuous-time flow defined on that space. In our framework, the latent representation is not fixed *a priori* and then evolved over time; instead, it is *jointly learned under temporal and perturbational supervision*, such that response dynamics actively shape the geometry of the embedding itself.

This is different from prior approaches such as PRESCIENT and CellOT. PRESCIENT models trajectory progression within predefined feature spaces in the same condition, typically in developmental settings, while CellOT learns perturbation-specific transport maps between endpoint distributions, and might not generalize readily to unseen perturbation conditions. In both cases, the representation space is assumed given, and modeling focuses on transitions or mappings defined on top of it. Specifically, ProteinTalks formulates temporal perturbation proteomics as a *representation learning problem*, where both the embedding and its evolution are learned jointly.

To provide more explanations, below we provide expanded discussion in three aspects: (1) how ProteinTalks transcends the original scope of neural ODEs through pretraining and representation learning (details provided in *Rebuttal-appendix 1*); (2) how we have revised the manuscript to clarify this contribution; and (3) how our work differs fundamentally from PRESCIENT and CellOT in task definition, supervision signal, and evaluation criteria (details provided in *Rebuttal-appendix 2*).

Key revisions to sharpen the novelty statement:

To address this concern, we have substantially revised the Abstract, Introduction, and Discussion.

In the **Abstract**, we now explicitly contrast our approach with static transcriptome-based methods and emphasize the pretraining strategy:

"Here, we generate over 38 million temporal protein abundance measurements from systematically perturbed breast cancer cell lines, and develop ProteinTalks, a dynamics-aware, proteomics-based operational virtual cell model. Central to ProteinTalks is the synergy of this large-scale dynamic proteomic resource and the model architecture, enabling a novel pretraining framework that learns transferable dynamical latent representations from

temporal proteome trajectories. By modeling how proteins respond conditionally to different perturbations, this approach enables the model to function as an operational tool for diverse drug discovery tasks..... By integrating large-scale perturbation proteomics with a generalizable pretraining framework for learning transferable dynamic representations, ProteinTalks demonstrates how data-driven virtual cells can serve as functional platforms for in silico drug discovery. "

Similar revisions have been made to the Introduction and Discussion to consistently emphasize that the novelty lies in operationalizing perturbation-conditioned dynamic pretraining at the proteome execution layer, rather than in the use of continuous-time modeling.

In the **Introduction**, we added "*However, the capacity of these models to represent and predict dynamic cellular behavior remains constrained by the data modalities and learning architectures. Representative foundation models, such as Geneformer⁴, scFoundation⁵, and UCE⁵, learn general-purpose embeddings from transcriptomic snapshots through self-supervised pretraining. Models trained on perturbation transcriptomic data generated by Perturb-seq, such as STATE⁷, STACK⁸, x-Cell⁹ and CellOT¹⁰, further extend this paradigm to model cellular responses to perturbations. In parallel, existing trajectory modeling approaches (e.g., PRESCIENT¹¹, trajectorynet¹², DeepRUOT¹³) are developed to better account for cellular dynamics. Together, these methods have substantially advanced the representation of cellular states and transitions¹⁴. However, they are optimized either for static cellular embeddings, perturbation endpoints, or trajectory interpolation within predefined latent spaces such as principal component analysis (PCA) embeddings, rather than for learning transferable representations directly from rich temporal perturbation dynamics across diverse biological contexts. ... Leveraging this large-scale dataset, we developed ProteinTalks, a virtual cell model that jointly learns a latent representation together with perturbation-conditioned dynamics through a neural ordinary differential equation (neural ODE) framework, where the dynamics themselves shape the learned representation (Figure 1).*"

In the **Discussion**, we added "*A key feature of ProteinTalks is the synergy between dataset collection and architecture design. Our perturbation proteomic dataset, comprising systematic time-course measurements across diverse drugs and cell lines, provides the temporally resolved supervision that is largely absent from existing proteomic resources. This rich temporal structure, in turn, enables a neural ODE-based pretraining framework which jointly learns both the latent representations and the perturbation-conditioned dynamics. In contrast to prior trajectory modeling methods that perform interpolation or optimal transport within predefined representations such as PCA embedding, ProteinTalks allows perturbation-conditioned trajectories to actively shape what features the model discovers. This integration of a uniquely structured dataset with a dynamic-aware architecture enables ProteinTalks in a parameter-efficient way to model cellular state transitions dynamically and to generalize from controlled cell line experiments to patient-derived tissues. ...*

By contrast, even very large-scale single-cell transcriptomic collections often remain limited in perturbational and contextual diversity, restricting cross-context generalization because the causal structure underlying cellular responses is only weakly identifiable from observational variation alone⁴⁵. The present dataset therefore provides a foundation for learning latent representations that transfer across drugs, cellular contexts, and disease states. ...

Although neural ODE has been previously applied in single-cell and omics analyses (e.g., trajectory inference and perturbation mapping), they remain largely task-specific rather than serving as a general pretraining strategy for transferable dynamic embeddings¹⁴. "

Overall, we respectfully wish to clarify that the primary novelty of ProteinTalks does not lie in algorithmic invention *per se*. Instead, its contribution stems from the integrated advances in data infrastructure, a tailored pretraining framework, and decision-relevant validation, which

together help realize the virtual cell concept in a practical manner for drug discovery. In addition, as explained below, the ProteinTalks model reveals biological insights inaccessible through existing approaches.

To strengthen the manuscript, we would encourage the authors to highlight the biological insights enabled by the model - for example, elaborate on the findings summarized in line 389-391 in the results section with experimental validation results. The key idea here is to demonstrate that the model reveals biological insights inaccessible through existing approaches.

Reply: Thanks for the comments. In this revision, we have strengthened the Results section by incorporating two complementary lines of evidence to address this point: support from public datasets, together with two new experimental validations based on clinical samples and *in vitro* functional assays. A summary is provided below:

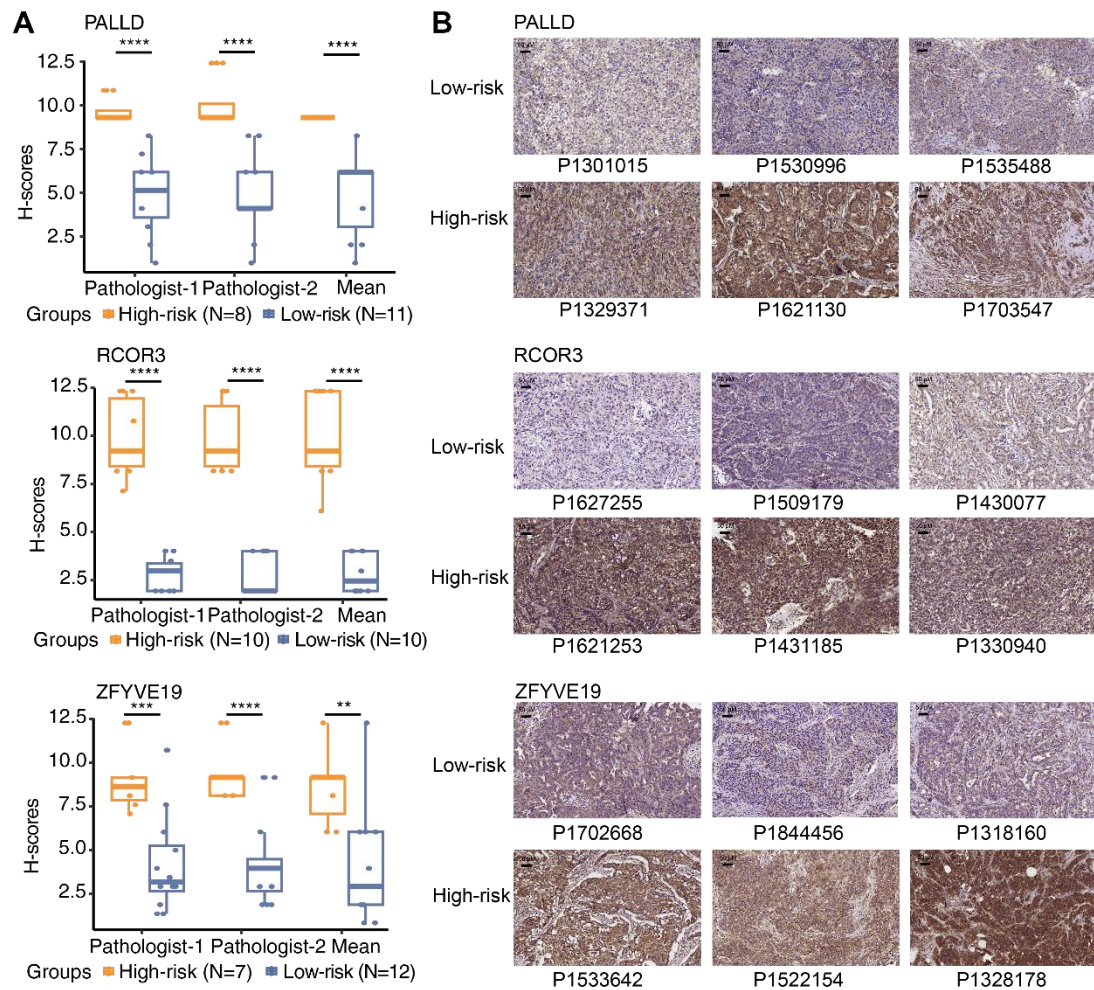
1. Public dataset support

We added this in the revised manuscript **Supplementary Information (SI)** section “*Biological insights and validation of top 100 proteins prioritized by ProteinTalks*”, as copied below.

*"The prognostic relevance of these three proteins was further validated across five independent public datasets and our in-house TNBC-HMU cohort, showing a consistent association between their expression levels and differential prognosis in breast cancer patients (SI Table 1). "(details provided in **Rebuttal-appendix 4**)*

2. Newly added experimental validation in clinical TNBC patients using IHC staining

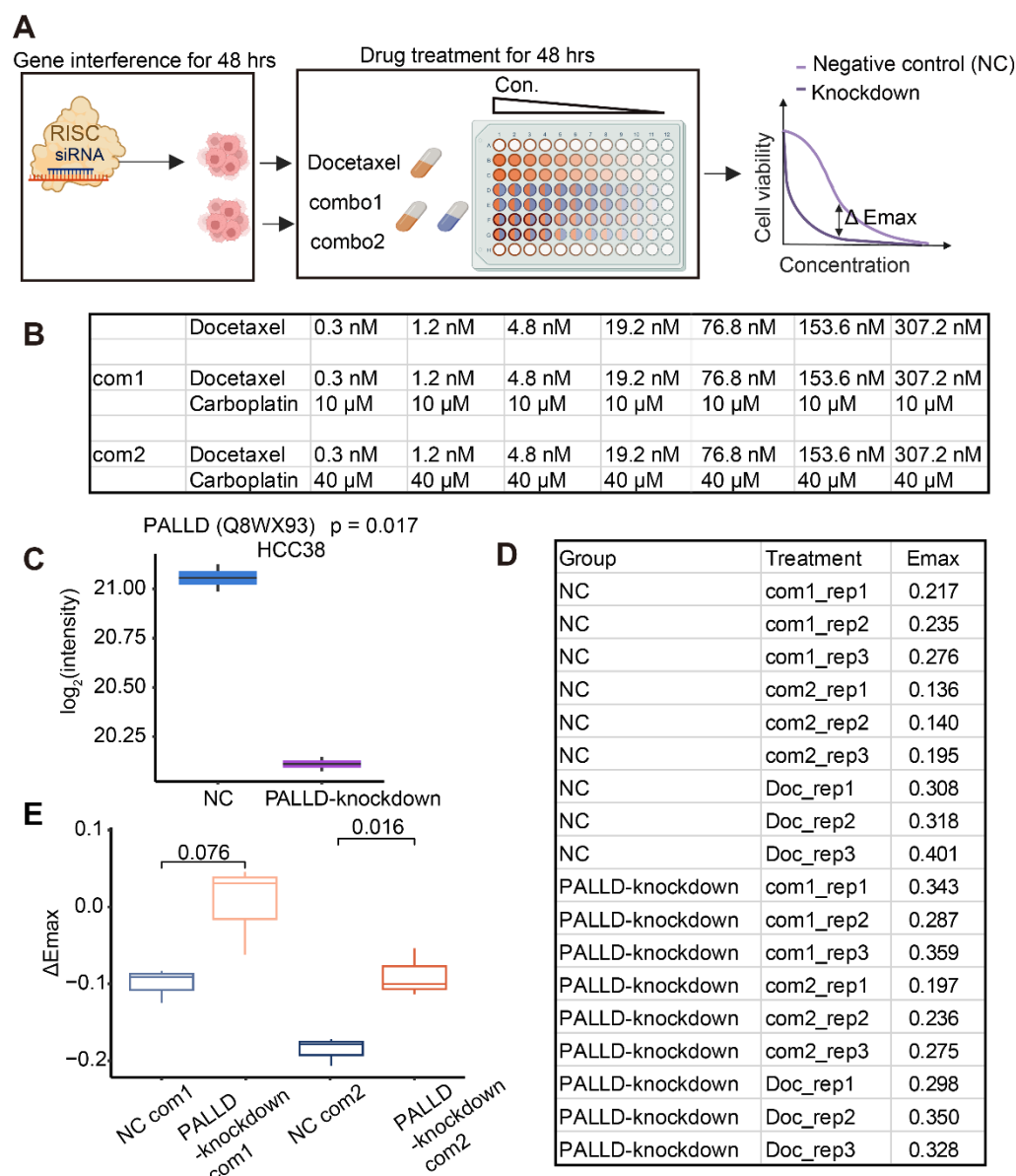
Through univariate analysis performed on the TNBC-HMU cohort and public datasets supplemented during the previous rebuttal, we found that high expression of ZFYVE19, PALLD, and RCOR3 was significantly and positively correlated with poor prognosis in specific subgroups of TNBC patients. To further corroborate these findings at the tissue level and validate the translational relevance of these proteins in a clinical setting, we further selected approximately 20 patients from both the high- and low-risk groups for each protein from the TNBC-HMU and performed immunohistochemical (IHC) staining. Scoring was conducted independently by two professional pathologists, and the results confirmed that protein expression levels were indeed significantly higher in the high-risk patient groups, providing histopathological evidence consistent with our in-silico findings (revised manuscript **SI** section “*Biological insights and validation of top 100 proteins prioritized by ProteinTalks*”, *SI Table 1*, and *SI Figure 2*). The detailed IHC staining and scoring procedures are provided in **Materials and Methods** section of the **SI**.



SI Figure 2 IHC validation of PALLD, RCOR3, and ZFYVE19 expression in the TNBC-HMU cohort. (A) Quantification of IHC staining scores for PALLD, RCOR3, and ZFYVE19 in the TNBC-HMU cohort. H-scores assigned by two pathologists (Pathologist-1 and Pathologist-2) and their mean are presented for each case. Statistical significance between groups was assessed using the two-sided t-test (*, $p < 0.01$; **, $p < 0.001$; ***, $p < 0.0001$). (B) Representative IHC images from three patients per risk group for each of the three proteins. Scale bar, 50 μ m.

3. Newly added experimental validation *in vitro*

Given that the top 100 proteins distinguished sensitive from resistant TNBC patients in response to neoadjuvant chemotherapy with docetaxel and carboplatin, we next investigated whether PALLD contributes to the synergistic effect of this drug combination *in vitro*. Using RNA interference to knock down PALLD in HCC38 cells, we observed no significant difference in cell viability under single-agent treatment conditions compared with control cells. However, PALLD knockdown significantly enhanced the synergistic effect of the docetaxel-carboplatin combination, suggesting that PALLD suppression potentiates the cooperative cytotoxic activity of this regimen. (Revised manuscript **SI** section “*Biological insights and validation of top 100 proteins prioritized by ProteinTalks*”, and **SI Figure 3**). The detailed Methods are provided in the **Methods** section in the main text, “*Gene interference in different cell lines*”, and “*Cell viability assay*”.



SI Figure 3 In vitro validation of PALLD biological function in modulating drug synergy. (A) Experimental workflow for drug synergy validation. HCC38 cells were subjected to siRNA-mediated PALLD knockdown, followed by treatment with serial dilutions of docetaxel alone or in combination with carboplatin. Each experimental condition was performed in three independent biological replicates. (B) Concentration matrix of drug combinations used in the synergy assay. (C) Box plot showing $\log_2$ -transformed protein abundance measured by mass spectrometry (MS) to assess RNA interference efficacy. PALLD knockdown efficiency was confirmed in HCC38 cells relative to siRNA control. (D) Dose-response curves for docetaxel combined with a fixed concentration of carboplatin (10 μ M or 40 μ M) in the PALLD-knockdown and the control HCC38 cells. The table summarizes the E_{max} values derived from the fitted dose-response curves for each treatment condition. (E) Efficacy shift (ΔE_{max}) quantifying the synergistic effect of the drug combination. ΔE_{max} is defined as the difference in maximum response between the combination treatment and docetaxel monotherapy; a larger ΔE_{max} indicates a stronger synergistic effect. Statistical significance was assessed by two-sided Student's t-test.

Referee #1 (Remarks to the Author):

The authors have successfully addressed my suggestions. I have no further comments.

Reply: Thank you sincerely for your time and effort in reviewing our manuscript, and for your positive feedback.

Referee #2 (Remarks to the Author):

The authors have addressed my comments. Experiments on organoids and biopsies are a nice addition.

Reply: We are truly grateful for your thorough evaluation of our manuscript and your encouraging comments.

I'm still hesitant to whether this paper represents a significant advance over current models that it warrants publication in Nature. The data is super nice, and it's obvious that the data produces a significant advantage in that proteines creates a closer proxy to drug response than transcriptome embeddings or other embeddings that the authors compare to. The innovation in terms of the AI model remains low, and calling it a "virtual cell model" is definately a stretch.

Reply: We thank the reviewer for this thoughtful assessment and for recognizing the value of our data. We sincerely apologize that those aspects of our original presentation may have led to a misunderstanding of the novelty and scope of this work. We address the reviewer's concerns on two fronts and have revised the *Abstract*, *Introduction*, and *Discussion* accordingly.

1. The novelty lies in the inseparable integration of data and model, not simply in either alone

We appreciate the reviewer's recognition that "the data produces a significant advantage". However, we respectfully clarify that our core innovation is not the data or the model in isolation, but rather their synergistic integration, which yields qualitative breakthroughs unattainable by either component alone. In contrast to most *in silico* cell models built on single-cell transcriptomic data, our study offers an alternative and promising approach using large-scale perturbation proteomics.

a) A new pretraining way for generalizable dynamic embeddings. On the algorithmic side, continuous-time modeling through ODEs was introduced by Weinan E (2017), operationalized in Neural ODEs (Chen et al., *NeurIPS*, 2018), and extended to latent continuous-time models for irregular time series (Rubanova et al., *NeurIPS*, 2019). Although widely applied in single-cell and omics tasks such as trajectory inference and perturbation mapping, these approaches remain task-specific rather than constituting a general pretraining paradigm for transferable dynamic embeddings. ProteinTalks reconceptualizes neural ODEs as a pretraining framework in which perturbation enters as an explicit first-class condition ($dz/dt = f_{\theta}(z, u)$), learning a shared perturbation manifold across drugs, doses, cell lines, and time points. Therefore, designing a shared dynamic pretraining paradigm that enables cross-drug transfer and synergy prediction from a single unified model is the our main rationale from the algorithmic perspective.

b) Temporal proteomics as the enabling substrate for dynamic modeling. Since public transcriptomic datasets often lack systematic time-series measurements, the current foundation models built on large-language-model-inspired Transformer architectures tend to learn static correlations rather than dynamic processes. Our large-scale temporal proteomic dataset (38 million measurements, 6 time points per condition, $>10\times$ the scale of prior perturbation proteomic datasets) captures the complex nonlinear dynamics inherent to real biological systems. The complex and diverse biological dynamics captured across contexts, when pretrained through neural ODEs, are encoded into a transferable dynamic causal mechanism that generalizes across settings, allowing ProteinTalks to achieve superior

generalization in a parameter-efficient way. In addition, recent benchmarks reveal that existing parameter-heavy foundation models remain insufficient for perturbation response prediction (Ahlmann et al., *Nat Methods*, 2025; Bioarsky et al., *Nature Machine Intelligence*, 2024, Kedzierska et al., *Genome Biol*, 2025). In this work, ProteinTalks demonstrates potential in tackling such problem in its robust applicability across predictive tasks spanning 18 breast cancer cell lines, PDX models, and a 504-patient cohort, extending from 63 training drugs to 3000 compounds and validating clinical translatability from cell lines to patient biopsies.

2. ProteinTalks as a proteome-grounded prototype toward AI virtual cells

We apologize for any ambiguity regarding our use of the term "virtual cell model." And we respectfully acknowledge that the term "virtual cell" may invite scrutiny, and we welcome the opportunity to clarify our usage. As with many emerging concepts in the history of science—from "systems biology" to "foundation models"—the notion of a "virtual cell" is still actively evolving, and different researchers have proposed different definitions and scopes (details provided in *Rebuttal-appendix 3*). We do not claim that ProteinTalks constitutes a fully realized digital twin of any cell in the most comprehensive sense; rather, we adopt this term "virtual cell model" to situate our work within a broader and rapidly growing conceptual framework aimed at building computational models that simulate cellular behavior. In essence, the "virtual cell model" we present here is a computational model that simulate certain, not all, cell behaviors.

There are several related terms used in the literature, including digital twins, digital cells, and virtual cells. We have discussed about this in response to a previous reviewer. In our view, the digital twin, literally meaning the digital "twin" of a cell, represents the ultimate and most advanced goal in computational biology—a personalized and continuously updated model of a biological system (e.g., a patient's cell or tumor) that evolves in real time with new experimental or clinical data, enabling individualized prediction and intervention. The digital cell, as a foundational layer toward this goal, refers to a comprehensive, data-centric collection of molecular and cellular information that integrates multi-omics measurements (e.g., genomics, transcriptomics, proteomics, and metabolomics) to describe the molecular composition and state of a cell. The more practical and currently achievable form, the virtual cell model, focuses on simulating and predicting specific cellular functions and responses to perturbations. Unlike a digital cell or digital twin, a virtual cell model does not require the integration of every molecular variable; rather, it aims to capture key cellular functions sufficient to reproduce and forecast biological behaviors under defined conditions.

In another word, the digital twin represents a long-term goal in computational biology, i.e. a continuously updated, personalized model of a living system. The digital cell refers to an integrative framework that ultimately incorporates multiple detailed molecular layers and cellular variables to reflect such biological complexity. Within this spectrum, the virtual cell model represents a more attainable step that prioritizes modeling the dynamic behaviors and functions of cells, particularly the proteomic changes that directly govern function to reconstruct and forecast cellular responses under defined perturbations.

Although building a true digital twin is a long-term goal, current virtual cell models have already leveraged single-cell transcriptomic data to develop foundation models that predict cellular behaviors. As Stephen Quake has noted, "In the future, biologists will spend less time wielding pipettes and more time relying on computational tools."¹ Many ongoing efforts, such as those led by CZI and Google DeepMind, have focused on constructing AIVCs based on single-cell RNA sequencing data, which provide snapshots of gene expression under specific conditions². Similarly, Arc Institute's first virtual cell model, State, also utilizes single-cell transcriptomics to predict cellular responses to drug or genetic perturbation³. The recent US\$175,000 virtual-cell competition launched by the Arc Institute, which challenges researchers to predict how human stem cells respond to genetic alterations⁴.

Our intent is not to claim a fully realized digital twin, but to demonstrate that modeling

proteomic dynamics is a necessary and foundational step toward it, particularly in the drug screening domain. ProteinTalks operates as a virtual cell model in this focused sense: it captures perturbation-driven proteomic dynamics to reconstruct and forecast cellular responses under defined conditions, while we explicitly acknowledge it does not constitute a comprehensive virtual cell.

We have revised the **Abstract**, **Introduction**, and **Discussion** to better convey that ProteinTalks represents our perspective on building virtual cell models through the synergy of uniquely structured temporal proteomic data and dynamics-aware model design, and to position ProteinTalks as a prototype embodying this vision rather than claiming it as a complete virtual cell model.

In the **Abstract**, we now explicitly contrast our approach with static transcriptome-based methods and emphasize the pretraining strategy:

"Here, we generate over 38 million temporal protein abundance measurements from systematically perturbed breast cancer cell lines, and develop ProteinTalks, a dynamics-aware, proteomics-based operational virtual cell model. Central to ProteinTalks is the synergy of this large-scale dynamic proteomic resource and the model architecture, enabling a novel pretraining framework that learns transferable dynamical latent representations from temporal proteome trajectories. By modeling how proteins respond conditionally to different perturbations, this approach enables the model to function as an operational tool for diverse drug discovery tasks..... By integrating large-scale perturbation proteomics with a generalizable pretraining framework for learning transferable dynamic representations, ProteinTalks demonstrates how data-driven virtual cells can serve as functional platforms for in silico drug discovery."

In the **Introduction**, we added "*However, the capacity of these models to represent and predict dynamic cellular behavior remains constrained by the data modalities and learning architectures. Representative foundation models, such as Geneformer⁴, scFoundation⁵, and UCE⁵, learn general-purpose embeddings from transcriptomic snapshots through self-supervised pretraining. Models trained on perturbation transcriptomic data generated by Perturb-seq, such as STATE⁷, STACK⁸, x-Cell⁹ and CellOT¹⁰, further extend this paradigm to model cellular responses to perturbations. In parallel, existing trajectory modeling approaches (e.g., PRESCIENT¹¹, trajectorynet¹², DeepRUOT¹³) are developed to better account for cellular dynamics. Together, these methods have substantially advanced the representation of cellular states and transitions¹⁴. However, they are optimized either for static cellular embeddings, perturbation endpoints, or trajectory interpolation within predefined latent spaces such as principal component analysis (PCA) embeddings, rather than for learning transferable representations directly from rich temporal perturbation dynamics across diverse biological contexts. ... Leveraging this large-scale dataset, we developed ProteinTalks, a virtual cell model that jointly learns a latent representation together with perturbation-conditioned dynamics through a neural ordinary differential equation (neural ODE) framework, where the dynamics themselves shape the learned representation (Figure 1).*"

In the **Discussion**, we added "*Current efforts toward virtual cell models predominantly rely on large-scale transcriptomic datasets. Here, we propose a complementary route grounded in large-scale, time-resolved perturbation proteomics and present ProteinTalks as a prototype virtual cell model. A key feature of ProteinTalks is the synergy between dataset collection and architecture design. Our perturbation proteomic dataset, comprising systematic time-course measurements across diverse drugs and cell lines, provides the temporally resolved supervision that is largely absent from existing proteomic resources. This rich temporal structure, in turn, enables a neural ODE-based pretraining framework which jointly learns both the latent representations and the perturbation-conditioned dynamics. In contrast to prior trajectory modeling methods that perform interpolation or optimal transport within predefined representations such as PCA embedding, ProteinTalks allows perturbation-*

conditioned trajectories to actively shape what features the model discovers. This integration of a uniquely structured dataset with a dynamic-aware architecture enables ProteinTalks in a parameter-efficient way to model cellular state transitions dynamically and to generalize from controlled cell line experiments to patient-derived tissues ...

By contrast, even very large-scale single-cell transcriptomic collections often remain limited in perturbational and contextual diversity, restricting cross-context generalization because the causal structure underlying cellular responses is only weakly identifiable from observational variation alone⁴⁵. The present dataset therefore provides a foundation for learning latent representations that transfer across drugs, cellular contexts, and disease states. ...

Although neural ODE has been previously applied in single-cell and omics analyses (e.g., trajectory inference and perturbation mapping), they remain largely task-specific rather than serving as a general pretraining strategy for transferable dynamic embeddings¹⁴

Moreover, as a module grounded in temporally resolved proteomic state transitions, ProteinTalks could be naturally integrated into broader agent-based architectures, such as those exemplified by CellForge⁴⁸—a multi-agent framework that autonomously designs and refines neural network architectures for predicting single-cell perturbation responses, incorporating trajectory-aware encoders, perturbation diffusion modules, and virtual instruments for phenotype decoding. In such systems, the generalizable representations learned by ProteinTalks would serve as a dynamic proteomic backbone, complementing other omics-specific modules while autonomous agents coordinate multi-omics perception, biological hypothesis generation, and closed-loop experimental validation. Together, these capabilities point toward a more comprehensive, interpretable, and actionable AI Virtual Cell capable of actively reasoning about and simulating cellular intervention strategies across biological scales."

References

- 1 Callaway, E. Can AI build a virtual cell? Scientists race to model life's smallest unit. *Nature* **643**, 13-14 (2025).
- 2 Leslie, M. The silicon cell. *Science* **390**, 436-439 (2025).
- 3 Abhinav K. Adduri, D. G., Beatrice Bevilacqua, Alishba Imran, Rohan Shah, Mohsen Naghipourfar, Noam Teyssier, Rajesh Ilango, Sanjay Nagaraj, Mingze Dong, Chiara Ricci-Tam, Christopher Carpenter, Vishvak Subramanyam, Aidan Winters, Sravya Tirukkovular, Jeremy Sullivan, Brian S. Plosky, Basak Eraslan, Nicholas D. Youngblut, Jure Leskovec, Luke A. Gilbert, Silvana Konermann, Patrick D. Hsu, Alexander Dobin, Dave P. Burke, Hani Goodarzi, Yusuf H. Roohani. Predicting cellular responses to perturbation across diverse contexts with State. *bioRxiv* (2025).
- 4 Roohani, Y. H. *et al.* Virtual Cell Challenge: Toward a Turing test for the virtual cell. *Cell* **188**, 3370-3374 (2025).

Referee #3 (Remarks to the Author):

Overall Assessment

The authors addressed the requested revisions, but we remain unconvinced by the proposed model and are therefore uncertain about recommending acceptance.

1. Summary of the key results

ProteinTalks is a neural ODE-based virtual cell model trained on 38 million perturbed protein measurements from 18 breast cancer cell lines treated with 63 FDA-approved drugs at three time points. The model outperforms transcriptomic foundation models (Geneformer,

GeneCompass, UCE) by 63–213% in AUPRC for cross-cell-line validation. Key validations include: knockdown confirmation of resistance proteins (AKR1C3, CMPK1, TYMS), clinical stratification of 504 TNBC patients, and prospective PDO screening of ~3,000 compounds identifying three candidates (CAY10603, citarinostat, THZ-1) that outperformed cisplatin.

2. Originality and significance

Strengths: First proteome-based virtual cell model with temporal dynamics; dataset ~10-fold larger than comparable studies; ~1000-fold fewer parameters than transcriptomic foundation models while achieving superior performance.

Addressed: Three time points justified theoretically (dynamical systems analysis) and empirically (10-time-point benchmark confirming 6h/24h/48h capture early/intermediate/late phases). Scaling law analysis supports foundation model claims.

3. Data & methodology

Addressed: DIA-MS with 1% FDR appropriate; rigorous QC (Pearson $r > 0.9$, CV $< 10\%$); systematic hyperparameter optimization for all baselines (Tables S4A-B); Results-Methods mapping table added; figure captions corrected.

4. Appropriate use of statistics and treatment of uncertainties

Addressed: Performance metrics now include SE/SD (Tables S5-S7, S9, S11); effect sizes reported in main text (e.g., ATG3 0.15-fold, CKS2 7.71-fold). SHAP uncertainty quantified. Minor limitation: PertScore sensitivity analysis (85% consistency) lacks formal FDR estimate.

5. Conclusions: robustness, validity, reliability

Robustness: Consistent superiority across seven baselines; scaling law confirmed; DU4475 poor performance explained by class imbalance.

Validity: Knockdown validations; drug synergies confirmed ($\Delta E_{\max}$ 0.28–0.74); clinical alignment with ongoing trials (NCT02286687); prospective PDO validation successful.

Reply: We sincerely thank you for your thorough re-evaluation and detailed feedback. We acknowledge your remaining reservations regarding the proposed model and respect your scientific rigor.

We believe the cumulative evidence, including cross-cell-line predictive performance, prospective PDO validation, and clinical cohort stratification, substantively supports the model's robustness and translational relevance. Additionally, we have provided detailed responses and supplementary information addressing each of your specific comments below. We remain available to provide any further clarification as needed.

6. Suggested improvements

Experimental validation of at least one novel protein (15 identified without prior cancer links)

Reply: Thanks for the comments. In this revision, we have strengthened the Results section by incorporating two complementary lines of evidence to address this point: support from public datasets, together with two new experimental validations based on clinical samples and *in vitro* functional assays. A summary is provided below:

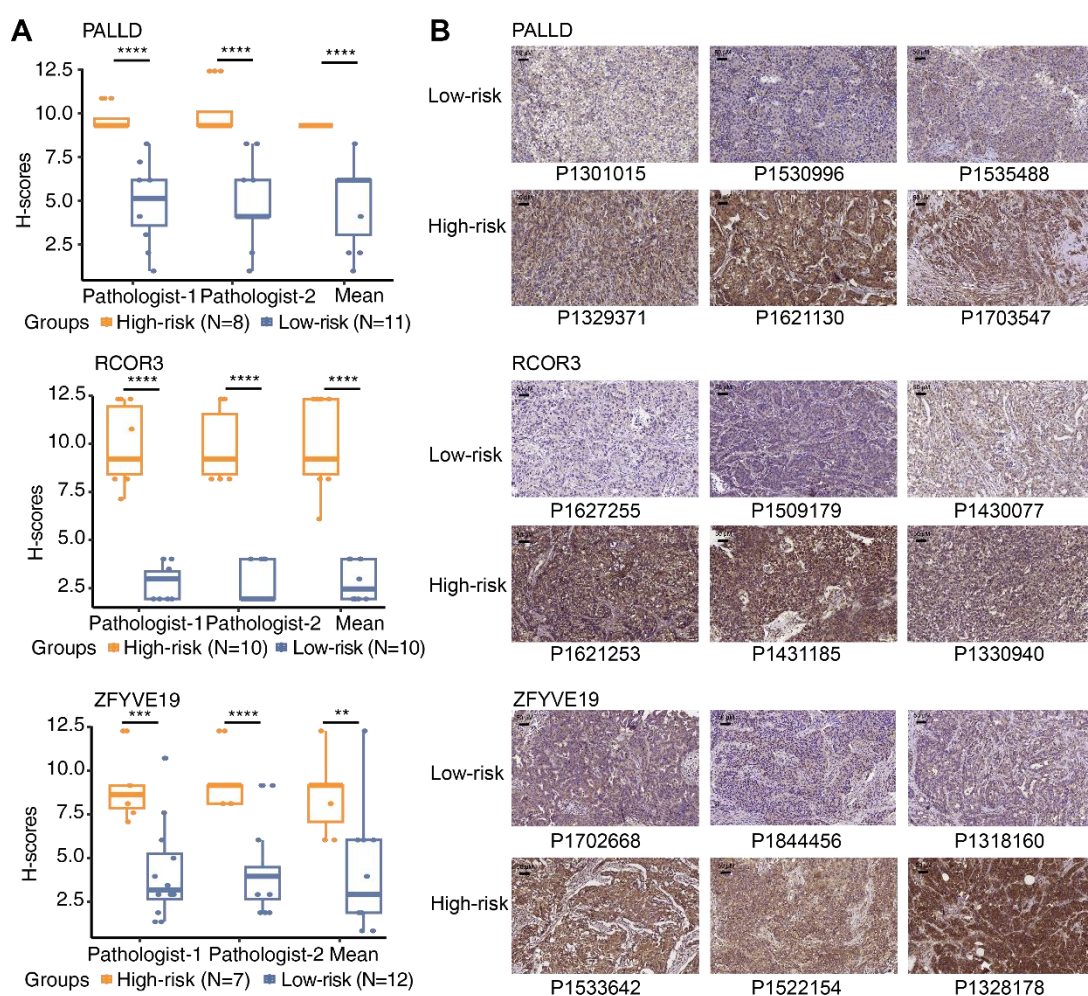
1. Public dataset support

We added this in the revised manuscript **Supplementary Information (SI)** section “*Biological insights and validation of top 100 proteins prioritized by ProteinTalks*”, as copied below.

"The prognostic relevance of these three proteins was further validated across five independent public datasets and our in-house TNBC-HMU cohort, showing a consistent association between their expression levels and differential prognosis in breast cancer patients (SI Table 1)." (details provided in *Rebuttal-appendix 4*)

2. Newly added experimental validation in clinical TNBC patients using IHC staining

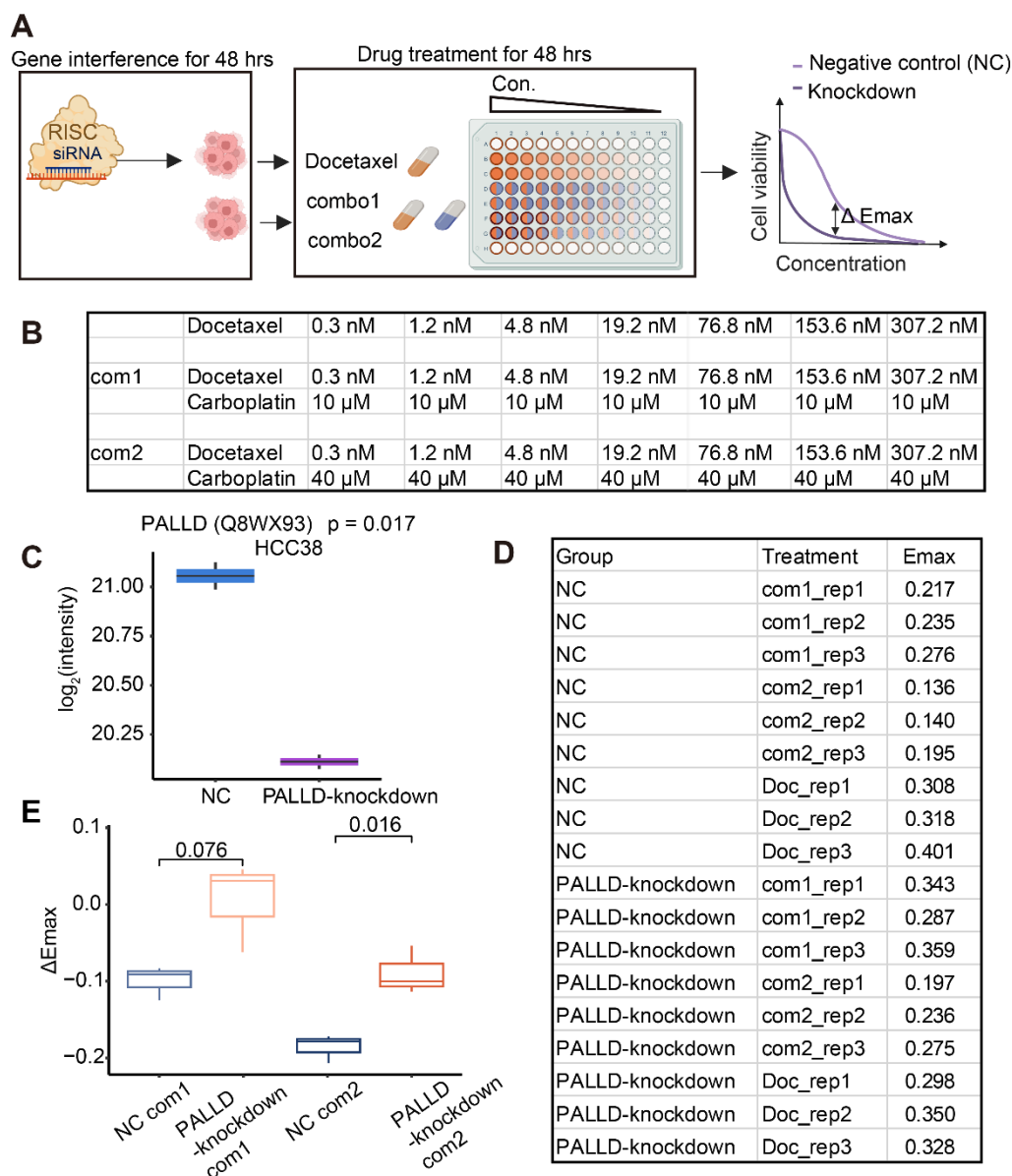
Through univariate analysis performed on the TNBC-HMU cohort and public datasets supplemented during the previous rebuttal, we found that high expression of ZFYVE19, PALLD, and RCOR3 was significantly and positively correlated with poor prognosis in specific subgroups of TNBC patients. To further corroborate these findings at the tissue level and validate the translational relevance of these proteins in a clinical setting, we further selected approximately 20 patients from both the high- and low-risk groups for each protein from the TNBC-HMU and performed immunohistochemical (IHC) staining. Scoring was conducted independently by two professional pathologists, and the results confirmed that protein expression levels were indeed significantly higher in the high-risk patient groups, providing histopathological evidence consistent with our in-silico findings (revised manuscript **SI** section “*Biological insights and validation of top 100 proteins prioritized by ProteinTalks*”, **SI Table 1**, and **SI Figure 2**). The detailed IHC staining and scoring procedures are provided in **Materials and Methods** section of the **SI**.



SI Figure 2 IHC validation of PALLD, RCOR3, and ZFYVE19 expression in the TNBC-HMU cohort. (A) Quantification of IHC staining scores for PALLD, RCOR3, and ZFYVE19 in the TNBC-HMU cohort. H-scores assigned by two pathologists (Pathologist-1 and Pathologist-2) and their mean are presented for each case. Statistical significance between groups was assessed using the two-sided t-test (**, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$). (B) Representative IHC images from three patients per risk group for each of the three proteins. Scale bar, 50 μ m.

3. Newly added experimental validation in vitro

Given that the top 100 proteins distinguished sensitive from resistant TNBC patients in response to neoadjuvant chemotherapy with docetaxel and carboplatin, we next investigated whether PALLD contributes to the synergistic effect of this drug combination *in vitro*. Using RNA interference to knock down PALLD in HCC38 cells, we observed no significant difference in cell viability under single-agent treatment conditions compared with control cells. However, PALLD knockdown significantly enhanced the synergistic effect of the docetaxel-carboplatin combination, suggesting that PALLD suppression potentiates the cooperative cytotoxic activity of this regimen. (Revised manuscript **SI** section “*Biological insights and validation of top 100 proteins prioritized by ProteinTalks*”, and **SI Figure 3**). The detailed Methods are provided in the **Methods** section in the main text, “*Gene interference in different cell lines*”, and “*Cell viability assay*”.



SI Figure 3 In vitro validation of PALLD biological function in modulating drug synergy. (A) *Experimental workflow for drug synergy validation. HCC38 cells were subjected to siRNA-mediated PALLD knockdown, followed by treatment with serial dilutions of docetaxel alone or in combination with carboplatin. Each experimental condition was performed in three independent biological replicates. (B) Concentration matrix of drug combinations used in the synergy assay. (C) Box plot showing log₂-transformed protein abundance measured by mass spectrometry (MS) to assess RNA interference efficacy. PALLD knockdown efficiency*

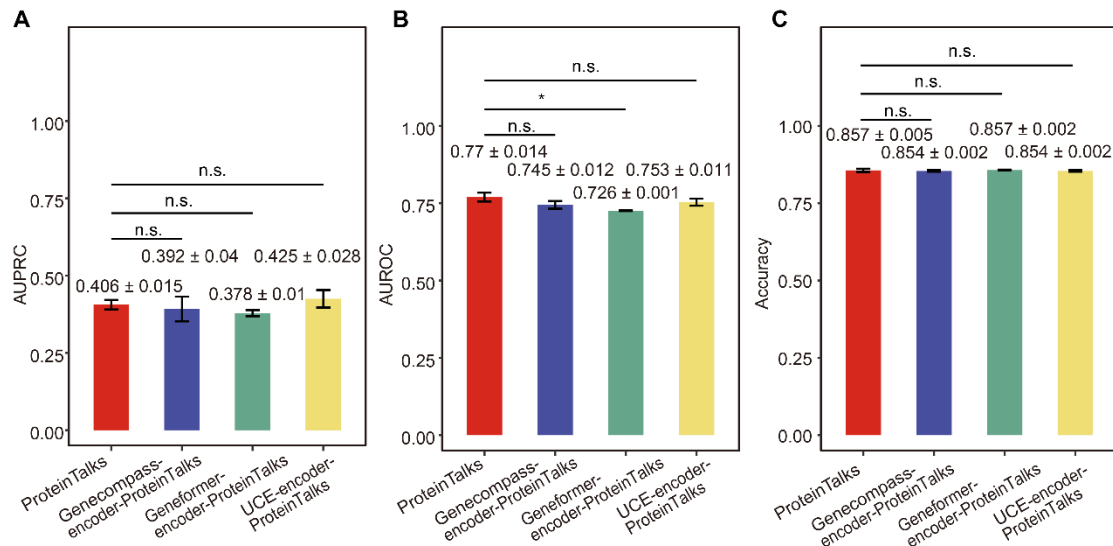
was confirmed in HCC38 cells relative to siRNA control. **(D)** Dose-response curves for docetaxel combined with a fixed concentration of carboplatin (10 μ M or 40 μ M) in the PALLD-knockdown and the control HCC38 cells. The table summarizes the E_{max} values derived from the fitted dose-response curves for each treatment condition. **(E)** Efficacy shift (ΔE_{max}) quantifying the synergistic effect of the drug combination. ΔE_{max} is defined as the difference in maximum response between the combination treatment and docetaxel monotherapy; a larger ΔE_{max} indicates a stronger synergistic effect. Statistical significance was assessed by two-sided Student's t-test.

Show negative transcriptomics integration result mentioned but not presented

Reply: Thank you for raising this point. We have supplemented the analysis to validate the effect of transcriptomic integration, with the detailed results included in the **SI**, as copied below.

"No significant improvement when incorporating embeddings from multiple transcriptomic foundation models into ProteinTalks"

To evaluate whether transcriptomics integration could further improve ProteinTalks, we incorporated pretrained transcriptomic foundation models as encoders within the ProteinTalks framework. Briefly, for each sample, baseline proteomics profiles were encoded through a transcriptomic foundation model to generate sample-specific embeddings, which were subsequently input to ProteinTalks. This design constitutes a form of transcriptomics integration, allowing transcriptomic and proteomic representations to inform the dynamic modeling of proteomic perturbation trajectories. The same data and training strategy as in the original ProteinTalks were used: pretraining on PTDS-1, followed by few-shot learning on PTDS-3 and testing. Among the three transcriptome-pretrained models evaluated, none demonstrated a statistically significant improvement over the original ProteinTalks across either AUPRC or AUROC metrics (**SI Figure 1**), indicating that transcriptomic embedding integration does not yield a meaningful performance gain relative to the proteomics-only framework under the current experimental conditions." The detailed methods are provided in **Materials and Methods** section of the **SI**.



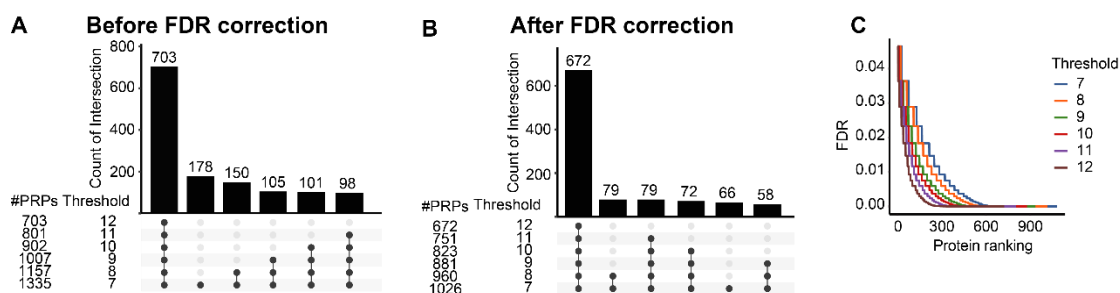
SI Figure 1 Transcriptome-pretrained per-protein features for ProteinTalks and transfer learning across five non-breast cancer cell lines. (A)-(C) Transcriptomics was integrated into ProteinTalks by changing the input representation from one scalar abundance per protein to one transcriptomics-informed feature vector per protein derived from transcriptome-pretrained foundation model checkpoints (GeneCompass, Geneformer, or UCE). Each measured protein was mapped to its gene symbol and assigned a gene embedding from the transcriptome-pretrained model, yielding a protein-by-feature matrix that served as input to

ProteinTalks; the transcriptomic model weights were used as frozen feature extractors. ProteinTalks architecture and training protocol were otherwise unchanged: pretraining on PTDS-1, followed by few-shot adaptation on PTDS-3 and evaluation using the standard test protocol. Comparative performance on five non-breast cancer cell lines in PTDS-3 by ProteinTalks, including AUPRC (A), AUROC (B), and Accuracy (C). Statistical significance assessed by t-test and horizontal brackets indicate pairwise comparisons (n.s., not significant; * $p < 0.05$).

We attribute these results to several fundamental limitations of transcriptomics integration in this setting. Existing transcriptomic foundation models are pretrained on single-cell RNA-seq data, whose gene-space representations do not naturally align with our proteomic feature space, introducing a systematic cross-modality mismatch. In addition, our task requires alignment between initial and subsequent time points and may require greater diversity in cell lines and protein measurements to achieve sufficient generalization.

PertScore sensitivity analysis (85% consistency) lacks formal FDR estimate.

Reply: Thank you for this important suggestion. We have now applied a formal FDR estimation for PertScore using a permutation-based approach, in which sample annotations were randomly shuffled 1000 times to construct a null distribution, enabling formal quantification of false discovery rates across PertScore thresholds. Following FDR correction, 79 proteins were excluded, retaining 823 high-confidence perturbation-related proteins (**Response letter Figure 1**). We revised manuscript **Results** section “The perturbation proteomics dataset captures drug-induced modulation of protein networks”, as copied below. “Beyond the proteins directly targeted by the drugs, we developed PertScore to quantify protein changes identifying 823 perturbation-related proteins (PertScore ≥ 10 , FDR < 0.05)”. Additionally, we have updated **Supplementary figure 3E-F** and **Supplementary table S3**. We also re-examined the data in **Figure 2A** and confirmed that the exclusion of these 79 proteins did not affect the results presented in **Figure 2A**. The detailed FDR estimation procedure has been added to **Methods**.



Response letter Figure 1 Robustness of PertScore across multiple thresholds assessed by FDR correction. (A) Number of overlapping perturbation-related proteins (PRPs) identified across PertScore thresholds (7, 8, 9, 10, 11, 12) prior to FDR correction. (B) Number of overlapping PRPs identified across PertScore thresholds (7, 8, 9, 10, 11, 12) following FDR correction (FDR<0.05) (revised manuscript **supplementary figure 3E**). (C) Distribution of FDR values for each protein across PertScore thresholds (7, 8, 9, 10, 11, 12) after FDR correction (FDR<0.05).

7. References

DIA-NN original paper now cited (reference 47). Appropriate coverage of recent foundation model critiques (Boiarsky et al., Ahlmann-Eltze et al.).

8. Clarity and context

Title appropriately revised to "virtual cell"; clear conceptual positioning; limitations acknowledged (no PTMs, cancer cell lines only, single-omics).

Referee #3 (Remarks on code availability):

Code reproducibility verified

Reply: We thank you for confirming that these points have been satisfactorily addressed

Rebuttal-appendix 1

From algorithmic invention to data infrastructure: How ProteinTalks transcends the original scope of neural ODEs

The continuous-time modeling perspective in machine learning was articulated early by Weinan E (a co-author of this manuscript) in 2017 (E, W., *Communications in Mathematics and Statistics*, 2017), who framed supervised learning through ODE flow maps and an optimal-control viewpoint, and was subsequently operationalized in the Neural ODE framework of Chen et al. (*NeurIPS* 2018), where hidden states evolve according to a learned differential equation and training is performed through scalable adjoint-based backpropagation. Chen and colleagues then extended this line to latent continuous-time modeling in Latent ODEs (*NeurIPS* 2019), enabling irregularly sampled time-series data to be modeled in a learned latent space.

Since then, Neural ODEs and related continuous-time models have indeed been applied in single-cell and omics settings, including trajectory generation and intervention modeling. However, *these studies are largely task-specific*: they use continuous-time dynamics for trajectory inference, interpolation, or perturbation mapping, rather than as a general pretraining principle for learning transferable dynamic embeddings. Therefore, systematic *Neural ODE-based pretraining* for transferable representation learning remains underdeveloped in omics, where current foundation-model practice has been dominated by static or self-supervised pretraining paradigms rather than shared continuous-time dynamic supervision.

Table 1. Comparison of continuous-time models for single-cell dynamics

Method	Year	Domain / task	Learns embedding?	Context-dependent heterogeneity	If pre-trained
TrajectoryNet	2020	scRNA-seq trajectory interpolation via dynamic OT / CNF	No; mainly works in low-dimensional PCA space	Explicit time dependence, but not perturbation-conditioned shared dynamics	No
PRESCIENT	2021	Time-series single-cell trajectory generation and intervention simulation	No in the strong sense; operates in predefined reduced feature space	Temporal progression / interventions	No
DeepVelo	2022	RNA velocity field learning with Neural ODEs	Partly yes; uses latent/denoising structure around velocity learning	Models state-dependent dynamics, not large-scale shared perturbation pretraining	No
RNAForecaster	2023	Transcriptomic forecasting in time-resolved single-cell data	No; emphasizes prediction in expression/count space rather than representation learning	Mostly temporal rollout, not shared perturbation-conditioned latent dynamics	No
CellOT	2023	Single-cell perturbation response prediction via neural OT	Not really; representation learning is not the core object	Not shared vector field dynamics	No
scNODE	2024	Generative modeling for temporal scRNA-seq; interpolation/extrapolation	Yes; VAE latent space jointly optimized with Neural ODE	Perturbation conditioning; mainly time-conditioned latent evolution	No
TIGON	2024	Dynamic non-equilibrium OT for single-cell trajectories and GRN inference	Not as core object; more PDE / OT field modeling than transferable latent representation learning	No perturbation	No
ProteinTalks	2025	Cross-drug, cross-context, and cross-disease generalization for drug discovery	Yes, learn transferable latent representations of cellular execution states	Share a latent space with dynamic meaning under perturbations	Large-scale pretraining

In this sense, our work extends the continuous-time program initiated by Weinan E and the latent dynamical modeling line developed by Chen and others from supervised or task-specific dynamics learning toward a pretraining setting, in which the embedding itself is learned through shared perturbation-conditioned temporal dynamics. Therefore, ProteinTalks demonstrated that continuous-time modeling, when combined with systematic perturbation proteomics at scale, can serve as a pretraining constraint to learn transferable latent representations of cellular execution states, enabling cross-drug, cross-context, and cross-disease generalization for drug discovery. In ProteinTalks, this framework is used *as a pretraining constraint* to shape a latent space with dynamic meaning under perturbations, rather than as an end in itself for trajectory fitting. This represents a conceptual progress from using neural ODEs for trajectory inference to using perturbation-conditioned dynamics as a foundation for operational virtual cells.

Rebuttal-appendix 2

A detailed comparison of ProteinTalks with PRESCIENT and CellOT

We would like to clarify the conceptual difference between these methods: PRESCIENT and CellOT are designed for interpolation and distribution alignment within predefined feature spaces (PRESCIENT operates in PCA space for trajectory simulation; CellOT learns optimal transport maps, typically using PC embeddings for tractability). *Both take a fixed representation as input and model dynamics or transport on top of it. In contrast, ProteinTalks provides a pretraining framework that learns transferable, perturbation-conditioned state representations* from systematic perturbation data, such that downstream predictions and hypotheses are both decision-relevant and experimentally testable. We now provide a point-by-point comparison with PRESCIENT and CellOT, followed by a broader discussion.

Concretely, the ProteinTalks model learns an encoder-defined latent state $z = E_\psi(x)$, together with a perturbation-conditioned vector field $\dot{z} = f_\theta(z, u, t)$. The key object is therefore not a perturbation-specific map or a trajectory fitted in a fixed space, but a *representation-dynamics pair* that is shared across perturbations, cell contexts, and time regimes. Continuous-time modeling here serves as an inductive bias that enforces temporal consistency; the primary contribution is the resulting *transferable dynamic embedding*, as well as in biologically meaningful predictions that can be validated experimentally. Therefore the computational advance of ProteinTalks lies not in proposing a new continuous-time architecture, but in establishing *dynamics-driven embedding learning* as a pretraining principle for temporal perturbation biology, moving beyond endpoint prediction and trajectory fitting toward transferable representations at the proteome execution layer.

Using a neural network to parameterize a potential function that drives continuous-time dynamics, PRESCIENT models differentiation as a specific stochastic process anchored to specific physical time points. The method is conceptually dynamic, but primarily addresses trajectory inference within that specific differentiation system and does not readily generalize to new conditions or contexts, whereas ProteinTalks targets systematic perturbation-driven pretraining that transfers across drugs and contexts.

CellOT learns optimal-transport maps that align unpaired control and perturbed populations, predicting perturbation responses as a state mapping between distributions rather than as an explicit continuous-time simulator. CellOT is not a neural ODE method *per se*, but rather an OT-based mapping framework, and it trained 34 models independently for 34 drug perturbations. This design is well-suited for distribution alignment, but it differs architecturally from learning a single shared dynamic engine that transfers across many perturbations and contexts, such as ProteinTalks.

ProteinTalks is designed to yield more decision-relevant and experimentally testable insights under perturbations than approaches optimized for static snapshots, endpoint alignment, or

within-process trajectory inference. Its core novelty lies in operationalizing perturbation-conditioned dynamic pretraining of the proteome execution layer at scale, producing transferable latent representations aligned to drug-response endpoints and enabling a closed loop from *in silico* prioritization to wet-lab validation.

We also provide a more detailed comparison of ProteinTalks with existing models (**Table 2**).

Table 2. Comparison of ProteinTalks with existing models.

	PRESCIENT (Yeo et al., <i>Nat Commun</i> , 2021)	CellOT (Bunne et al., <i>Nat Methods</i> , 2023)	Static RNA foundation models (e.g., scGPT /Geneformer/UCE)	Our dynamic proteome model (ProteinTalks)
Primary scientific task	Continuous-time trajectory simulation for cell-fate dynamics with interventions.	Perturbation response prediction via OT mapping between unpaired distributions.	Large-scale static representation learning on snapshots.	Perturbation-conditioned proteome dynamics representation learning for drug response prediction + mechanism (experimentally testable hypotheses) discovery + cross-context generalization
Data modality / Observed variables	scRNA-seq time series $x \in \mathbb{R}^n$ (gene expression snapshots)	scRNA-seq pre/post perturbation; $x \in \mathbb{R}^n$ (control/perturbed snapshots; unpaired)	Predominantly RNA-centric snapshot corpora (self-supervised pretraining). $x \in \mathbb{R}^n$ (static)	Time-resolved perturbational proteomics (functional execution layer) under drug perturbations $p(t, u) \in \mathbb{R}^m$ (protein abundance over time under perturbation (u))
Mathematical object learned	Trajectory generator / simulator ($t \mapsto x(t)$) with interventions.	Transport map ($T_k: \mu_{control} \rightarrow \mu_{perturbed}$) (OT mapping).	Embedding function ($z = g_\theta(x)$, static); pretraining via generative/self-supervised losses.	Perturbation-conditioned transition operator / vector field for proteome state evolution: learn $z = \phi(p)$ and $\frac{dz}{dt} = f_\theta(z, u)$
Training objective (prototype)	Match simulated trajectories to observed progression + regularizers.	OT objective via parameterized dual potentials / ICNNs (input convex neural networks) to induce (T_k).	Self-supervised generative pretraining (masked/generative transformer) + downstream tasks.	(i) dynamic pretraining loss across time points & perturbations + (ii) drug-response aligned downstream objective.
Cross-context generalization (virtual-cell criterion)	No	No	Yes	Yes
Mechanistic interpretability	Hypotheses about fate/trajectory modulators.	Hypotheses via mapped state changes; mechanism often indirect.	Mostly correlational embeddings.	Protein-level dynamic signatures $\rightarrow$ experimentally testable targets/markers under drug perturbation
Clinical validation	No	No	No	Yes

Rebuttal-appendix 3

Concepts and roadmap for AI virtual cell

term "virtual cell" may invite scrutiny, and we welcome the opportunity to clarify our usage. As with many emerging concepts in the history of science—from "systems biology" to "foundation models"—the notion of a "virtual cell" is still actively evolving, and different researchers have proposed different definitions and scopes. We do not claim that ProteinTalks constitutes a fully realized virtual cell in the most comprehensive sense; rather, we adopt this term to situate our work within a broader and rapidly growing conceptual framework aimed at building computational models that simulate cellular behavior. In essence, the "virtual cell model" we present here is a computational model that simulate certain, not all, cell behaviors.

There are several related terms used in the literature, including digital twins, digital cells, and virtual cells. We have discussed about this in response to a previous reviewer. In our view, the digital twin, literally meaning the digital "twin" of a cell, represents the ultimate and most advanced goal in computational biology—a personalized and continuously updated model of a biological system (e.g., a patient's cell or tumor) that evolves in real time with new experimental or clinical data, enabling individualized prediction and intervention. The digital cell, as a foundational layer toward this goal, refers to a comprehensive, data-centric collection of molecular and cellular information that integrates multi-omics measurements (e.g., genomics, transcriptomics, proteomics, and metabolomics) to describe the molecular composition and state of a cell. The more practical and currently achievable form, the virtual cell model, focuses on simulating and predicting specific cellular functions and responses to perturbations. Unlike a digital cell or digital twin, a virtual cell model does not require the integration of every molecular variable; rather, it aims to capture key cellular functions sufficient to reproduce and forecast biological behaviors under defined conditions.

In another word, the digital twin represents a long-term goal in computational biology, i.e. a continuously updated, personalized model of a living system. The digital cell refers to an integrative framework that ultimately incorporates multiple detailed molecular layers and cellular variables to reflect such biological complexity. Within this spectrum, the virtual cell represents a more attainable step that prioritizes modeling the dynamic behaviors and functions of cells, particularly the proteomic changes that directly govern function to reconstruct and forecast cellular responses under defined perturbations.

Although building a true digital twin is a long-term goal, current virtual cell models have already leveraged single-cell transcriptomic data to develop foundation models that predict cellular behaviors. As Stephen Quake has noted, "In the future, biologists will spend less time wielding pipettes and more time relying on computational tools."¹ Many ongoing efforts, such as those led by CZI and Google DeepMind, have focused on constructing AIVCs based on single-cell RNA sequencing data, which provide snapshots of gene expression under specific conditions². Similarly, Arc Institute's first virtual cell model, State, also utilizes single-cell transcriptomics to predict cellular responses to drug or genetic perturbation³. The recent US\$175,000 virtual-cell competition launched by the Arc Institute, which challenges researchers to predict how human stem cells respond to genetic alterations⁴.

By synthesizing perspectives on AIVC development and existing AIVC models, the field is progressively advancing toward a fully realized AIVC through the following trajectories:

Path I, State-space representation (omics foundation models)

Models such as scGPT (Cui et al., *Nat Methods*, 2024), Geneformer (Theodoris et al, *Nature*, 2023), and UCE (Yanay et al, *bioRxiv*, 2024) learn static embeddings derived from transcriptomic snapshots through self-supervised pretraining. These approaches effectively capture cellular identity and correlations among genes within individual samples. Their design

focuses on representing cell states as points in a learned embedding space, offering valuable insights into cellular heterogeneity and relationships.

Path II, *In silico* perturbation in state space (continuous trajectory and transport models)

Some models incorporate perturbation information into trajectory or transport frameworks (Zhang et al., *NPJ Syst Bio Appl*, 2025; Yeo et al., *Nat Commun*, 2021, Bunne et al., *Nat Methods* 2023). These models capture dynamic cellular changes within specific experimental settings by modeling continuous-time trajectories or transitions between control and perturbed states, with evaluation mainly focused on accuracy within these observed regimes.

Path III, Cross-context generalization with experimental validation (ProteinTalks).

Unlike previous two paths that concentrate on within-static modeling or context dynamic modeling, ProteinTalks emphasizes generalizable representations of cellular state transitions grounded in temporally resolved proteomic data. Its dynamics-aware training paradigm enables generalizable representations of cellular state changes that transfer across diverse biological contexts, including unseen drug treatments and cell lines, without requiring task-specific retraining. Critically, ProteinTalks not only predicts cellular responses but also generates experimentally testable intervention strategies, bridging computational inference and laboratory validation. This capacity for cross-context generalization and experimental grounding distinguishes Path III from prior approaches, which remain largely confined to interpolation within observed data regimes, and positions ProteinTalks as a foundational dynamic modeling component upon which more complex agentic systems, as described in Path IV, can be constructed.

Path IV, Agentic virtual cell systems

Building upon the generalizable dynamic modeling foundations established in Path III, an emerging direction involves embedding such cellular modeling components within broader autonomous agent-based frameworks. Representative efforts in this space, such as CellForge (Tang et al, *arXiv*, 2025), explore how multi-agent systems can autonomously design and refine neural network architectures for predicting single-cell perturbation responses, incorporating mechanisms such as trajectory-aware encoders, perturbation diffusion modules, and virtual instruments for phenotype decoding. In this paradigm, specialized agents coordinate multi-omics perception, hypothesis generation, and iterative experimental feedback, enabling the system to move beyond passive prediction toward active reasoning about cellular intervention strategies. Dynamic modeling modules grounded in temporally resolved data, such as ProteinTalks, could naturally serve as specialized components within such architectures, contributing proteomic state transition representations to a larger integrated system. While this path remains nascent and many technical challenges persist, including harmonization of heterogeneous multi-omics data streams and robust closed-loop experimental validation, it points toward a longer-term vision of a comprehensive, interactive AI virtual cell capable of reasoning and acting across biological scales.

Rebuttal-appendix 4

SI Table 1 Associations of ZFYVE19, RCOR3, and PALLD with breast cancer prognosis

Gene	Datasets	Types	Patients	No. of Patients	Prognosis	Median		P-value
						Time	HR (95% CI)	
ZFYVE19	TCGA 2015 Cell	Transcriptomics	TNBC	47	DFS	35.22	4.41 (1.1-17.66)	0.0362
ZFYVE19	TCGA 2015 Cell	Transcriptomics	TNBC	51	OS	35.91	7.34 (1.55-34.66)	0.0119
ZFYVE19	cBioPortal (METABRIC)	Transcriptomics	BCa+chemo+radio	59	RFS	1619	3.22 (1.07-9.65)	0.037
ZFYVE19	cBioPortal (METABRIC)	Transcriptomics	BCa+chemo	83	RFS	1550	3.87 (1.68-8.91)	0.001
ZFYVE19	cBioPortal (METABRIC)	Transcriptomics	BCa	89	RFS	1528	3.37 (1.46-7.79)	0.004
ZFYVE19	cBioPortal (METABRIC)	Transcriptomics	TNBC+chemo	165	RFS	65.867	2.2 (1.24-3.92)	0.007
ZFYVE19	cBioPortal (METABRIC)	Transcriptomics	TNBC+chemo	165	OS	75.4	1.76 (1.04-3)	0.037
ZFYVE19	cBioPortal (METABRIC)	Transcriptomics	TNBC	319	RFS	88	1.66 (1.14-2.42)	0.009
ZFYVE19	TNBC-HMU	Proteomics	TNBC Stage 3	243	OS	3095	3.03(1.26-7.25)	0.013
ZFYVE19	TNBC-HMU	Proteomics	TNBC Stage 3	239	RFS	3018	2.22(1.13-4.37)	0.021
RCOR3	TCGA 2015 Cell	Transcriptomics	BCa	728	DFS	25.165	1.85 (1.04-3.28)	0.036
RCOR3	TCGA 2015 Cell	Transcriptomics	BCa	803	OS	29.8	0.46 (0.25-0.87)	0.016
RCOR3	Jiang, et al., 2024, Nature Cancer	Proteomics	BCa	310	DMFS	83.384	0.41 (0.22-0.76)	0.005
RCOR3	Jiang, et al., 2024, Nature Cancer	Proteomics	BCa	321	RFS	82.891	0.39 (0.22-0.68)	0.0009
RCOR3	Jiang, et al., 2024, Nature Cancer	Proteomics	BCa	320	OS	84.090	0.28 (0.14-0.56)	0.0003
RCOR3	Mo, et al., 2025, Science Advances	Transcriptomics	TNBC	48	RFS	61.5	0.26 (0.07-0.95)	0.042
RCOR3	Mo, et al., 2025, Science Advances	Transcriptomics	BCa	136	RFS	64	0.25 (0.1-0.64)	0.004
RCOR3	TNBC-HMU	Proteomics	BCa TNBC treated with two-drug combination	101	OS	3296	3.19(0.97-10.46)	0.043*
PALLD	TNBC-HMU	Proteomics	TNBC stage 2-3	72	OS	3095	4.253(4.25-1.61)	0.004
PALLD	TNBC-HMU Gong, et al., 2022, Cell Reports	Proteomics	TNBC stage 2-3	70	RFS	3095	4.00(1.60-9.98)	0.003
PALLD	Gong, et al., 2022, Cell Reports	Transcriptomics	BCa+chemo	83	RFS	1550	5.92 (1.39-25.2)	0.016
PALLD	Gong, et al., 2022, Cell Reports	Transcriptomics	BCa	89	RFS	1528	6.3 (1.49-26.67)	0.012

BCa, breast cancer; TNBC, triple-negative breast cancer; IDC, invasive ductal carcinoma; BCa+ chemo, breast cancer with receipt of chemotherapy; BCa+chemo+radio, breast cancer with receipt of chemotherapy and radiotherapy; DFS, disease-free survival; OS, overall survival; RFS, recurrence-free survival, DMFS, distant metastasis-free survival; HR, hazard ratio; CI, confidence interval.

** P-value was derived from Kaplan–Meier survival analysis (log-rank test). All other p-values were calculated by univariate Cox proportional hazards regression based on individual protein expression levels.*

Dear Editors,

Thank you for the detailed editorial feedback and the opportunity to submit a revised version of our manuscript entitled “An operational perturbation proteomics-based virtual cell model”. We appreciate the thorough checklist provided to ensure our study meets the high publication standards of Nature.

In this revision, we have meticulously addressed every editorial point, with a particular focus on enhancing the transparency of our proteomics data deposition and ensuring full compliance with ethical reporting and figure formatting guidelines. We have reorganized the manuscript structure to include the Methods section within the main article file, verified all third-party rights for our schematics, and consolidated our Extended Data to meet the specified limits. All datasets have been updated in their respective repositories to facilitate seamless access upon publication.

We believe these comprehensive refinements have further strengthened the clarity and accessibility of our work. Below, we provide a point-by-point response to the editorial comments.

Thank you for your continued consideration of our research.

Sincerely,

Tiannan

Point-by-point reply

1. Please make use of our Extended Data format by moving the most important SI figures. We allow up to 10 Extended Data items (tables and figures), which must be submitted in .jpg, .tif or .eps format. See 'EXTENDED DATA' below for detailed guidance.

Reply: We have promoted 9 key figures to Extended Data (EDF1-9, submitted as .tif files). To stay within the 10-item limit while maximizing information density, we removed figures presenting data redundant with existing tables: original Figures S5, S6, S7E-G, and S10, which duplicated Supplementary Tables 5, 7, 9, and 12, respectively. All data remain accessible in the tables. Remaining SI figures have been renumbered.

2. Please reorganize your main article file. For reference, the Word file should be in the order: title, authors, affiliations (plus present addresses), bold first paragraph, main text, references, tables, figure legends, Methods (will be online-only) plus any Methods-associated references (numbering continuous with main references, i.e. do not start at 1 again); data and code availability statements included at end of Methods, acknowledgements, funding statement, author contributions, competing interest declaration (encompassing all authors), additional information (containing supplementary information line (if any) and corresponding author line), Extended Data figure legends and Extended Data table titles and footnotes (any references unique to the Extended Data should be added to the end of the online-only reference list).

Reply: We have reorganized the main article file to follow Nature's required structure. All figures have been removed from the manuscript and will be submitted as separate files. All sections are now arranged in the specified order.

Changes in manuscript:

- Removed all embedded figures from manuscript; figures submitted separately
- Formatted first paragraph in bold
- Separated Methods-associated references and placed them after Methods section with continuous numbering from main text
- Added separate Data Availability and Code Availability statements at end of Methods
- Separated Acknowledgements and Funding Statement into distinct sections
- Added Extended Data figure legends at end of manuscript

3. Please reduce the length of the title to 75 characters (with spaces) or less, so that it fits on two lines in the final layout.

Reply: We have revised the title to 75 characters.

4. Please provide the manuscript in .docx format; it is currently in PDF format.

Reply: We have provided the manuscript in .docx format as requested.

5. Please add references to the abstract.

Reply: We have added appropriate references to the abstract.

6. Please create a separate reference list for any Methods references, ensuring that the numbering continues from the main text references.

Reply: We have created a separate Methods reference list placed immediately after the Methods section, with numbering continuing sequentially from the main text references.

7. Please remove the main figures from the article file and re-supply them individually in an acceptable format such as EPS, AI, PS, PDF, PPT, PSD or XLS (for graphs) with editable vector files.

Reply: We have removed all main figures from the article file and are submitting them individually in PDF format with editable vector elements.

8. Please ensure that the legends for the main figures do not exceed 300 words; if this is not possible, please make them as close as possible to 300 words.

Reply: We have revised all main figure legends to 300 words or fewer.

9. Please reduce subheadings to 40 characters (with spaces) or less.

Reply: All subheadings have been revised to meet the 40-character limit.

10. Please provide a supplementary information guide (also called SI guide) as a separate Word document (see SUPPLEMENTARY INFORMATION below).

Reply: We have provided the supplementary information guide as a separate Word document.

11. Please check the sources of all illustrations for potential third-party rights issues and clarify whether permissions are needed to adapt or reproduce them. Please ensure that the relevant details are included in the third-party rights table when you resubmit. If BioRender or a similar software has been used, please also provide the relevant licenses. In particular, please check the following figures: Figure 1, 3a, 3f, 4c, 5a, 5f, Supplementary figure 3a.

Reply: We have checked all illustrations for third-party rights issues and obtained the necessary permissions. All relevant details, including BioRender licenses where applicable, have been included in the third-party rights table.

12. Please provide a third-party rights table when you resubmit (see list of required files near the end of this email, which includes a link to the template). Even if no third-party rights

issues exist, an empty table must be submitted. If Biorender or similar software has been used, please also ensure the relevant licenses are provided.

Reply: We have submitted the completed third-party rights table with all relevant permissions and licenses.

13. Please revise figure panels such that they do not contain tables; instead, present these as separate tables or in an alternative format.

Reply: We have removed the table from Figure 3g and relocated it to *Supplementary Table 10*.

14. Please remove SI information from the article file and re-supply it as a separate SI file.

Reply: We have removed all supplementary information from the main article file and provided it as a separate SI Word document.

15. Please ensure that the text size in all figures is at least 5 pt Arial.

Reply: We have verified that all text in figures meets the minimum 5 pt Arial requirement

16. Please provide the funding information as a separate section rather than merging it with the acknowledgements, in accordance with current policy.

Reply: We have separated the Funding Statement from the Acknowledgements section as required.

17. Please organize public release of the data deposited in PRIDE (dataset identifier PXD053384) in a timely manner, as it is currently not available.

Reply: We have released a backup dataset in iProx (IPX0007409000) to ensure immediate data availability. We are currently working with PRIDE to resolve technical issues preventing public release of PXD053384 and will complete this as soon as possible.

18. Please provide separate 'Data availability' and 'Code availability' statements instead of a combined statement.

Reply: We have separated the *Data Availability* and *Code Availability* statements into two distinct sections.

19. Please revise the Data Availability Statement so that it is not a review version and includes full information on the data, which we strongly encourage you to make fully available with

the paper (also see below)

Reply: We have updated the Data Availability Statement in the manuscript as follows:

“Data are available upon request for non-commercial purposes. Raw mass spectrometry proteomics data have been deposited in the ProteomeXchange Consortium (iProX: <https://www.iprox.org/>, project ID: IPX0007409000). The PTDS protein matrix is publicly available at db.prottalks.com.”

20. Please check and confirm for all Supplementary Figures that:

- If data is presented as bar charts, individual data points are shown using overlaid dot plots.
- The n number (i.e. the sample size used to derive statistics) is provided and defined as a precise value (not a range), using the wording “n=X samples/cells/independent experiments” etc. where applicable.
- Any chart axis, error bars, scale bars, symbols and colour scales are defined.
- Any statistical tests used for data analysis are specified and exact p-values are provided either on the figures themselves, in the legend or in the Source Data file.
- Wherever representative data such as micrographs are shown, the legend indicates how many times the experiment was repeated with the same result.

Reply: We have reviewed and revised all Supplementary Figures to meet these requirements:

Individual data points

We have revised Supplementary Figures 1, 2A, 3B-C, and 4 to show individual data points overlaid on bar charts.

Sample size

We have added precise n values in Supplementary Figure 1, 3B-C, and 4.

Supplementary Figure 2A: the number of patients in each group is indicated by N in the high-risk and low-risk groups, with the figure legend stating “N indicates the number of patients in each group (high-risk vs. low-risk).”

Definitions

We have added error bar definitions (SE) in Supplementary Figure 1 and ensured all axes, scale bars, symbols, and color scales are properly defined throughout.

Statistical information

We have displayed exact P-values on the figures, including Supplementary Figures 1A-C, 2A and 3B-C.

Representative images

For IHC images in Supplementary Figure 2B, each patient had one tissue section stained and scored. The images shown are from three representative patients per risk group (one image per patient). Each section was independently scored by two pathologists who evaluated the overall staining pattern of the entire tissue section. Approximately 10 patients per group were included in the quantitative analysis (Supplementary Figure 2A), and statistical significance

was observed for both individual pathologist scores and their mean values.

Details are provided in the **SI figure legend**: "*Quantification of IHC staining scores for PALLD, RCOR3, and ZFYVE19 in the TNBC-HMU cohort. H-scores assigned by two pathologists (Pathologist 1 and Pathologist 2) and their mean are presented for each case.*"

The H-score methodology is described in the **Methods** section: "*Staining was independently scored by two experienced pathologists blinded to clinical information, using the H-score method. Each section was evaluated for staining intensity on a scale of 0–3 (0, negative; 1, weak; 2, moderate; 3, strong) and the percentage of positive tumor cells on a scale of 1–4 (1, ≤25%; 2, 26–50%; 3, 51–75%; 4, >75%). The H-score was calculated as the product of these two parameters, yielding a total score ranging from 0 to 12. Samples were classified as negative (0), low (1–4), moderate (5–8), or high (9–12) expressors.*"

TRANSPARENT PEER REVIEW: Nature uses a transparent peer review system for original research manuscripts. We encourage increased transparency in peer review by publishing the reviewer comments, author rebuttal letters and editorial decision letters. Such peer review material is made available as a supplementary peer review file.

Please note: we allow redactions to authors' rebuttal and reviewer comments in the interest of confidentiality. If you are concerned about the release of confidential data, please let us know specifically what information you would like to have removed. Please note that we cannot incorporate redactions for any other reasons. Reviewer names will be published in the peer review files if the reviewer signed the comments to authors, or if reviewers explicitly agree to release their name. For more information, please refer to our [website](#).

Reply: We agree to transparent peer review and have no confidential information requiring redaction.

ORCID--IMPORTANT: All authors identified as 'corresponding author' on the manuscript must have an ORCID associated with their Nature account before submitting the final version of the manuscript. While non-corresponding authors do not have to link their ORCIDs, they are encouraged to do so. Please note that it will NOT be possible to add/modify ORCIDs at the proof stage. Thus, if they wish to have their ORCID added to the paper they must follow the above procedure prior to acceptance. If you have any issues attaching an ORCID identifier to your Nature account, please contact the [Platform Support Helpdesk](#).

Reply: We confirm that the ORCID identifiers of all corresponding authors have already been linked to their Nature accounts.

In order to avoid delays with publication of your manuscript, please read the guidelines below carefully before resubmission of your manuscript.

STATISTICS: When revising your manuscript, you should ensure that any statistical analysis used is sound and that it conforms to our [guidelines](#). A collection of articles explaining the basics of statistical analysis and advice on how to best present it can be found [here](#).

Reply: We confirm that all statistical analyses used in the revised manuscript are appropriate and have been conducted in accordance with the journal's guidelines.

REPRODUCIBILITY: To ensure that the quality and transparency of methods and statistical reporting (as discussed [here](#)) are sound before the paper is published, we have reviewed your Reporting summary editorially. I have attached two documents: one listing specific issues related to your manuscript and one containing an annotated version of the Reporting summary. Please ensure that, as well as the more general points below, the points highlighted in the attached documents are addressed in full, both on these forms and within the manuscript. Both forms should be uploaded as a "Related Manuscript" file type. The Reporting summary will be published with your paper.

Reply: We uploaded the Reporting summary as a "Related Manuscript"

LENGTH: In print, biological sciences papers do not normally exceed 8 pages on average; the final print length, however, is at the editor's discretion. The typical length of an 8-page article with 5 modest (quarter-page) display items is 4300 words. If a composite figure (with multiple panels) must occupy at least half a page in order for all the elements to be visible, the text length may need to be reduced accordingly to accommodate such figures. Essential but technical details can be moved into the Methods or Supplementary Information (see below).

In this case, we feel the current length of the paper is appropriate, so no further shortening is necessary; you should not significantly add to the text when revising.

Reply: Thank you for this guidance. We have not significantly increased the length of the manuscript in the revised version.

TITLE: Titles cannot exceed 75 characters (including spaces); they must not contain punctuation.

Reply: We confirm that the title complies with these requirements.

AUTHOR NAMES USING NON-ROMAN CHARACTERS: We can support presentation of author names using non-Roman characters in the HTML version of the published article. Currently supported scripts include Arabic, Chinese, Cyrillic, Devanagari, Greek, Hebrew,

Hangul, Japanese and Persian. If this is relevant to your authors, please include the author names in parentheses after the Roman-character spelling (see an example [here](#)). You will be asked to verify the rendering is correct at the proof stage.

Reply: We have already included the Chinese names in parentheses after the Roman-character spellings of the authors' names.

SUMMARY PARAGRAPH: Papers start with a fully referenced, bold paragraph, ideally of about 200 words, aimed at readers in other disciplines. Numbers, abbreviations, acronyms or measurements should be avoided unless essential. The summary paragraph consists of 2 to 3 sentences of basic-level introduction to the field; a brief account of the background and rationale of the work; a statement of the main conclusions (introduced by the phrase 'Here we show' or its equivalent); and a conclusion of 2 to 3 sentences putting the main findings into general context so it is clear how the results described in the paper have moved the field forward. A downloadable, annotated example is available [here](#).

Reply: We have carefully checked the summary paragraph and confirm that it is within 200 words.

MAIN TEXT: If further introductory material is necessary, the main text can begin with up to 500 words of introduction expanding on the background to the work (some overlap with the summary is acceptable), before proceeding to a concise, focused account of the findings, and ending with 1 or 2 short paragraphs of discussion. Sections are separated with subheadings (up to 40 characters including spaces) to aid navigation.

Reply: We have checked the main text and confirm that it is within 200 words.

REFERENCES: As a guideline, most papers should include no more than 50 main text references; all additional references can be cited in (and listed after) the Methods section, as detailed below.

Reply: We have separated the references cited in the main text and Methods, and we confirm that the main text includes no more than 50 references.

FIGURE LEGENDS: These should be listed sequentially after the main text references and not in the figure files. Each legend should begin with a brief title for the whole figure and continue with a short description of each panel and the symbols used. Legends should not exceed 300 words each. Each figure legend should contain, for each panel where relevant, the following information:

* the exact sample size (n) for each experimental group/condition, given as a number, not a

range; ,

- * a description of the sample collection allowing the reader to understand whether the samples represent technical or biological replicates (including how many animals, litters, cultures, etc);
- * a statement of how many times the experiment shown was replicated;
- * definitions of statistical methods and measures:
- * very common tests (e.g. t-test, simple Chi-square tests, Wilcoxon and Mann-Whitney tests) can be identified by name only, but more complex techniques should be described in the Methods;
- * whether tests are one-sided or two-sided;
- * whether there are adjustments for multiple comparisons;
- * the statistical test results (e.g., P values);
- * the definition of 'center values' as median or average;
- * the definition of error bars as s.d. or s.e.m.

Descriptions that are too long for the figure legend should be included in the Methods section.

Reply: We have revised the figure legends and related sections to ensure compliance with Nature's requirements. This information has been synchronized in the Figure legends in manuscript, the Reporting Summary, and the accompanying checklist file provided with the editorial materials, "07_03_489234_4_attach_24_16281.docx". Specifically:

- * Figure legends are listed sequentially after the main text references rather than embedded in the figure files. Each legend now begins with a brief title for the whole figure, followed by concise descriptions of individual panels and the symbols used. We have also checked that all legends remain within the 300-word limit.
- * Exact sample sizes (n) are now provided as precise values wherever relevant. Specifically, n values are indicated directly in the figures for Fig. 2C-D and Fig. 5D-E, and are provided in the figure legends for Fig. 3C-E.
- * The nature of the samples and replicates has been clarified to distinguish biological from technical replication. For Fig. 2C-D, the description of sample types is too long, which are provided in the Methods section. For Fig. 3C-E, the sample numbers are provided in the figure legends and Methods section. For Fig. 5D-E, the samples are independent patient specimens and are labeled in the figure. For Fig. 5G-I, each concentration was tested in three independent biological replicates. The plotted points in Fig. 5G-I represent model-fitted values at each concentration, resulting in one fitted value per concentration point. The corresponding replicate-level measurements used for model fitting are shown in SI Figure 4. We have also described this in the Methods section of the SI.
- * Statements of experimental replication have been added where applicable. In particular, Fig. 3C-D now specifies in the figure legend and Methods section.

* Statistical methods and measures have been clarified in the figure legends and Methods. For Fig. 3C-E and Fig. 5D-E, the statistical tests used are specified in the corresponding figure legends. For more complex analyses, including the PertScore-related testing framework, detailed descriptions are provided in the Methods.

* Whether statistical tests were one-sided or two-sided is now explicitly stated in the figure legends. This information has been added for Fig. 3C-E and Fig. 5G-I.

* Adjustments for multiple comparisons have been clarified in the Methods. Specifically, the PertScore analysis and its correction procedure are described there, and the *p* values used in the survival analyses for Fig. 5D-E are specified as Benjamini-Hochberg-adjusted in the Methods.

* Statistical test results, including *P* values, are now provided where relevant. These have been added directly to the figures for Fig. 3C-E, Fig. 5D-E, and Fig. 5g-i.

* The definition of center values has been clarified in the figure legends. Specifically, for Fig. 3C-E and Fig. 5D-E, we now define whether center values represent the median or the average, as appropriate.

* The definitions of error bars have also been added. For Fig. 3C-E, the error bar definition is provided in the figure legend, and for Fig. 5D-E, the error bars are defined in the figure/legend accordingly.

* Where additional detail would make the legends overly long, the relevant methodological descriptions have been included in the Methods section, in line with Nature's guidance.

METHODS: The Methods section, which provides the full, step-by-step instructions that would allow other researchers to replicate the results, is included after the main text figure legends. The Methods section will not appear in print but will appear online in the full-text HTML and PDF versions. The Methods section should be written as concisely as possible but should contain all elements necessary to allow interpretation and reproduction of the results. If there are additional references (in the Methods section, Supplementary Information, etc), their numbering should continue from the last entry in the main text reference list, and they should be listed following the Methods section. Specialized methods that require chemical structures, figures, or tables cannot be accommodated in the Methods section of the main text file. If such information is part of the Methods, the entire Methods section must instead be included within a Supplementary Information text file.

Reply: We confirm that the Methods section has been prepared in accordance with Nature's requirements.

ETHICS STATEMENT: For research involving human research participants, the Methods

section must include an ethics statement. This statement should provide the name of the committee that approved the study; confirm that the research was performed in accordance with all relevant guidelines and regulations; and confirm that informed consent was obtained from all participants. If the study was granted an exemption from requiring ethics approval, details of the committee granting the exemption must be included.

Reply: We have now included this information in a dedicated “*Ethics statement*” subsection in the Methods section. This standalone section states the name of the ethics committee that approved the study, confirms that all research involving human participants was performed in accordance with relevant guidelines and regulations, and confirms that informed consent was obtained from all participants, where applicable.

Research involving human embryos or gametes, or human stem cells in contexts requiring ethical oversight, also must include an ethics statement in the Methods section. This statement should provide the name of the committee that approved the study and confirm that the research was performed in accordance with all relevant guidelines and regulations. We encourage authors to follow the principles laid out in the [2021 ISSCR Guidelines for Stem Cell Research and Clinical Translation](#). Where necessary, the ethics statement should also describe the conditions of donation of materials, such as human embryos or gametes, and confirm that informed consent was obtained from all donors of cells or tissues.

Reply: We confirm that this study did not involve research using human embryos, human gametes, or human stem cells.

MAIN TEXT STATEMENTS: Several statements (which will not appear in print but will appear online in the full-text HTML and PDF) are required after the Methods (and additional references, if present). First, there should be an Acknowledgements section. Second, any relevant funding should be declared in a separate Funding statement; our funding statement guidelines can be found [<https://www.nature.com/nature-portfolio/editorial-policies/funding>];>here. Third, we require a detailed Author Contribution statement; the specific contributions of each author, particularly in terms of which authors performed which specific experiments, must be listed. Fourth, a Competing Interest statement noting financial and non-financial interests, as well as any patents, must be provided; patent information should include at a minimum the patent number, what is covered by the patent, and who submitted the patent application. Finally, an Additional Information statement should include information regarding reprints and permissions and name the author(s) to whom correspondence and requests for materials should be addressed. Formatting details and an example are available [<https://www.nature.com/nature/for-authors/formatting-guide>];>here.

Reply: We confirm that the required main text statements have been provided as separate sections, including *Acknowledgements*, *Funding*, *Author Contributions*, and *Competing Interests*. Each section has been prepared in accordance with Nature's requirements and includes the necessary information.

DATA AND CODE AVAILABILITY STATEMENTS: Any manuscript reporting original research must include a Data Availability statement that makes transparent to the reader the conditions of access to the “minimum dataset” that is necessary to interpret, verify and extend the research in the article. This minimum dataset may be provided through deposition in public community/discipline-specific repositories, custom proprietary repositories (for certain types of datasets), or general repositories like Figshare, Zenodo and Dryad. We strongly discourage providing large datasets in Supplementary Information; the preferred approach is to make data available in repositories. More information on Nature Portfolio's reporting standards and guidance on preparing your Data Availability statement can be found [here](#).

For all studies using custom code or mathematical algorithms that are deemed central to the conclusions, a Code Availability statement must be included, indicating whether and how the code or algorithm can be accessed, including any restrictions to access. The Code Availability statement is listed as a separate section after the Data Availability statement but before any additional references. Code should be deposited in a DOI-minting repository such as Zenodo, Gigantum or Code Ocean and cited in the reference list. Authors are encouraged to manage subsequent code versions and to use a license approved by the open source initiative.

Additional details can be found [here](#).

Reply: We have separated the *Data Availability* and *Code Availability* statements into two distinct sections. And we have made all data fully publicly available in iProx (IPX0007409000). And the project data analysis codes are deposited in GitHub (<https://github.com/guomics-lab/PTV-1>).

DISPLAY ITEMS: We suggest that you take stock of all data that have been generated throughout the review process and ensure that only the data most central to the conclusions are presented in the main text figures. Any figures included within the main text file during the review process must be removed from the final main text file and uploaded as separate, individual files; they will be integrated into the main paper in print and online. An overview of the key features of this presentation may be found [here](#).

Figures should be comprehensible to readers in other disciplines and assist in understanding of the paper. Main text figures (but **not** Extended Data) must be provided in production-quality versions in an editable format (i.e., .ai, .cmx, .cdr, .doc, .eps, .pdf, .ppt, .ps, .psd, .svg

and .xls); we cannot accept figures in .cvs, .gif, .jpg, .png and .tif formats. We highly encourage you to consult our [artwork guidelines](#). They should be as small and simple as is compatible with clarity. All panels of a figure should be logically connected and assembled on a single page in a rectangular shape; any essential alignments (parts horizontal, vertical, spacings, etc) should be indicated. Each panel of a multipart figure should be sized so that the whole figure can be proportionally reduced and reproduced on the printed page at the smallest size at which essential details are visible. Nature's standard figure sizes are either 9 or 18 cm wide; the maximum permitted height is 17 cm. Panels should be arranged to fit these widths while minimizing excess space around the panels. Tables should be prepared using the Table menu in Word. As we must be able to edit the figures so that they conform to our house style, the submission of files that are incorrectly formatted, flattened, or of insufficient resolution may delay final acceptance of your manuscript.

Reply: We have revised the figures in accordance with Nature's requirements. In particular, we have removed the table from Figure 3G and relocated it to *Supplementary Table 10*.

THIRD PARTY RIGHTS: You must provide proof that you have secured permission to use any third party materials that appear in any part of your manuscript, including Extended Data and Supplementary Information. Please fill out a [Third Party Rights Table](#), and upload this with the final version of your manuscript. Third party materials include any figures, tables, images, videos or text boxes that are reproductions or adaptations of items that have previously been published elsewhere and/or are owned by a third party. This includes pictures taken by professional photographers, maps and images downloaded from the internet. You will need to obtain the right to use each of these items before your paper can be accepted for publication. You will also need to give proper attribution to the copyright holders in your paper. Please ensure you upload any necessary grants of rights alongside the final version of your manuscript. More information is available on our [Rights and permissions page](#). Failure to obtain the appropriate rights and to supply a completed third party rights table will delay the publication of your article. The editorial assistant (cc'd) can help with any questions.

Reply: We have checked all illustrations for third-party rights issues and obtained the necessary permissions. All relevant details, including BioRender licenses where applicable, have been included in the third-party rights table.

COVER ARTWORK: We welcome submissions of artwork for consideration for our cover. More information can be found in our [guide for cover artwork](#). The file name(s) should include the manuscript reference number and be labelled as a cover suggestion; a short description is also preferred. Illustrations should be selected more for their aesthetic appeal than for their scientific content. We cannot promise that your suggestions will be selected for

the cover, as competition is intense.

Reply: Thank you for the invitation. We do not wish to submit a cover artwork suggestion for consideration.

CHEMICAL STRUCTURE PRESENTATION: Any chemical structures in the main text or Extended Data figures must conform to our [chemical structure style guide](#). This guide lists the ChemDraw preferences and [stylesheet](#) that must be used to draw all structures. The style and size of chemical structures should not be modified from the default settings in the template, unless absolutely necessary (see the guide for examples), in which case 80% size and 5 pt font is the smallest size possible. Please export any ChemDraw (.cdx) files as a PDF, retaining editing capabilities — we find that ‘print to pdf’ works well for this — and upload this with your manuscript.

Reply: We confirm that this study does not include any chemical structures in the figures

IMAGE INTEGRITY: We strongly advise that you go carefully through all the data (including Extended Data and Supplementary Information) to ensure there are no accidental image/data duplications, other image manipulations or data errors. Such issues generally require correction after publication. Any image provided for publication, either in print or online (including Extended Data and Supplemental Information), may be subject to a quality control process to check for image integrity and manipulation. A discussion of our standards regarding how images should be prepared and presented can be found [here](#).

Reply: We confirm that all images and data have been prepared and presented in accordance with the journal’s image integrity standards.

EXTENDED DATA: Extended Data do not appear in print but are included online within the full-text HTML and integrated in the downloadable PDF. Extended Data are an integral part of the paper, and only data that directly contribute to the main message should be included. All Extended Data must be referred to in the main text, and their legends should be listed sequentially at the end of the main text file, not in the Extended Data files. Extended Data should be assembled into a maximum of 10 A4 size, multi-panelled display items. They must be supplied as individual files in .jpg, .tif or .eps format **only**. They should be of the same quality as the main figures, but there are important differences in their formatting. More specific instructions are provided [here](#). If you need to describe a complex process, we encourage you to add a schematic of the main finding as part of the Extended Data to aid readers unfamiliar with the immediate discipline.

Reply: We confirm that Extended Data have been prepared and presented in accordance with the journal’s standards.

SUPPLEMENTARY INFORMATION: Supplementary Information (SI) is online-only, peer-reviewed material that is essential background to the study (e.g., large data sets, more complex methods, and calculations), but which is too large or impractical, or of interest only to a few specialists, to justify inclusion in the print version of the paper (see [here](#) for further details). While SI should not typically contain data figures (any figures additional to those appearing in the main text should be formatted as Extended Data), we require that the raw, uncropped data for gels be presented as an SI figure (see below). Tables may be included in SI, but only if they are unsuitable for formatting as Extended Data (e.g., tables containing large data sets that cannot fit a single page or raw data tables that are best suited to Excel files). If a manuscript has SI, each discrete SI item (e.g., videos, tables) must be referred to at an appropriate point in the main text file. You must also provide a Word file entitled “SI Guide”, containing a cover page with manuscript title and author information; a table of contents (preferably with page numbers); and then any SI text, notes, figures, and titles and legends for any separate SI files; for additional information see [here](#).

Reply: A standalone "*SI Guide*" document has been provided as requested.

We recommend that you pay careful attention to the formatting of the SI because it is not subedited. After the paper has been accepted, SI files can only be amended for critical changes to the scientific content, not for style.

Reply: We have carefully reviewed the Supplementary Information for formatting, consistency, and clarity, and have made the necessary style and formatting checks before resubmission.

CELL LINE IDENTIFICATION: To help curb the inadvertent use of cross-contaminated or misidentified cell lines, we ask that you check your reagents against the [list of commonly misidentified cell lines](#) maintained by the International Cell Line Authentication Committee, which is also accessible through the NCBI BioSample [database](#). If you have used a cell line that is on this list, you must provide a scientific justification and state the identity issue in the Methods. The editors reserve the right to demand that the data be removed from the paper if the justification is deemed unsatisfactory. In addition, authors must identify the source of cell lines (with catalog number if obtained from a vendor or cell bank) and report whether the cell lines have been authenticated, including the method used, the results, and the date authentication testing was last performed for that cell line. You should be able to provide the test results upon request. Mycoplasma contamination testing status must also be reported. These requirements will be particularly scrutinized for cancer studies, where the issue of cell

line misidentification has been well documented. Resources on cell line authentication are available [here](#).

Reply: We confirm that none of the cell lines used in this study appear on the International Cell Line Authentication Committee's list of commonly misidentified cell lines. Cell line sources and catalog numbers are provided in **Supplementary Table 1**. This information has been synchronized in the Reporting Summary and **Methods of SI**.

“All cell lines were authenticated by short tandem repeat (STR) profiling, most recently performed on June 12, 2026. Mycoplasma contamination testing was negative, assessed by PCR-based detection using Genetic Testing Biotechnology (GTB)'s Mycoplasma Detection Kit, which detects over 500 mycoplasma species with a detection limit of 0.1 ng template DNA. The method was performed as previously described [Uphoff CC, Drexler HG. Detection of mycoplasma contaminations. Methods Mol Biol. 2013;946:1-13].”

SOURCE DATA (GRAPHS): To increase transparency, we strongly encourage you to provide, in spreadsheet form, the data underlying the graphical representations used in figures. For all experiments presenting data from animal models, this is a requirement and is not optional. This is in addition to our well-established data-deposition policy for specific types of experiments and large datasets. Online readers of the manuscript will be able to access the graphical source data directly from the figure legend. Spreadsheets must be submitted in .xls, .xlsx or .csv formats. One file per figure is permitted. If there is a multi-panelled figure, the source data for each panel should be clearly labeled in the file; alternatively the source data for a figure can be included in multiple, clearly labeled sheets within an Excel file. File sizes of up to 30 MB are permitted, but it is expected that the vast majority of graphical source data files will be considerably smaller than this. When submitting these files with your manuscript, you should select the "Source Data" file type and use the title field in the file description tab to indicate the figure(s) to which the source data pertain. Source data should not be provided as Extended Data.

Reply: We confirm that this study does not include any experiments using animal models.

RAW DATA (GELS): You must provide the original source images for all data obtained by electrophoretic separation (e.g., EMSA, northern/Southern/western blots, etc). The raw images must be assembled into a single .pdf or .tif file (multiple gels on a single page is encouraged). The file should be uploaded as Supplementary Figure 1. The full scanned images must be in uncropped form and contain labeled size/molecular weight markers and loading controls. There should be an accurate indication of how the gels were cropped for the final figure. The figure legends and raw data files should indicate whether controls (such as beta-actin) were run on the same gel as loading controls, or on separate gels as sample

processing controls (see [here](#) for guidance). While the data can be displayed in a relatively informal style, there must be a correspondence between each source data image and a specific main text or Extended Data figure. The main text or Extended Data figure legends should refer to the uncropped scans explicitly (e.g., “For gel source data, see Supplementary Figure 1.”). For examples, see [here](#) or [here](#).

Reply: We confirm that this study does not include any data obtained by electrophoretic separation (e.g., EMSA, northern/Southern/western blots). Therefore, no gel source data are applicable.

TUMOUR DATA: The Methods must detail the maximal tumour measurements/volume/other endpoints that were permitted by your IACUC and state that these limits were not exceeded in any of the experiments. You must also provide the source data for any figures presenting an analysis of tumour growth. Where appropriate, these data should be displayed as the longitudinal measurements per mouse. They should be uploaded as a “Source Data” file type.

Reply: We confirm that this study does not include any animal experiments or tumour models.

DATA DEPOSITION: The following specific points may be relevant to your paper, so please ensure that you provide the following information:

* Sequences for any RNAi/small RNA constructs must be included.

Reply: We have provided all siRNA sequences

* Accession numbers for gene expression data or RNA sequencing data must be listed.

Reply: We confirm that this study does not include gene expression microarray data or RNA sequencing data.

* Papers reporting protein structures must conform to our standards listed in the [Guide to Authors](#). The Data Availability statement must state that the X-ray crystallographic coordinates and structure factor files (or comparable NMR or cryoEM data) have been deposited in the appropriate, named, public database, along with all relevant accession number(s). You must use the standard Nature [templates for structural data](#); there are separate links to tables for X-ray crystallographic, NMR and cryoEM structures. These tables must be presented as Extended Data; if the number of entries causes the table to exceed a page, it must be divided into two Extended Data items. The contour level of any electron density maps presented, as well as the type of map (i.e., Fo-Fc or 2Fo-Fc), should be explicitly stated in the figure legend.

Reply: We confirm that this study does not report any protein structures (X-ray crystallography, NMR, or cryo-EM data).

* For every new chemical compound, a complete description of the synthesis and the physical characterization (i.e., NMR, MS, etc) must be included in the Supplementary Information (see [here](#)).

Reply: We confirm that this study does not involve the synthesis or characterization of any new chemical compounds.

* Papers containing new or revised formal taxonomic nomenclature for animals, whether living or extinct, are accepted conditional on the provision of LSIDs (Life Science Identifiers) by means of registration of such nomenclature with ZooBank, the online registration system for the International Code of Zoological Nomenclature (ICZN). ZooBank LSIDs can be resolved and the associated information viewed through any standard web browser by appending the LSID to the prefix "<http://zoobank.org/>".

Reply: We confirm that this study does not include any animals.

* We strongly encourage deposition of 3D morphological data in a suitable repository such as MorphoBank, MorphoSource or similar; the relevant accession numbers should be listed in the Data Availability statement.

Reply: We confirm that this study does not include 3D morphological data

* For animal experiments, you must confirm that all experiments were performed in accordance with relevant guidelines and regulations. There should be a statement identifying the institutional and/or licensing committee approving the experiments, including any relevant details. Sex and other characteristics of animals that may influence the results must be described. Details of housing and husbandry must be included if they are likely to influence experimental results. Further details can be found [here](#).

Reply: We confirm that this study does not include any animal experiments.

* Human genotype data (e.g., SNP array data) should be deposited into a public database (dbGAP or EGA) with a controlled access policy.

Reply: We confirm that this study does not include any human genotype data.

* A full clinical and pathological characterization of patients/human subjects and samples

should be provided in tabular format, including the magnitude of response for each patient (partial, complete, stable disease), the site of the biopsy, whether or not that lesion was progressing and mutational status if appropriate.

Reply: We would like to clarify that the patients included in our study were not treated in the neoadjuvant setting. Instead, they were postoperative patients who subsequently received adjuvant therapy. Therefore, parameters such as treatment response magnitude for the biopsied lesion (for example, partial response, complete response, or stable disease) and whether the biopsied lesion was progressing are not applicable in this cohort.

In addition, the specimens used for our analyses were derived from surgically resected tumor tissues obtained prior to any drug treatment. As described in the Methods section, we selected tissue sections containing more than 70% tumor content for downstream proteomic analysis. We further clarify no genomic testing was performed for these patients as part of this study; therefore, mutational status is not available.

To comprehensively characterize the cohort, we have provided the available clinical and pathological information in *Supplementary Table 13* for each patient, including TNM stage, recurrence status, recurrence-free survival time, survival status, overall survival time, treatment drug combinations, and the corresponding proteomic data.

CONSORTIA: For papers containing one or more consortia, all members of the consortium who contributed to the paper must be listed in the paper (i.e., print/online PDF). If necessary, individual authors can be listed in both the main author list and as a member of a consortium listed at the end of the paper. When submitting your revised manuscript via the online submission system, the consortium name should be entered as an author, together with the contact details of a nominated consortium representative. See [here](#) for our authorship policy and [here](#) for further consortia formatting guidelines, which should be adhered to prior to acceptance.

Reply: This manuscript does not include any consortium authorship or consortium contributors.

We will not send your revised paper for further review. If the revised paper is in our format (as detailed above), in accessible style and of appropriate length, we shall begin the acceptance process.

In order to accept your paper, we require the following electronic files:

* A cover letter describing your response to any editorial comments and detailing any format

changes during revision, particularly if the overall length is affected.

* A point-by-point response (preferably in Word) to any remaining issues raised by our referees.

* The final version of your text as a Word document. Word Equation Editor/MathType should be used only for formulae that cannot be produced using normal text or symbol font. If this is not possible, the manuscript can be supplied as a single plain vanilla TeX or LaTeX file that includes all references and abbreviations, with no special formatting, as well as a PDF version that is uploaded as a 'related manuscript file'.

* Production-quality versions of all figures (see above).

* The final version of the Extended Data.

* The final version of any Supplementary Information, presented as one file (ideally a PDF) if feasible, as well as a separate SI Guide.

* Source Data, if appropriate.

* For optimal quality videos we encourage H.264 encoding and a standard aspect ratio of 16:9 (4:3 is second best), without compression.

* Completed and signed copies of the following **four (or five) forms**, uploaded as a "Related Manuscript File" file type:

- 1) [Biology editorial checklist](#);
- 2) [Manuscript checklist](#);
- 3) [Reporting summary](#);
- 4) [Third-party rights table](#);
- 5) [Code and software submission checklist](#) (if applicable).

Reply: We have uploaded these files.